Pothole impacts on rental e-scooters and the case for a larger front wheel: measurements in Oxford and a dynamics model

Kai Yi Ji

Draft, 10th October 2026

Abstract

A small front wheel drops further into a pothole and strikes its far edge harder, which may make e-scooters more likely to fall. We asked whether a larger front wheel would help, using the Voi Voiager 5 rental scooter, whose tyre markings show an 11.1 in front wheel and a 10.1 in rear wheel. First, we recorded three legal rides in Oxford (6.08 km in total) with a smartphone fixed to the handlebar. It picked up 100 sharp jolts, each at least as strong as the pull of gravity (1 g), or about one every 61 m. Ten places jolted the scooter on more than one ride, and 6 of these 10 had no public road-surface report within 25 m in the year before. Second, we ran a physics simulation (MuJoCo) of a scooter and rider crossing potholes, comparing the Voiager 5's wheels with a smaller 10 in reference, with bigger front wheels only and with bigger wheels front and back. With the Voiager 5's front suspension, the simulated rider fell in 9 of 81 runs with its 11/10 in wheels, as often as with 10 in wheels, but in 6 with a 12 in front wheel and in 1 with a 14 in front wheel. A bigger front wheel on its own matched bigger wheels at both ends in every case of the main set of runs, and stayed within one fall of it when the pothole edges were sloped, which supports our hypothesis that the front wheel matters most. These results come from one rider and a simplified model that has not been checked against physical tests; they do not show that today's scooters are unsafe, and the 1 g level is our own threshold, not a danger limit. They suggest that front wheels larger than the Voiager 5's, around 12–14 in, deserve physical testing, and that current wheel-size rules for rental trials (11 in in London, none elsewhere in Britain) may be set too low to make a difference.

1 Introduction

1.1 Background

Rental e-scooters are ridden on British roads under trials authorised by the Department for Transport (DfT). They are the only e-scooters that may legally be used on public roads; privately owned e-scooters may be ridden only on private land with the landowner's permission (Department for Transport 2026a). The design of every legal e-scooter on a British road is therefore set by a small number of rental operators, within the vehicle rules of the national trials and of the local areas that host them.

In 2025, 10 e-scooter riders were killed and about 410 seriously injured in reported road collisions in Great Britain, and 384 of the 1,377 reported collisions involving an e-scooter (28%) involved no other vehicle (Department for Transport 2026b). A recent systematic review found falls to be the most common type of e-scooter accident, with human, vehicle and environmental factors all contributing (Nathania et al. 2026). Riders themselves name infrastructure hazards among the risks they perceive (Anke et al.

2025), and in a study of injured riders in Washington, DC, injuries sustained on the road were about twice as likely to be of moderate or greater severity as those sustained elsewhere (Cicchino et al. 2021).

Small wheels are a plausible part of the vehicle side of this problem. A wheel that drops into a pothole meets the far edge at a steeper angle the smaller it is, and the front wheel of a scooter both steers and takes the first impact. Using a multibody model of an e-scooter with 10 in wheels, Posirisuk et al. (2022) found that 3 cm deep potholes caused no falls, while 6 cm deep potholes caused falls in 41 of 60 simulated cases, with the rider pitching forward over the handlebar. Other models have examined falls caused by striking a kerb (Fournier et al. 2023), and experimental tests of small-wheeled personal vehicles found compact, small-wheeled devices less stable than those with larger wheels; their authors recommend wheels of at least 14 in and front suspension (Paudel et al. 2025). Instrumented tests have related e-scooter stability and comfort to vehicle specifications (Leoni et al. 2022), and vibration and comfort on e-scooters have been measured and modelled (Cano-Moreno et al. 2022; Ventura et al. 2024; Vaca Michilena and Cano-Moreno 2026). Smartphones and other low-cost sensors have also been used to map potholes and road roughness from moving vehicles (Eriksson et al. 2008; Zang et al. 2018).

1.2 The regulatory gap

Neither the national trial guidance (Department for Transport 2024) nor the legal definition of a trial e-scooter (UK Government 2020) sets a minimum wheel size. The guidance's stability tests include riding through a depression 1 m long and 5 cm deep with vertical walls. London's trial goes further: since November 2022, its rental e-scooters must have a front wheel of at least 11 in and a rear wheel of at least 9 in (Transport for London 2022), and Transport for London describes its vehicle standards, including larger wheels, as going beyond the national ones (Transport for London 2023). Outside London the national rules set no wheel size; London's 11 in is the only minimum we found, although we did not check the contracts of individual trial areas, which could add their own.

The Voi Voiager 5, the model ridden in this study in Oxford (Voi Technology 2026a; Voi Technology 2026b; Oxfordshire County Council n.d.), sits at that minimum. Published sources give its wheels as 10 in without separating front and rear (Levy Electric n.d.; Alsalkini 2023), but the author's photographs of a parked Voiager 5 in Oxford show a 60/55-8.5 front tyre and a 60/55-7.5 rear tyre (Figure 1). Their nominal outer diameters are 282 mm (11.1 in) and 257 mm (10.1 in): the front wheel is larger than the rear and meets London's 11 in minimum by about 2.5 mm. We use these photographed sizes, written 11/10, throughout. The model was introduced in 2022 with a larger front wheel, deeper tyre tread and greater shock absorption than its predecessor (Bowden 2022), and was still being deployed in new trial areas in 2025 (Essex Highways 2025). The open question is whether a front wheel at London's minimum is enough, and what a larger one would change.

1.3 Research question and contributions

This paper asks whether pothole impacts on the Voiager 5 reach levels associated with loss of stability, and whether a front wheel larger than its current 11 in would reduce them. It tests a specific hypothesis that could turn out to be false:

> Because the front wheel both steers and takes the peak load at impact, enlarging the front wheel alone gives most of the stability benefit of enlarging both wheels.

If the hypothesis held, a front-wheel minimum such as London's would capture most of the benefit of larger wheels, at lower cost than enlarging both. If it failed, for example because the rear wheel's

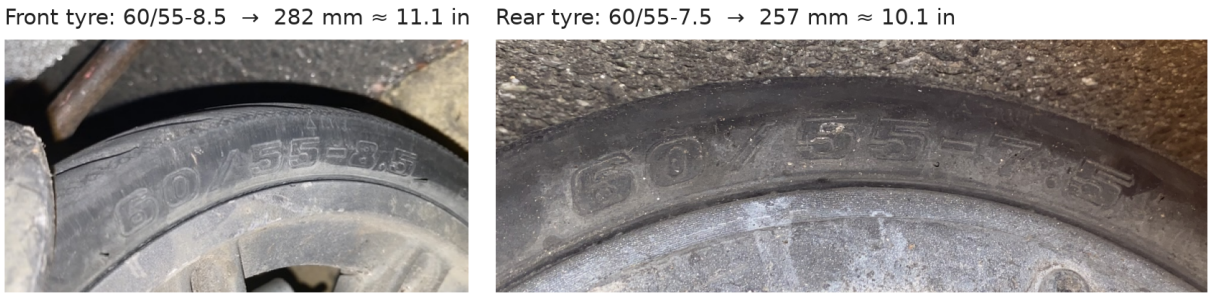


Figure 1: Tyre sidewall size codes of a Voi Voiager 5 parked in Oxford, photographed by the author on 9 October 2026 (crops). Metric code width/aspect ratio-rim diameter; nominal outer diameter = rim + 2 × width × aspect ratio. The photographed scooter is the same model as the three ridden scooters (Voiager 5).

impact also caused falls, a front-only rule would be insufficient.

The work has two parts. **Part A** (Section 2) measures what a rider experiences: three legal rides on Voiager 5 scooters in Oxford, recorded with a handlebar-mounted smartphone, give the number, size and location of sudden vertical impacts, which are compared with public pothole reports. The recording method itself is checked, including whether it can tell riding from walking. **Part B** (Sections 3 and 4) uses a geometry model and a planar dynamics model of scooter and rider to compare the current 11/10 wheels with a smaller 10/10 reference, with larger front wheels alone and with larger wheels front and rear. The measured impacts provide a consistency check for the model, not a validation.

Compared with earlier work, the contributions are: measurement of discrete impacts on a rental e-scooter in ordinary use, rather than comfort vibration; repeated impacts at the same places across rides, compared with what the public had reported; and a direct test of whether a larger front wheel alone performs like larger wheels front and rear. Section 5 discusses what the results do and do not show and what they suggest for trial rules.

2 Measuring what riders experience on the road

2.1 Data

Rides. One rider recorded three rides on Voi rental e-scooters in Oxford, ridden legally within the national e-scooter trial (Department for Transport 2024; Oxfordshire County Council n.d.): one on 3 October 2026 (vehicle SZ22) and two on 7 October 2026 (vehicles UPAR and BTLZ). Voi customer support confirmed that all three vehicles were Voiager 5 scooters (Voi Technology 2026a; Voi Technology 2026b). The rides covered parts of Marston Road, Morrell Avenue, Old Road (Headington), Headington Road and connecting streets. Only the rider's own rides were recorded.

Recording. An iPhone 11 Pro Max was mounted on the handlebar and recorded with the Sensor Logger app, version 1.67.1 (Choi 2026). The app logged user acceleration (gravity removed) and the gravity vector from the iOS sensor fusion (Apple Inc. n.d.[b]), the raw accelerometer, the gyroscope (all at 100 Hz) and GPS (about 1 Hz). The phone was set up before each ride and not touched while riding.

Table 1: The three rides. Riding time and distance count only travel at 7 km/h or faster (section 2.3).

Ride	Date, start (local)	Vehicle (Voiager 5)	Recorded	Riding	Distance ridden
1	3 Oct 2026, 18:30	SZ22	13.7 min	9.6 min	2.89 km
2	7 Oct 2026, 10:37	UPAR	35.3 min	5.9 min	1.48 km
3	7 Oct 2026, 13:38	BTLZ	11.4 min	6.3 min	1.71 km
All			60.4 min	21.8 min	6.08 km

Ride 2 also includes a walk after the rental ended and a long period with the phone at rest; these parts are excluded from the impact analysis by the speed rule and were used to check the recording method (section 2.2).

2.2 Checking the recording method

Before using the phone data, we checked that they are internally sound and that riding can be told apart from walking. The full method and results are given in Appendix B; the main points are summarised here.

Data quality. In all three recordings the motion sensors ran at 99.72 Hz with sampling intervals of 10.03 ms and no gaps, duplicated or out-of-order timestamps. During the one period when the phone lay still (ride 2, after the walk), the acceleration noise was about 0.001 g and the raw acceleration magnitude was 0.994 g, both small against the 1.0 g impact threshold used below. The phone's own estimate of GPS horizontal accuracy had a median of 11–14 m per ride. These checks show internal consistency and plausibility, not absolute accuracy, which would need a reference instrument.

Riding versus walking. Walking and riding were distinguished with simple, fixed rules on 4 s windows. The rules used established features: the vertical acceleration obtained by projecting onto the gravity direction (Mizell 2003); the share of its power in the walking step band of 1.2–2.8 Hz, estimated with Welch's method (Welch 1967), around measured mean step frequencies of 1.8–2.0 Hz (Pachi and Ji 2005); the autocorrelation at one step period, a measure of step regularity (Moe-Nilssen and Helbostad 2004); and GPS speed. Similar signals are used for activity and transport-mode detection with phones (Kwapisz et al. 2011; Reddy et al. 2010); our aim was not a new classifier but a check that this app and phone, mounted on a rental scooter, separate the two.

In ride 2 the method found sustained walking from 10:46:47, the first step of a 275 s walk at 2.04 steps per second (Figure 2). This agrees with two references, both with one-minute precision: the Voi app recorded the rental ending at 10:46, and the rider recalled walking "after 10 minutes" of the recording (10:47). In rides 1 and 3, which the rider confirmed contained no walking, no walking bout was detected; isolated walking labels covered 6 s and 8 s, all after the ride had ended. In 1,126 windows where GPS alone showed riding speed (12 km/h or more), none met the walking criteria. With one walking episode this shows that the data separate riding from walking; it does not give a detection rate.

Implications for impact detection. Two findings bear on the method in section 2.3. First, walking alone produced 11 peaks of 1.0 g or more (maximum 1.19 g) in ride 2's walk; the speed rule excluded all

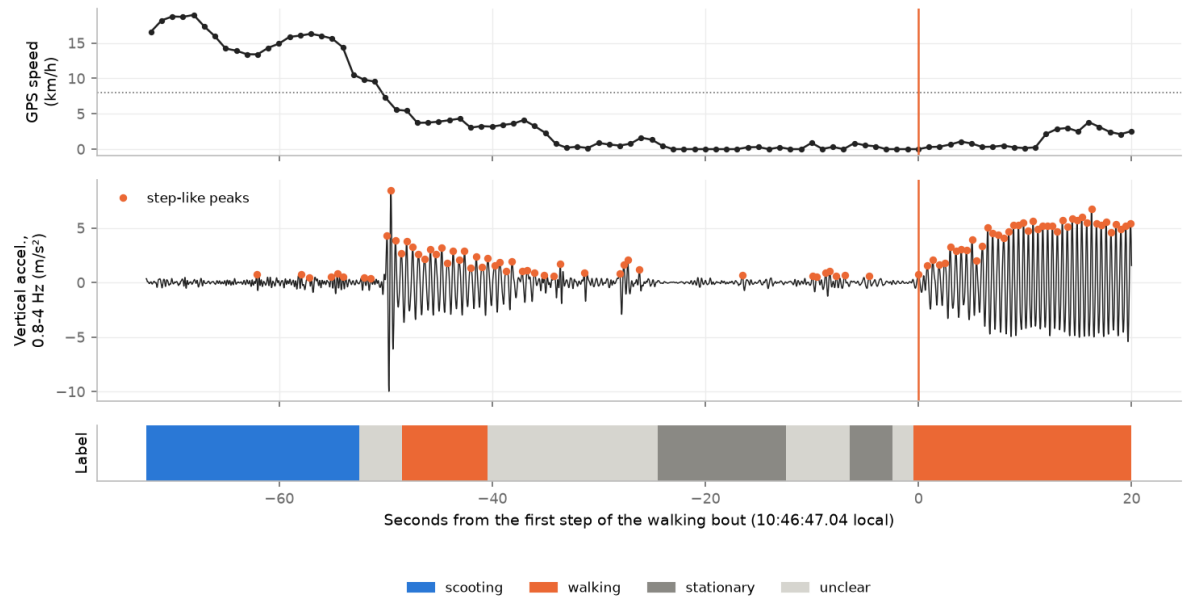


Figure 2: Transition from riding to walking in ride 2. Top: GPS speed; middle: vertical acceleration (0.8–4 Hz) with step-like peaks; bottom: labels from the window rules. Time 0 is the first step of sustained walking (10:46:47). GPS speed rose only about 12 s after walking began.

of them, and all impacts in ride 2 occurred during the ride. Second, the phone's GPS speed reading was sometimes too low. While walking it showed 3.0 km/h, although the distance between GPS positions showed at least 4.6 km/h. While riding it read too low in 4 of 6 stretches, by 0.2–3.2 km/h. Since impacts were only counted above 7 km/h of reported speed, some impacts at low riding speeds may have been missed; we did not measure how many. Dedicated GPS receivers have also been found to underestimate speed on curved paths (Witte and Wilson 2004).

2.3 Impact detection

The user acceleration was projected sample by sample onto the upward direction given by the gravity vector, so that the result does not depend on how the phone sat in its mount (Mizell 2003). An **impact** is a peak in the absolute vertical acceleration of at least **1.0 g**, at least 0.5 s after the previous one (SciPy `find_peaks` (Virtanen et al. 2020)), while the GPS-reported speed was at least **7 km/h**. The speed rule excludes getting on and off, stops, walking and handling the phone. Each impact was located by interpolating the GPS track, using only fixes with a reported accuracy of 20 m or better. For each impact, the rotation rate of the handlebar (gyroscope) in the 1.5 s after the impact was divided by that in the 2 s before it; a ratio of 1.5 or more is reported as **marked wobble**.

The 1.0 g threshold, the 7 km/h speed rule and the 1.5 wobble ratio are analyst choices, not established criteria. A fixed threshold in g was used, rather than one set per ride, so that rides and vehicles can be compared directly.

2.4 Roads and repeated spots

Each impact and each second of riding was assigned to a study road if it lay within 20 m of the road's centreline in OpenStreetMap (OpenStreetMap contributors 2026; Olbricht n.d.), and to "other roads" otherwise. Rates per km are given only where at least 0.2 km was ridden on a road (an analyst choice; shorter distances give meaningless rates).

To find places that produce impacts repeatedly, the impacts of all three rides were pooled and grouped by single-linkage clustering (Sibson 1973) with a 30 m distance, which is larger than the typical GPS error. Each group is a **spot**, ranked in three tiers: impacts from two or more separate rides (tier 1), two or more impacts from one ride (tier 2), and single impacts (tier 3). The rides share one rider, phone and route, so repeated hits on a spot are repeated measurements of one place, not independent confirmations.

2.5 Comparison with FixMyStreet

Public road-surface reports were taken from FixMyStreet (mySociety 2026) for each ride's own area (potholes, carriageway defects, manholes and drain ironwork; reports in two catch-all categories were kept only if their title described the road surface). An impact counts as near a report if a report made in the 365 days before that ride lies within 25 m. Because reports are dense, each rate is compared with a **route baseline**: the share of all points along the ridden route that lie within 25 m of such a report, which is what chance alone would give. With reports from all years instead of the last 365 days, 70–87% of route points lie near a report, which is why only recent reports are used.

2.6 Results

2.6.1 Impacts

The three rides produced **100 impacts** over 6.08 km of riding, 16.4 per km, or one every 61 m (Table 2). Their median peak was 1.27 g (interquartile range 1.13–1.73 g); 32 reached 1.5 g, 12 reached 2.0 g, and 95% were at or below 2.68 g (Figure 3). One impact, in ride 3, reached 6.28 g; all others were 2.97 g or less. Thirty-two impacts were followed by marked wobble, 12 of them also at 1.5 g or more.

Table 2: Impacts per ride and pooled. Pooled values describe the three rides together; the rides are not independent samples.

Ride	Impacts	Per km	Median (g)	Max (g)	≥ 1.5 g	≥ 2.0 g	Marked wobble
1	40	13.8	1.19	2.80	6	4	13
2	39	26.4	1.34	2.87	15	3	11
3	21	12.3	1.67	6.28	11	5	8
All	100	16.4	1.27	6.28	32	12	32

Rates differed between rides (12.3–26.4 per km). Ride 2's higher rate comes largely from one stretch where 13 impacts followed each other every 0.6–1.2 s at about 8 km/h over roughly 20 m; ride 3 also produced a cluster there (spot P3, Table 3). Whether this is a rough surface, such as setts or a broken patch, has not yet been checked.

Speed and severity. In ride 1 the impact peak was not related to speed (Spearman $\rho = 0.03$, $p = 0.88$ (Spearman 1904)). In rides 2 and 3, slower impacts were stronger ($\rho = -0.63$ and -0.68), and pooled $\rho = -0.48$. This pooled association comes mostly from the two dense clusters at about 8 km/h (Figure 4): without the spots of 10 or more impacts, $\rho = -0.12$ ($p = 0.33$, $n = 71$), and with one value per spot, $\rho = -0.36$ ($p = 0.04$, $n = 33$). Because impacts within a spot are not independent, these p-values overstate the evidence; no conclusion about speed is drawn.

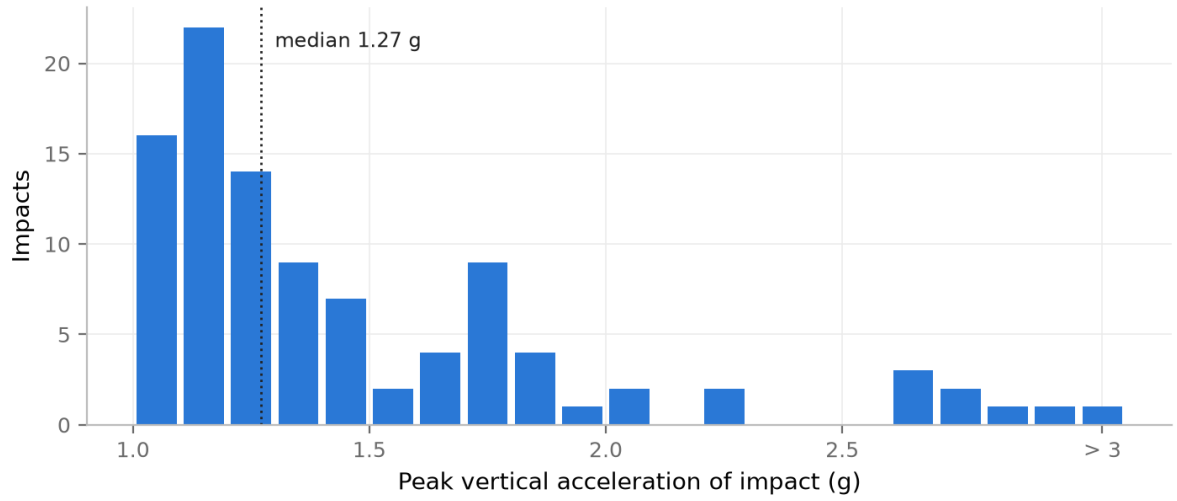


Figure 3: Peak vertical acceleration of the 100 impacts from all three rides. The detection threshold is 1.0 g; impacts above 3 g are shown as one bar (one impact, 6.28 g).

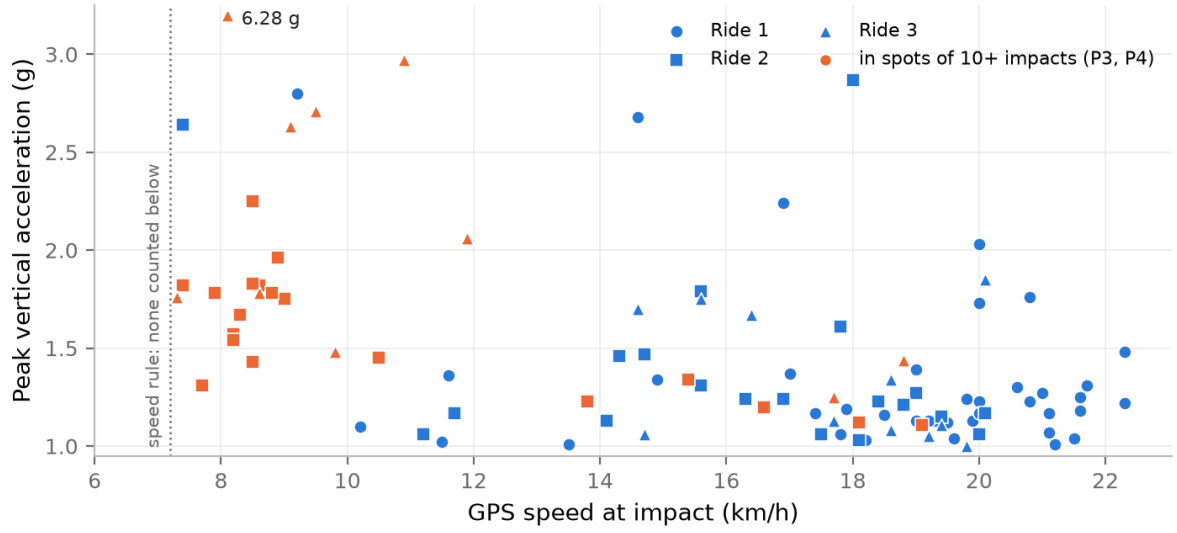


Figure 4: GPS speed at impact against peak vertical acceleration, all 100 impacts (marker shape: ride). Orange: the 29 impacts in the two spots with 10 or more impacts (P3, P4, Table 3), which produce most of the pooled speed–severity association. The 6.28 g impact is drawn at the top edge. Dotted line: speed rule (7.2 km/h); impacts at lower speeds are not counted.

2.6.2 Where the impacts were

Per km ridden, the study roads gave 12.3 impacts (Marston Road, 2.52 km), 14.3 (Morrell Avenue, 1.05 km) and 8.3 (Old Road, 0.24 km), and other roads 22.7 (2.25 km) (Figure 5). Headington Road was ridden for only 0.02 km, too little for a rate.

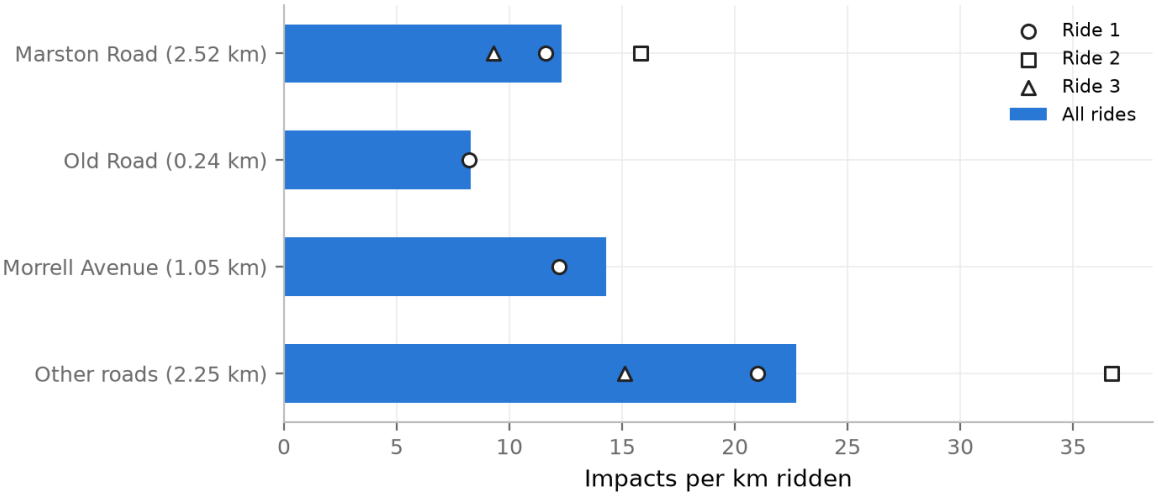


Figure 5: Impacts per km ridden by road, all rides (bars) and per ride (markers). Rates are shown only where at least 0.2 km was ridden on the road by that ride; the distance ridden on each road over all rides is given in brackets.

The impacts formed 33 spots: **10 hit on two or more separate rides**, 10 with several impacts from one ride, and 13 single impacts (Figure 6). Two spots on Marston Road, P1 and P2, were hit on all three rides (Table 3). The strongest single impact, 6.28 g, lies in spot P4, which was hit on two rides.

Table 3: Spots hit on two or more rides, with the nearest FixMyStreet road-surface report made in the 365 days before the ride.

Spot	Rides	Impacts	Max (g)	Road	Recent report within 25 m	Nearest recent report
P1	3	6	2.24	Marston Road	No	60 m (closed)
P2	3	6	1.47	Marston Road	No	44 m (closed)
P3	2	19	2.25	Other	Yes	5 m (closed)
P4	2	10	6.28	Other	Yes	8 m (fixed)
P5	2	9	1.79	Marston Road	No	121 m (closed)
P6	2	4	1.85	Other	Yes	8 m (closed)
P7	2	3	1.17	Marston Road	Yes	8 m (closed)
P8	2	2	2.80	Other	No	38 m (closed)

Spot	Rides	Impacts	Max (g)	Road	Recent report within 25 m	Nearest recent report
P9	2	2	2.03	Marston Road	No	125 m (fixed)
P10	2	2	1.34	Marston Road	No	87 m (closed)

2.6.3 Comparison with FixMyStreet

At the level of single impacts, FixMyStreet reports did not single out the places where impacts occurred. A recent report lay within 25 m of 2 of 40 impacts in ride 1 (5%, route baseline 14%), 12 of 39 in ride 2 (31%, baseline 33%) and 9 of 21 in ride 3 (43%, baseline 34%): close to what chance would give in each ride. At the level of spots, 6 of the 10 spots hit on two or more rides had no recent report within 25 m, including both spots hit on all three rides (Table 3). Spot P4, with the 6.28 g impact, had a report within 8 m marked "fixed"; its report and repair dates have not been checked. A missing FixMyStreet report does not mean that a defect was unknown to the highway authority, which also receives reports through other channels and inspects roads itself.

2.6.4 Summary of Part A

Riding three Voiager 5 scooters over 6.08 km produced 100 jolts of 1 g or more at the handlebar, about one every 61 m, a third of them 1.5 g or more. Ten places produced impacts on two or more separate rides, and most of these had no recent public report. These results come from one rider, one phone and short routes, and use analyst-chosen thresholds; they describe exposure on these rides, not the risk to riders in general.

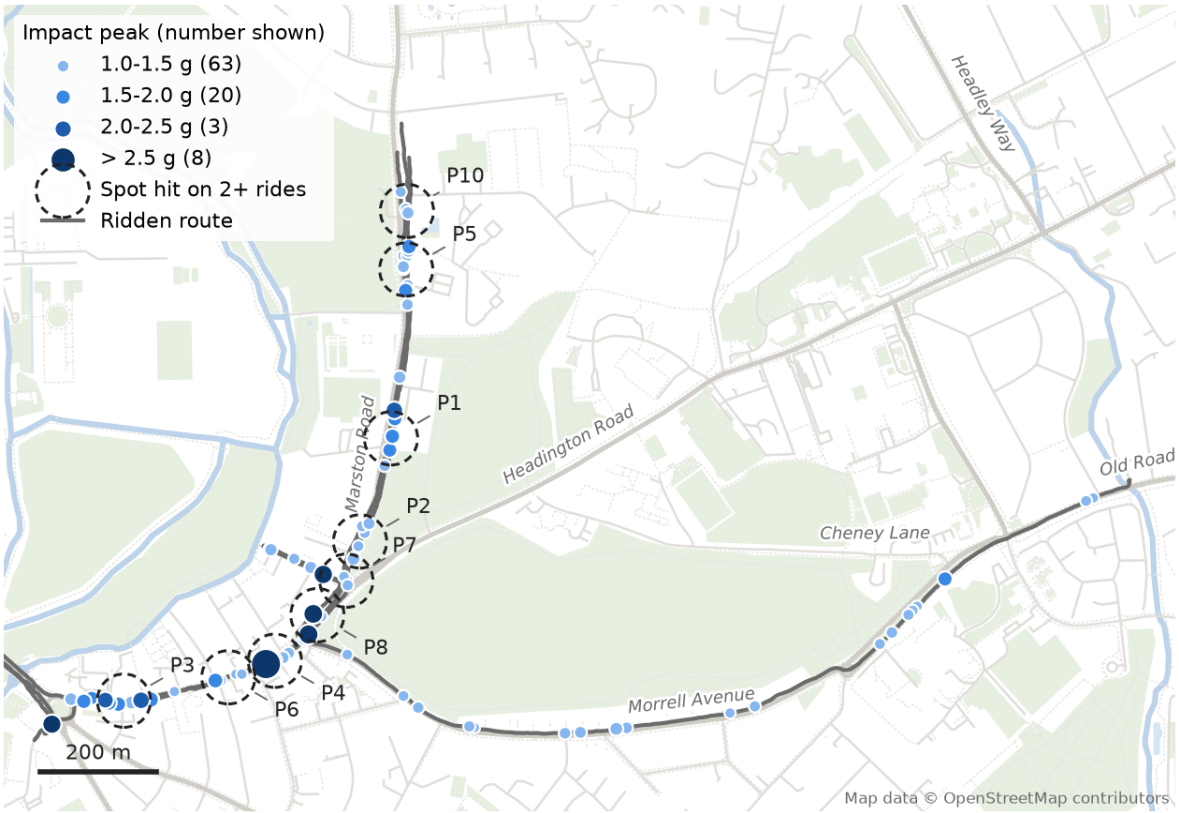


Figure 6: Impacts from all three rides on the ridden routes, coloured and sized by peak vertical acceleration; dashed circles mark the 10 spots hit on two or more rides (Table 3). For privacy, the first and last 200 m of each route are not drawn; the 6 impacts in those parts are included in all statistics. Background: streets, paths, parks and water from OpenStreetMap (OpenStreetMap contributors 2026), map data © OpenStreetMap contributors (ODbL).

3 Methods: modelling the effect of wheel size

3.1 Overview

Rental e-scooters are the only e-scooters that may legally be ridden on UK public roads, so a scooter with modified wheels cannot be tested on the road. We therefore used two models, in the same way as the Imperial College study of e-scooter falls this work builds on (Posirisuk et al. 2022):

1. a **geometry model**, which describes how far a single wheel drops into a pothole and how hard it strikes the far edge, and
2. a **planar multibody dynamics model** of scooter and rider in MuJoCo (Todorov et al. 2012), which predicts whether the scooter pitches over.

Both models compare wheel configurations under identical assumptions. They are not validated against physical tests, so we report relative differences between configurations, not real-world fall thresholds.

3.2 Wheel configurations

Configurations are written front/rear diameter in inches (nominal size taken as tyre outer diameter).

Configuration	Meaning
10/10	Smaller reference
11/10	**Current vehicle**: Voi Voiager 5, the model of all three ridden scooters as confirmed by Voi customer support (Voi Technology 2026a; Voi Technology 2026b), with tyre sizes from the author's photographs (Figure 1; nominally 11.1/10.1 in, modelled as 11/10)
12/10	Larger front wheel only
12/12	Larger wheels front and rear
14/10	Much larger front wheel only

A front/rear ratio study added 16/10, 20/10, 16/8 and 20/8 (ratios up to 2.5).

3.3 Geometry model

A rigid wheel of radius R rolls at speed v across a rectangular pothole of depth d and length w with sharp edges. The wheel's lowest point sinks below road level by

$$h = \min\left(d,\ R - \sqrt{R^2 - (w/2)^2},\ \tfrac{1}{2} g \left(\frac{w}{v}\right)^2\right),$$

the smallest of the depth, the sag of a wheel resting on both edges, and the free-fall distance in the time taken to cross (an upper bound). The far edge then meets the tyre at an angle θ from vertical, with $\cos\theta = (R - h)/R$. Treating the impact as plastic, the velocity lost into the edge is $v \sin\theta$ and the forward speed loss of the axle is $v \sin^2\theta$. A larger wheel gives a smaller θ for the same pothole, so both quantities fall. Each wheel is analysed separately, so this model cannot say whether the scooter falls; that requires the dynamics model.

The geometry model was evaluated for 8–16 in wheels, speeds of 15, 20 and 25 km/h, depths of 0–120 mm and lengths of 150–1000 mm (`scripts/wheel_geometry.py`).

3.4 Dynamics model

Structure. The model is planar (side view) with six degrees of freedom: forward position, height and pitch of the chassis; rotation of each wheel; and forward lean of the rider. It follows the structure of Posirisuk et al. (2022), who modelled the scooter as the front wheel and the rest of the scooter joined at the axle, but adds a separate rear wheel, so that front-only and front-and-rear changes can be compared, and a rider. The rider is a single rigid body pinned to the deck at the feet; a rotational spring (1500 N·m/rad) and damper (150 N·m·s/rad) represent the rider bracing with legs and arms, chosen so that braking at 0.3 g leans the rider about 8° forward.

Parameters. Voi does not publish the vehicle's mass properties, so most parameters are assumed (Table 4). The rider's mass and height match the median male model of Posirisuk et al. (2022).

Table 4: Main model parameters.

Parameter	Value	Source
Scooter mass	32 kg	Assumed (Voiager 7 reported at 36 kg (Startups Magazine 2024))
Wheel mass	2.5 kg each	Assumed; scaling with diameter did not change results
Wheelbase	0.95 m	Assumed
Deck height	0.16 m (10 in wheels), rising with wheel size; deck kept level	Assumed
Handlebar height	1.05 m	Assumed
Rider	76 kg, 1.70 m, centre of mass at 55% of height	Posirisuk et al. (2022); centre-of-mass fraction assumed
Tyre compliance	Contact time constant 0.025 s (≈80 kN/m, ≈6 mm static deflection)	Assumed, typical small pneumatic tyre
Front fork	25 kN/m, damping ratio 0.3, 30 mm travel	Assumed; the Voiager 5 has a front spring (Levy Electric n.d.)
Friction	0.9	Dry asphalt, assumed
Integration	Implicit-fast, 0.5 ms time step	Numerical choice

Tyre contact. Wheels are rigid cylinders whose contact with the road is softened through MuJoCo's soft-contact parameters to represent tyre compliance.

Front suspension. Because the Voiager 5 has a front spring, the main results use a sprung front fork (Table 4). Its values were fixed before any run and tested in the sensitivity analysis. Results without suspension are reported for comparison.

Pothole. The main experiments use a rectangular pothole with vertical walls and sharp edges, 5 m after the start, giving about 1 s of run-up. To test potholes with broken edges, a second contact model computes tyre–road contact directly against a sampled road profile in which each edge falls over a width b with a raised-cosine shape (0, 25, 50 or 100 mm). It applies one radial spring-damper force per

wheel. Before use, it was checked on a flat road (0.01 g at the handlebar), for convergence with profile resolution, and against MuJoCo's own contact on sharp edges, where fall outcomes agreed in 30 of 30 cases. Its absolute accelerations are too high because a point contact cannot envelop an edge, so it is used only to compare falls between edge shapes.

Procedure. Each run settles the model at rest for 0.5 s, sets the forward speed and matching wheel spin, and lets the scooter coast (no motor or braking, no steering) across the pothole.

Outcomes. Speed loss, maximum scooter pitch and rider lean, and peak acceleration at the handlebar, where the phone was mounted. The handlebar acceleration is averaged over 10 ms windows to match the phone's 100 Hz sampling.

Fall definition. A run counts as a fall if the rider leans more than 30° forward, the scooter pitches more than 30° nose-down, or forward speed drops below 0.5 m/s. These thresholds are analyst-defined and were fixed before any experiment. A fall is attributed to the rear wheel only if the rear wheel dipped more than 5 mm into the pothole before the fall; otherwise it is attributed to the front-wheel impact.

3.5 Experiments

Table 5: Simulation experiments.

Experiment	Design	Runs
Main sweep	5 configurations × 9 depths (20–100 mm) × 3 lengths (300, 600, 1000 mm) × 3 speeds (15, 20, 25 km/h); with and without front suspension	405 each
Sensitivity	Tyre stiffness (τ = 0.015, 0.035 s), rider bracing (1000, 3000 N·m/rad) and, with suspension, fork stiffness (15, 40 kN/m) and damping (ratio 0.15, 0.5), varied one at a time	5 settings (9 with suspension) × 5 configurations
Front/rear ratio	8 configurations from 10/10 to 20/8 on the main grid	648 each
Soft edges	Profile contact with suspension; edge widths 0, 25, 50, 100 mm × main sweep	1620
Consistency check	Current 11/10 at 20 km/h on assumed common potholes (20–40 mm deep, 300 and 600 mm long)	6 per model (with and without suspension)

The depth grid has a 10 mm resolution, so a reported threshold of "80 mm" means a first fall somewhere between 70 and 80 mm. All runs are deterministic; rerunning the main sweep and sensitivity analysis reproduced every outcome exactly, and when the 11/10 configuration was added, every earlier run of the other configurations was reproduced unchanged. Analysis used NumPy, pandas and Matplotlib

(Harris et al. 2020; McKinney 2010; Hunter 2007); animations were drawn with Matplotlib and encoded with Pillow (Clark et al. 2026).

Consistency check. The criterion, fixed before running, was that simulated handlebar peaks for the current scooter on common potholes should fall within roughly 1–3 g, the range of the measured impacts. With the three rides of Part A, 99 of 100 measured peaks were 1.0–2.97 g (median 1.27 g) and one was 6.28 g; the first ride alone gave 1.0–2.8 g. Pothole depths along the route were not measured, so this is a consistency check, not a validation.

4 Results: what a larger front wheel would change

4.1 Geometry

At 20 km/h on a 600 mm long pothole, compared with the 10 in reference wheel, the Voiager 5's 11 in front wheel reduced the edge impact velocity by 3.4–4.1% and the front-wheel speed loss by 6.7–8.0%; a 12 in wheel reduced them by 6.5–7.7% and 12.6–14.8%, and a 14 in wheel by 12.1–13.9% and 22.6–25.8% (Table 6). At speed, wheels barely dropped into short potholes (about 3 mm into a 150 mm pothole at 20 km/h), so long potholes are the more dangerous ones.

Table 6: Front-wheel edge impact at 20 km/h, 600 mm pothole (geometry model).

Front wheel	Depth 30 mm: impact velocity / speed loss	Depth 60 mm: impact velocity / speed loss
10 in (reference)	3.59 m/s / 41.7%	4.64 m/s / 69.8%
11 in (Voiager 5)	3.44 m/s / 38.3% (−4.1% / −8.0%)	4.48 m/s / 65.1% (−3.4% / −6.7%)
12 in	3.31 m/s / 35.5% (−7.7% / −14.8%)	4.34 m/s / 61.0% (−6.5% / −12.6%)
14 in	3.09 m/s / 30.9% (−13.9% / −25.8%)	4.08 m/s / 54.0% (−12.1% / −22.6%)

At 60 mm and deeper, the drop is limited by the time available to fall (57 mm), so the 60 and 90 mm results are identical.

Figure 7 shows the same comparison over all depths.

4.2 Falls with the Voiager 5's front suspension

With front suspension, falls occurred in 9 of 81 runs for the current 11/10 configuration (the Voiager 5), 9 for the 10/10 reference, 6 for 12/10, 6 for 12/12 and 1 for 14/10 (Table 7). The Voiager 5's front wheel, 1 in larger than the reference, fell at the same depths as the reference in every case; the first fall was delayed only from 12 in. No 300 mm long pothole caused a fall at any depth up to 100 mm; 600 mm potholes caused falls only at 15 km/h, where the slower wheel has more time to drop.

Figure 8 shows one such case as still frames: the 80 mm deep, 600 mm long pothole at 15 km/h, the shallowest falling depth for the current scooter (Table 7). With the 10/10 reference the front wheel drops into the hole and strikes the far edge, and the scooter pitches forward about it; the fall criterion is reached 0.15 s after the front wheel reaches the edge. The Voiager 5 (11/10) also pitches over, more slowly (fall criterion at 0.59 s). With a 12 in front wheel, alone or with a 12 in rear wheel, the scooter

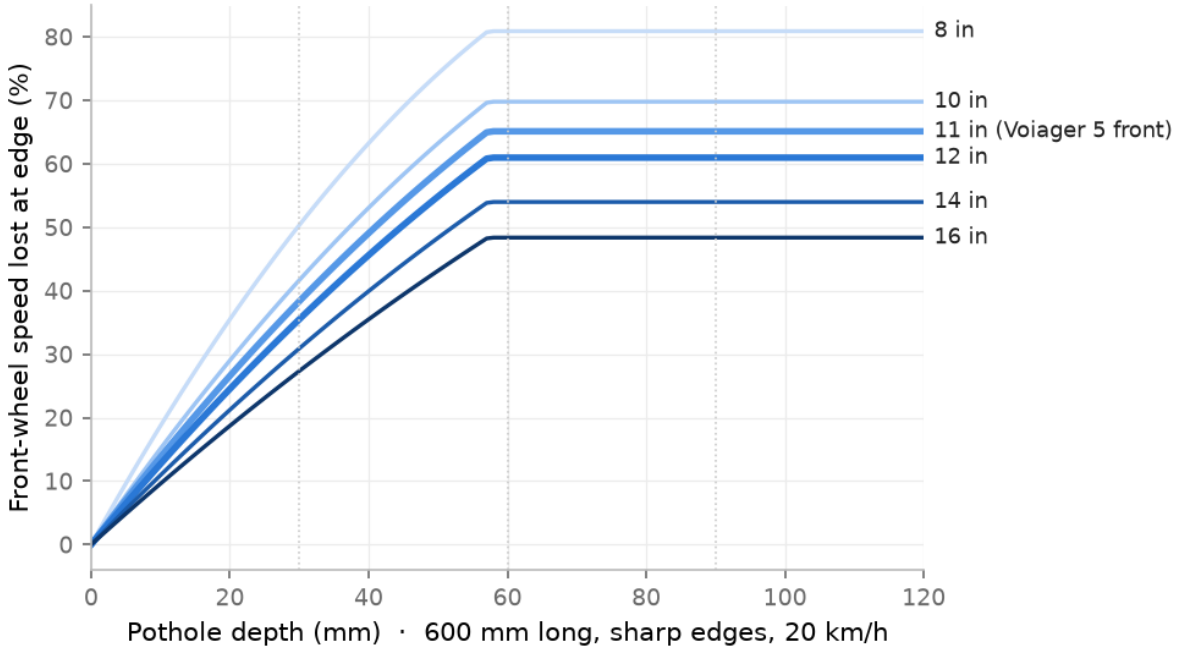


Figure 7: Geometry model: share of the front wheel's speed lost at the far edge of a 600 mm long, sharp-edged pothole at 20 km/h, by wheel diameter. Rigid wheel, no tyre or suspension; the curves flatten at 57 mm, the deepest the wheel can drop in the time available. Schematic and further geometry results in Appendix E.

Table 7: Shallowest pothole depth (mm) causing a fall, with front suspension (– = no fall up to 100 mm). 11/10 = the Voiager 5 that was ridden; 10/10 = smaller reference wheels. Rows with no falls in any configuration are omitted.

Length, speed	10/10 (ref.)	11/10 (V5)	12/10	12/12	14/10
600 mm, 15 km/h	80	80	90	90	–
1000 mm, 15 km/h	90	90	100	100	–
1000 mm, 20 km/h	90	90	90	90	100
1000 mm, 25 km/h	90	90	100	100	–
Falls (of 81 runs)	**9**	**9**	**6**	**6**	**1**

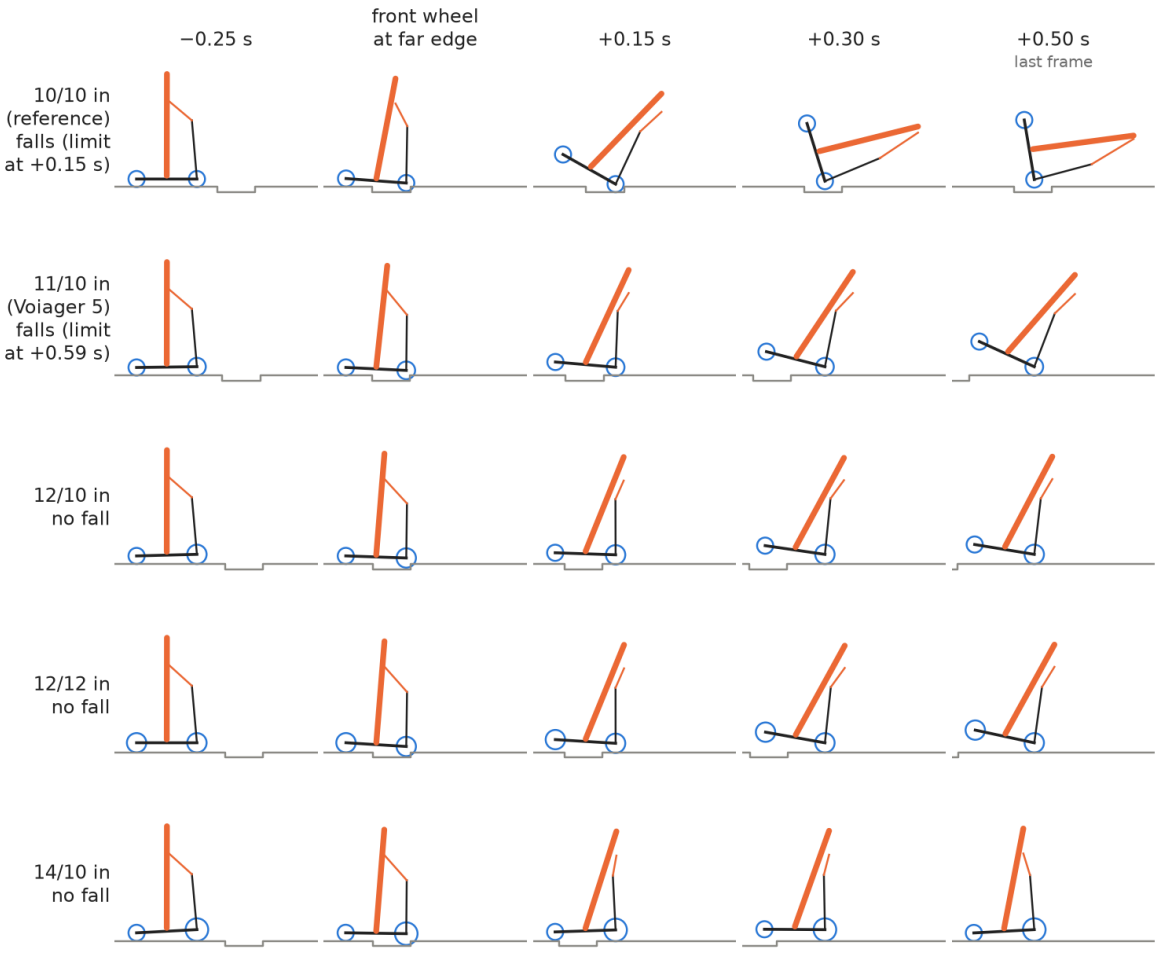


Figure 8: Key frames from the simulation of the model with its front spring crossing a sharp-edged pothole 80 mm deep and 600 mm long at 15 km/h. Rows: wheel configuration (front/rear diameter; 11/10 = the Voiager 5 that was ridden, 10/10 = smaller reference). Columns: the same moments in each row, relative to the front wheel reaching the pothole's far edge; the view follows the front wheel. Blue: wheels; black: deck and stem; orange: rider. The rider is pinned to the deck at the feet, so scooter and rider rotate together; a real rider would separate from the scooter. Planar model with assumed parameters (Table 4); one run per configuration, illustrating the results of Table 7.

rolls out with almost identical motion (maximum chassis pitch 13–14°, speed loss 71–72%). With a 14 in front wheel it rolls out with less pitch (7°) and less speed loss (56%).

All but one of the falls, in every configuration, were forward pitch-overs of scooter and rider about the front wheel, started by the front-wheel impact, with the rear wheel lifted clear of the pothole. The exception is one 11/10 run (90 mm deep, 1000 mm long, 25 km/h), in which the rear wheel dipped 5.8 mm into the pothole before the fall criterion was met, just over the 5 mm rule for attributing a fall to the rear wheel; it was the same forward pitch-over. This matches the mechanism reported by Posirisuk et al. (2022).

The benefit of a 12 in front wheel over the Voiager 5's 11 in was modest with suspension: it delayed the first fall by 10 mm in three of the four cases above, and speed loss on a 60 × 600 mm pothole at 20 km/h was 25.5% against 26.8% for 11/10 (27.6% for 10/10). A 14 in front wheel almost eliminated falls in the tested range.

Without suspension, falls were more frequent (15, 15, 14, 14 and 11 of 81 runs for 10/10, 11/10, 12/10, 12/12 and 14/10), began at 70–80 mm for the current scooter as for the reference, and the 12 in front wheel delayed the fall by 10 mm on a 1000 mm pothole at 20 km/h (70 ⟶ 80 mm). The front suspension itself therefore had a large effect, comparable to enlarging the front wheel.

4.3 A larger front wheel alone performs like larger wheels front and rear

The central comparison of the hypothesis is 12/10 against 12/12. In the main sweep they had the same number of falls and the same shallowest falling depth in every case, both with suspension (Table 7) and without. The reason is visible in the simulations: with 10–12 in wheels, the front-edge impact lifts the rear of the scooter so that the rear wheel skips over the pothole, so its size makes almost no difference to whether the scooter falls. In Figure 8 the 12/10 and 12/12 rows are almost indistinguishable (speed loss 70.9% and 71.7%).

4.4 Sensitivity

Without suspension, the 12/10 configuration fell at the same depth as 12/12, and 10 mm deeper than the 10/10 reference, under all five tyre and bracing settings; the current 11/10 scooter fell at the same depth as the reference in two settings and 10 mm deeper in three (stiffer tyre, softer tyre, weaker bracing), its first fall on a 1000 mm pothole at 20 km/h lying between 70 and 80 mm. With suspension, 12/10 and 12/12 fell at the same depth in 8 of 9 settings (the exception, more fork damping, differed by 10 mm), and 11/10 fell at the same depth as 10/10 in 8 of 9 (10 mm deeper with the softer tyre). The ordering of configurations was therefore robust, but absolute fall depths depend on assumed parameters and should not be read as real-world thresholds.

4.5 Potholes without sharp edges

Table 8: Number of falls (of 81 runs) by edge width, profile contact with suspension. 11/10 = the Voiager 5 that was ridden; 10/10 = smaller reference wheels.

Edge width	10/10 (ref.)	11/10 (V5)	12/10	12/12	14/10
0 mm (sharp)	8	8	6	6	4
25 mm	8	7	5	6	4
50 mm	7	5	5	5	4

Edge width	10/10 (ref.)	11/10 (V5)	12/10	12/12	14/10
100 mm	4	4	3	3	2

Sloped edges reduced falls (11/10: 8 to 4 from sharp to 100 mm edges). At every edge width, the configurations with a 12 in or 14 in front wheel fell no more often than the current 11/10 scooter, and 14/10 fell less often at every width; 12/10 stayed within one fall of 12/12. Figure 9 shows the same 80 × 600 mm pothole with 50 mm sloped edges: the 10/10 reference still pitches over, more slowly (fall criterion at 0.42 s), while the Voiager 5 (11/10) and the 12 in and 14 in front wheels roll through; the Voiager 5 pitches more than they do (maximum chassis pitch 22°, speed loss 54%). As with sharp edges, 12/10 and 12/12 move almost identically (maximum chassis pitch 16°, speed loss 50%); 14/10 pitches less (11°, speed loss 44%).

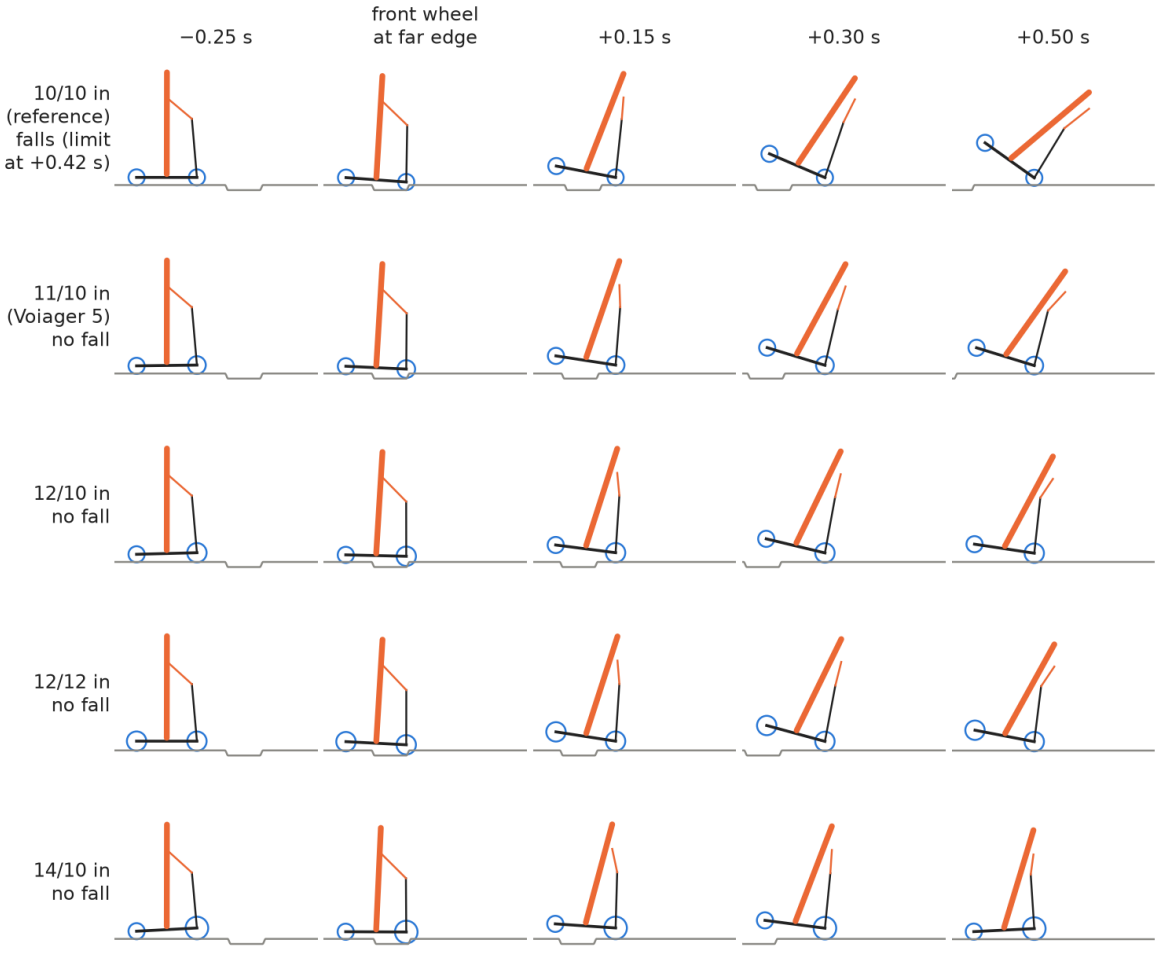


Figure 9: As Figure 8, but each pothole edge slopes over 50 mm (raised-cosine profile, custom tyre contact; section 3.4). Jolt sizes from this contact model are not used as results; fall outcomes are.

4.6 How much larger should the front wheel be?

The ratio study asked whether a front wheel can be too large relative to the rear. It repeated the main sweep (81 runs per configuration) for front/rear diameter ratios from 1.0 (10/10) to 2.5 (20/8); the full results are in Appendix E.3. Falls kept decreasing as the front wheel grew beyond 11 in (Table 9): from 15 of 81 runs for 10/10 and 11/10 to 1 for 20/10 and 20/8 without suspension, and from 9 to none with suspension. No ratio up to 2.5 made falls more likely: with a larger front wheel the shallowest depth causing a fall stayed the same or rose, in 10 mm steps (Figures E10 and E12).

Table 9: Front/rear ratio study: runs meeting the fall criteria (of 81 per configuration). Planar model with assumed parameters (Table 4); relative comparison of wheel configurations only.

Front/rear (in)	Ratio	Without suspension	With suspension
10/10	1.0	15	9
11/10	1.1	15	9
12/10	1.2	14	6
14/10	1.4	11	1
16/10	1.6	9	1
20/10	2.0	1	0
16/8	2.0	9	1
20/8	2.5	1	0

Very large front wheels did change what the rear wheel does (Figures E11 and E13). With a small front wheel, the front impact lifts the rear so that it skips over the pothole. With a very large front wheel it no longer does, so the rear wheel drops into the pothole and causes a second jolt: without suspension, the largest speed loss at the rear crossing (20 km/h, runs without a fall) rose from 19% for 10/10 to 27% for 20/10 and 33% for 20/8. With suspension the configurations differed little. This second jolt caused no falls in the model, but it would affect comfort and control.

In this model a larger front wheel was never worse: falls kept decreasing up to the largest front wheel tested (20 in). With front suspension, however, most of the reduction was reached by 14/10 (from 9 to 1 of 81 runs), and larger front wheels added little. The model cannot represent steering, frame geometry, weight or cost, all of which limit how large a front wheel can practically be, and the rear-wheel jolt above would affect comfort. We therefore suggest a moderate increase over the Voiager 5's 11 in front wheel, to 12–14 in. London's tender requires a front wheel of at least 11 in (Transport for London 2022), which the Voiager 5 just meets; in this model a front wheel at that size fell as often as the 10 in reference. This limit is a judgement based on factors outside the model, not a result of it.

4.7 Consistency with the measured rides

With front suspension, simulated handlebar peaks for the current 11/10 scooter on 300 mm long potholes were 1.9–2.4 g, inside the range of 99 of the 100 measured impacts (1.0–2.97 g). On 600 mm potholes they were 4.3–6.5 g, about 1.5–2.2 times the top of that range; the highest exceeds even the single 6.28 g outlier, and one value is not taken as agreement. Without suspension they were 5.3–8.7 g, about two to three times the top of the range. The suspension model is therefore the more faithful representation of the Voiager 5 and is used for the main results, with the caveat that its absolute accelerations are reliable only for short potholes. Figure 10 shows the comparison.

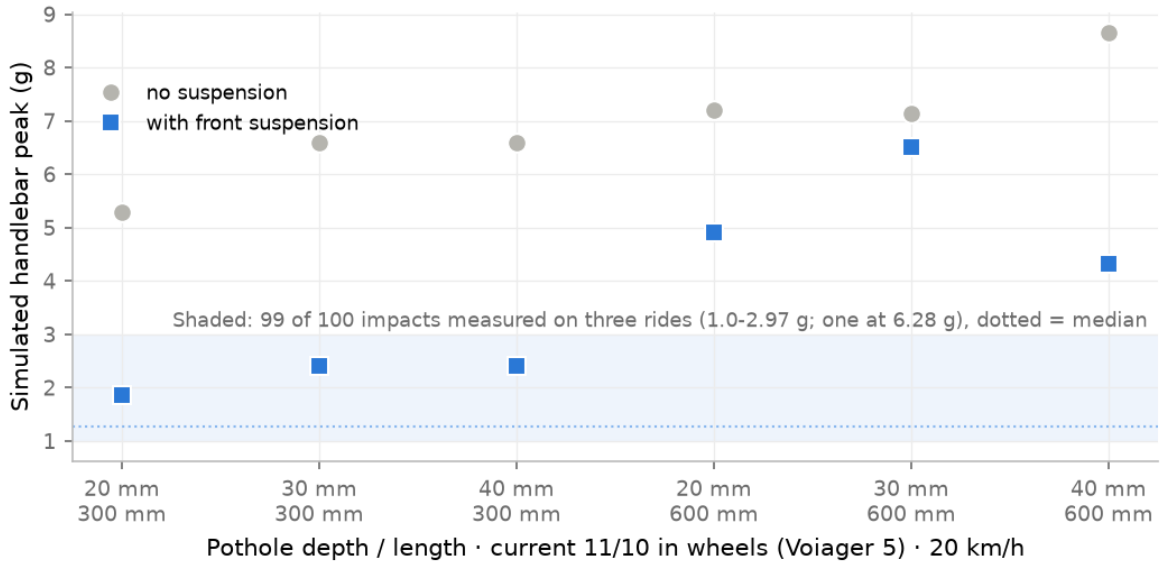


Figure 10: Consistency check: simulated handlebar peaks for the current 11/10 configuration (the Voiager 5 that was ridden) at 20 km/h, with and without front suspension, against the range of 99 of the 100 impacts measured on the three rides (shaded; dotted line: median). Pothole depths on the route were not measured, so this is a consistency check, not a validation.

4.8 Limitations of the models

- **Planar:** no steering, handlebar twist, roll or sideways falls. Loss of steering control, which a front-wheel impact can also cause, is not represented.
- **Assumed vehicle:** mass, inertias, wheelbase, tyre stiffness and fork are not published by Voi. The wheel sizes come from photographs of a Voiager 5's tyre sidewalls (nominal sizes, rounded to 11/10 in).
- **Simple rider:** one rigid body with a fixed spring; no active balance or reaction.
- **Idealised potholes:** rectangular, or with smooth sloped edges; real potholes are irregular.
- **Potholes only:** other obstacles were not part of the main study. An exploratory extension to raised bumps (Appendix D) found fewer falls with larger front wheels without suspension, but with the front spring the few modelled bump falls started at the rigid rear wheel, so the front-wheel result may not hold for every type of obstacle.
- **Analyst-defined falls** and a 10 mm depth resolution.
- **Not validated** against physical tests; the consistency check is only a partial pass.

Within these limits, the models consistently show that the front wheel determines whether the scooter pitches over, that the Voiager 5's 11 in front wheel behaves almost like the 10 in reference, and that enlarging the front wheel alone, to 12 in or more, gives the same benefit in the model as enlarging both wheels.

5 Discussion

5.1 Main findings

On three rides totalling 6.08 km, a handlebar-mounted phone recorded 100 vertical impacts of 1 g or more on Voiager 5 scooters, about one every 61 m, and a third of them reached 1.5 g (Section 2.6.4). Ten

places produced impacts on two or more separate rides, two of them on all three, and most of these had no FixMyStreet report in the preceding year. The recording method itself held up: the motion sensors sampled at a steady rate with no gaps, and the walk at the end of one recording was separated from riding to within the minute-level precision of the available references (Appendix B).

The Voiager 5 that was ridden has, from its tyre markings, an 11.1 in front wheel and a 10.1 in rear wheel (Figure 1). In the model with its front suspension, this 11/10 configuration fell in 9 of 81 runs, the same number and at the same depths as a 10/10 reference, while a 12 in front wheel fell in 6 and a 14 in front wheel in 1 (Table 7). In this model, then, the benefit of a larger front wheel began above the Voiager 5's current size. The difference between 10 and 11 in may be smaller than the model's 10 mm depth resolution can show, so "no benefit" should be read as "no benefit detected".

The hypothesis survived the test that could have refuted it: enlarging the front wheel alone (12/10) gave the same number of falls and the same falling depths as enlarging both wheels (12/12) in every case of the main sweep, with and without suspension, and in 8 of the 9 settings of the sensitivity analysis with suspension (the exception, more fork damping, differed by one 10 mm step; Section 4.4). The mechanism in the model is that the front-edge impact lifts the rear of the scooter, so that the rear wheel largely skips the pothole. All but one of the modelled pothole falls were forward pitch-overs started by the front-wheel impact, the mechanism also reported by Posirisuk et al. (2022); the exception, one 11/10 run, was credited to the rear wheel only because the rear dipped 5.8 mm, just over the 5 mm attribution rule. With sloped pothole edges, 12 and 14 in front wheels fell no more often than the Voiager 5 at any edge width, and 12/10 stayed within one fall of 12/12 (Table 8).

Suspension also mattered. Without it, falls were more frequent for every configuration (15, 15, 14, 14 and 11 of 81 runs), and the suspended model's handlebar accelerations matched the measured range on short potholes where the rigid model's did not (Section 4.7). This agrees with the recommendation of Paudel et al. (2025) that small-wheeled vehicles should have both larger wheels and front suspension.

5.2 What the results do not show

The results describe exposure on these rides and the behaviour of a model; they do not measure risk. The measurements come from one rider, one phone and short routes in one city, and the 1.0 g threshold, the severity bands and the wobble ratio are analyst choices rather than established criteria. No impact was linked to a fall or near-fall, and nothing here supports a statement about the safety of Voi scooters, about injury risk, or about other riders.

The model is planar: it has no steering, roll or sideways fall, so it cannot represent the loss of steering control that a front-wheel impact may also cause. Its mass, inertias, wheelbase, tyre and fork properties are assumptions, and the rider is a single body on a fixed spring. Its absolute falling depths (60–100 mm, depending on suspension, speed and tyre stiffness) are not real-world thresholds and should not be compared directly with the 6 cm of Posirisuk et al. (2022), whose model differs in rider, scooter and contact representation. The comparison between wheel configurations under the same assumptions is the result; the consistency check against the rides supports the suspended model's jolt sizes on short potholes only and is not a validation, because the depths of the potholes on the route were not measured.

The front-only hypothesis was tested on potholes. In an exploratory extension to raised, sharp-edged bumps (Appendix D), larger front wheels again reduced falls in the rigid model, but with the front spring the few modelled bump falls followed the strike of the rigid rear wheel. Whether a front-only rule covers every kind of obstacle is therefore an open question. Finally, the wheel sizes come from

the sidewall markings of a Voiager 5, the model ridden, and are nominal sizes of new tyres; worn tyres under load are slightly smaller.

5.3 Implications for trial vehicle rules

The national trial guidance and the legal definition of a trial e-scooter set no minimum wheel size, while London requires a front wheel of at least 11 in (Department for Transport 2024; UK Government 2020; Transport for London 2022). The Voiager 5 in service in Oxford meets London's minimum, by about 2.5 mm. Outside London, the national rules do not require even that; whether individual trial areas set their own wheel-size conditions in their contracts, as London does, was not checked. Within the limits above, the model suggests two things: that a larger front wheel delays pothole falls, and that a front wheel at London's 11 in minimum did not fall less often than a 10 in one, whereas 12 and 14 in front wheels did. For potholes, a front-wheel minimum also captured the benefit of larger wheels without requiring both wheels to be enlarged. Because only trial rental scooters are legal on public roads, a small number of actors can act on this:

1. **The Department for Transport** could add a front-wheel minimum to the national trial vehicle requirements, at least London's, and consider whether a larger one (12 in or more, with front suspension) would be justified once physically tested. It could also consider whether its 5 cm stability test is deep enough to separate wheel sizes: in the model, no configuration fell in a 5 cm deep pothole, so a test at that depth would not have shown the differences in falls between them.

2. **Transport for London** could review whether its 11 in front-wheel minimum is sufficient, since in this model a front wheel at that size fell as often as a 10 in one.

3. **Operators**, when renewing fleets, could prioritise models with a front wheel larger than 11 in and with front suspension.

4. **Local authorities hosting trials**, such as Oxfordshire County Council, could ask for at least London-equivalent vehicle standards when awarding or extending trial contracts, and could use repeated impacts at the same places as one input to road inspection.

5. **Research groups with test facilities** could test front-wheel size on instrumented scooters over pothole-shaped obstacles; this is the step needed to confirm or refute the model.

5.4 Future work

The most useful next steps are: recording more rides, riders and phones with the same method, under an ethics approval if other riders are involved; measuring the depth of the defects at repeatedly hit spots so that the model can be validated rather than checked for consistency; refining the model's depth resolution around the 10–12 in front sizes; extending the model to three dimensions with steering; and physical tests of the kind described above.

6 Conclusion

A smartphone on the handlebar of a rental e-scooter recorded frequent, repeated vertical impacts on ordinary Oxford roads, often at places with no recent public report. The Voiager 5 ridden has an 11 in front wheel, at London's minimum. A dynamics model of it suggests that this front wheel falls in potholes about as often as a 10 in one, that 12 and 14 in front wheels reduce pothole falls, and that

enlarging the front wheel alone gives the same benefit as enlarging both, a hypothesis the model could have refuted but did not. These are results from three rides and an unvalidated model, not proof that current scooters are unsafe or that larger wheels prevent falls. They do justify extending a front-wheel minimum to trials outside London, asking whether London's 11 in minimum is high enough, and testing larger front wheels physically.

Ethics

All recordings are of the author's own rides on rental e-scooters hired through the operator's app within the official trial, ridden legally on public roads. The phone was mounted on the handlebar before each ride and not handled while riding. No data about other people were collected. A study recording other riders would need university ethics approval and GDPR-compliant handling.

Use of AI tools

This work was carried out with assistance from an AI system, Claude (Anthropic; model Claude Opus 5.5, used through Claude Code). The AI wrote and ran the analysis and simulation code, searched the literature and checked sources against their abstracts or official pages, and drafted the text. The author conceived the project and research question, collected all data on their own rides, confirmed the scooter model with the operator, photographed the scooter's tyre markings that give its wheel sizes, and made the main decisions on scope, methods and wording. The author also originated several elements that the AI then implemented: animations of the simulated scooter and their key-frame figures, the map overlays of the rides, impacts and walk (including a combined map of all rides), the use of that walk to test whether the recordings can separate walking from riding, the extension to road bumps, and the use of street-level imagery to view the impact spots, which led to the linked and printable list of impact spots. The author reviewed and edited the text and checked each cited source against the claims made in it. The author takes full responsibility for the content, including any errors.

Data and code availability

In this paper. Every impact is listed in Table A5 (time, peak acceleration, speed, road, spot), and every spot in Table C1. The model results behind each figure are summarised in Appendices D and E.

To be deposited [DOI to be added]:

- *Code:* the analysis and simulation scripts, which regenerate every figure and table from the data below.
- *Impact data:* the impact list with spot coordinates, the per-spot table and the FixMyStreet comparison, with the first and last 200 m of each route removed.
- *Model outputs:* all simulation results (one row per run) for the main sweep, sensitivity analysis, ratio study, soft-edge, bump and consistency experiments, and the geometry model sweep.
- *Recording quality:* the per-window features and labels used for walking detection and the data-quality summary.

Not shared publicly. The raw Sensor Logger recordings include the full GPS track, including the start and end of each ride, and are therefore withheld for the author's privacy. A version with the route ends removed is available from the author on request.

Third-party data. FixMyStreet reports (mySociety) and OpenStreetMap map data (© OpenStreetMap contributors) are subject to their providers' terms; the deposit gives the query used and the download date instead of redistributing them where those terms require it.

References

Alsalkini, A. (2023). "The influence of e-scooters on urban mobility and transport related emissions". MA thesis. Chalmers University of Technology. URL: https://odr.chalmers.se/server/api/core/bitstreams/2d2b9c13-ea55-44e3-aa07-ac1eb8a65968/content.

Anke, J., M. Ringhand, D. Schackmann and T. Petzoldt (2025). "How e-scooter riders navigate road safety hazards – Understanding the perceptions and strategies of regular riders". In: *Journal of Cycling and Micromobility Research* 4, p. 100065. DOI: 10.1016/j.jcmr.2025.100065.

Apple Inc. (n.d.[a]). *CMAcceleration*. Apple Developer Documentation. Acceleration values in G; 1 G = 9.81 m/s². URL: https://developer.apple.com/documentation/coremotion/cmacceleration (visited on 7th Oct. 2026).

Apple Inc. (n.d.[b]). *CMDeviceMotion*. Apple Developer Documentation. URL: https://developer.apple.com/documentation/coremotion/cmdevicemotion (visited on 7th Oct. 2026).

Bowden, A. (June 2022). *Voi rolls out Voiager 5 e-scooter with "magic wheels"*. ebiketips (road.cc). URL: https://road.cc/ebiketips/content/news/voi-rolls-out-voiager-5-e-scooter-with-magic-wheels-3827 (visited on 8th Oct. 2026).

Butterworth, S. (1930). "On the theory of filter amplifiers". In: *Experimental Wireless and the Wireless Engineer* 7. No DOI; details from the journal's 1930 index and secondary sources, pp. 536–541.

Cano-Moreno, J. D., J. M. Cabanellas Becerra, J. M. Arenas Reina and M. E. Islán Marcos (2022). "Analysis of E-Scooter Vibrations Risks for Riding Comfort Based on Real Measurements". In: *Machines* 10.8, p. 688. DOI: 10.3390/machines10080688.

Choi, K. T. H. (2026). *Sensor Logger*. Version 1.67.1. iOS app, published by Logger Labs Ltd; recorded on iPhone 11 Pro Max, iOS 26.6.2. DOI: 10.5281/zenodo.20277979. URL: https://www.tszheichoi.com/sensorlogger.

Cicchino, J. B., P. E. Kulie and M. L. McCarthy (2021). "Severity of e-scooter rider injuries associated with trip characteristics". In: *Journal of Safety Research* 76, pp. 256–261. DOI: 10.1016/j.jsr.2020.12.016.

Clark, A. et al. (2026). *Pillow (PIL Fork)*. Version 12.3.0. URL: https://python-pillow.org.

Department for Transport (2024). *E-scooter trials: guidance for local areas and rental operators*. Updated 17 January 2024. URL: https://www.gov.uk/government/publications/e-scooter-trials-guidance-for-local-areas-and-rental-operators/e-scooter-trials-guidance-for-local-areas-and-rental-operators.

Department for Transport (2026a). *E-scooter trials: guidance for users*. Updated 7 July 2026. URL: https://www.gov.uk/guidance/e-scooter-trials-guidance-for-users.

Department for Transport (July 2026b). *Reported road casualties Great Britain: e-scooter factsheet, 2025*. Official statistics, GOV.UK. URL: https://www.gov.uk/government/statistics/reported-road-casualties-great-britain-vulnerable-road-user-factsheets/reported-road-casualties-great-britain-e-scooter-factsheet-2025 (visited on 8th Oct. 2026).

Department of Transport (Apr. 1996a). *75mm High Road Humps*. Traffic Advisory Leaflet 2/96. Traffic Advisory Unit. URL: https://tsrgd.co.uk/pdf/tal/1996/tal-2-96.pdf.

Department of Transport (June 1996b). *Highways (Road Humps) Regulations 1996*. Traffic Advisory Leaflet 7/96. Traffic Advisory Unit. URL: https://tsrgd.co.uk/pdf/tal/1996/tal-7-96.pdf.

Eriksson, J., L. Girod, B. Hull, R. Newton, S. Madden and H. Balakrishnan (2008). "The pothole patrol: using a mobile sensor network for road surface monitoring". In: *Proceedings of the 6th International Conference on Mobile Systems, Applications, and Services (MobiSys '08)*, pp. 29–39. DOI: 10.1145/1378600.1378605.

Essex Highways (Aug. 2025). *Voi announced as new e-scooter trial operator in Essex*. URL: https://www.essexhighways.org/news/voi-announced-as-new-e-scooter-trial-operator-in-essex.

Fournier, M., N. Bailly, A. Schäuble and Y. Petit (2023). "Head impact kinematics and injury risks during E-scooter collisions against a curb". In: *Heliyon* 9.9, e19254. DOI: 10.1016/j.heliyon.2023.e19254.

Google (n.d.). *Maps URLs*. Google Maps Platform documentation. URL: https://developers.google.com/maps/documentation/urls/get-started (visited on 7th Oct. 2026).

Harris, C. R., K. J. Millman, S. J. van der Walt et al. (2020). "Array programming with NumPy". In: *Nature* 585.7825, pp. 357–362. DOI: 10.1038/s41586-020-2649-2.

Hunter, J. D. (2007). "Matplotlib: A 2D graphics environment". In: *Computing in Science & Engineering* 9.3, pp. 90–95. DOI: 10.1109/MCSE.2007.55.

KartaView (n.d.). *KartaView (formerly OpenStreetCam)*. Open street-level imagery, CC BY-SA 4.0; API queried 2026-10-08 (api.openstreetcam.org/2.0/photo). URL: https://kartaview.org (visited on 8th Oct. 2026).

Kwapisz, J. R., G. M. Weiss and S. A. Moore (2011). "Activity recognition using cell phone accelerometers". In: *ACM SIGKDD Explorations Newsletter* 12.2, pp. 74–82. DOI: 10.1145/1964897.1964918.

Leoni, J., M. Tanelli, S. C. Strada and S. M. Savaresi (2022). "Assessing e-scooters safety and drivability: a quantitative analysis". In: *IFAC-PapersOnLine* 55.24, pp. 260–265. DOI: 10.1016/j.ifacol.2022.10.294.

Levy Electric (n.d.). *Voiager 5 fleet specification*. Aggregator; secondary source. URL: https://fleets.levyelectric.com/vehicles/voi-voiager-5.

McKinney, W. (2010). "Data Structures for Statistical Computing in Python". In: *Proceedings of the 9th Python in Science Conference*, pp. 56–61. DOI: 10.25080/Majora-92bf1922-00a.

Merry, K. and P. Bettinger (2019). "Smartphone GPS accuracy study in an urban environment". In: *PLOS ONE* 14.7, e0219890. DOI: 10.1371/journal.pone.0219890.

Mizell, D. (2003). "Using gravity to estimate accelerometer orientation". In: *Proceedings of the Seventh IEEE International Symposium on Wearable Computers (ISWC 2003)*, pp. 252–253. DOI: 10.1109/ISWC.2003.1241424.

Moe-Nilssen, R. and J. L. Helbostad (2004). "Estimation of gait cycle characteristics by trunk accelerometry". In: *Journal of Biomechanics* 37.1, pp. 121–126. DOI: 10.1016/S0021-9290(03)00233-1.

mySociety (2026). *FixMyStreet*. Online reporting service; data downloaded 3 October 2026. URL: https://www.fixmystreet.com.

Nathania, C. J., H. Zhou, T. Daimon and J. Lee (2026). "Understanding Electric Scooter Fall Accidents Through Human–Vehicle–Environment Interactions: A Systematic Literature Review Using the Haddon Matrix". In: *Applied Sciences* 16.10, p. 4855. DOI: 10.3390/app16104855.

Olbricht, R. (n.d.). *Overpass API*. URL: https://wiki.openstreetmap.org/wiki/Overpass_API (visited on 7th Oct. 2026).

OpenStreetMap contributors (2026). *OpenStreetMap*. Extract retrieved via the Overpass API on 3 October 2026, Open Database License. URL: https://www.openstreetmap.org/copyright.

Oxfordshire County Council (n.d.). *E-scooter trial in Oxford*. URL: https://www.oxfordshire.gov.uk/residents/roads-and-transport/connecting-oxfordshire/active-travel/e-scooter-trial (visited on 7th Oct. 2026).

Pachi, A. and T. Ji (2005). "Frequency and velocity of people walking". In: *The Structural Engineer* 83.3. Page numbers to verify. URL: https://www.istructe.org/journal/volumes/volume-83-(published-in-2005)/issue-3/frequency-and-velocity-of-people-walking/.

Panoramax (n.d.). *Panoramax: open street-level imagery*. Federated open imagery; meta-catalogue search API (api.panoramax.xyz) queried 2026-10-08. URL: https://panoramax.fr (visited on 8th Oct. 2026).

Paudel, M., F. F. Yap, F. Zhu, H. J. Xu and Y. D. Wong (2025). "Evaluating the safety of small-wheeled micromobility devices: A design-agnostic, performance-based experimental approach". In: *Transportation Research Interdisciplinary Perspectives* 31, p. 101376. DOI: 10.1016/j.trip.2025.101376.

Posirisuk, P., C. Baker and M. Ghajari (2022). "Computational prediction of head-ground impact kinematics in e-scooter falls". In: *Accident Analysis & Prevention* 167, p. 106567. DOI: 10.1016/j.aap.2022.106567.

Reddy, S., M. Mun, J. Burke, D. Estrin, M. Hansen and M. Srivastava (2010). "Using mobile phones to determine transportation modes". In: *ACM Transactions on Sensor Networks* 6.2, pp. 1–27. DOI: 10.1145/1689239.1689243.

Sibson, R. (1973). "SLINK: An optimally efficient algorithm for the single-link cluster method". In: *The Computer Journal* 16.1, pp. 30–34. DOI: 10.1093/comjnl/16.1.30.

Spearman, C. (1904). "The Proof and Measurement of Association between Two Things". In: *The American Journal of Psychology* 15.1, pp. 72–101. DOI: 10.2307/1412159.

Startups Magazine (Apr. 2024). *Voi unveils new scooter: the Voiager 7*. URL: https://startupsmagazine.co.uk/article-voi-unveils-new-scooter-voiager-7.

Todorov, E., T. Erez and Y. Tassa (2012). "MuJoCo: A physics engine for model-based control". In: *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems*, pp. 5026–5033. DOI: 10.1109/IROS.2012.6386109.

Transport for London (Nov. 2022). *Electric Scooters (E-Scooters) Rental Trial 2 in London*. Find a Tender notice 2022/S 000-033381. URL: https://www.find-tender.service.gov.uk/Notice/033381-2022.

Transport for London (July 2023). *TfL awards contracts to Dott, Lime and Voi to run the next phase of London's rental e-scooter trial*. Press release. URL: https://tfl.gov.uk/info-for/media/press-releases/2023/july/tfl-awards-contracts-to-dott-lime-and-voi-to-run-the-next-phase-of-london-s-rental-e-scooter-trial.

UK Government (1999). *The Highways (Road Humps) Regulations 1999*. SI 1999/1025, regulation 4 (dimensions). URL: https://www.legislation.gov.uk/uksi/1999/1025/regulation/4/made.

UK Government (2020). *The Electric Scooter Trials and Use of Powered Transporters Order 2020*. SI 2020/663; definition of "electric scooter" in article 2. Checked 2026-10-10: no wheel size. URL: https://www.legislation.gov.uk/uksi/2020/663.

Vaca Michilena, E. X. and J. D. Cano-Moreno (2026). "Multibody Dynamic Analysis of an E-Scooter Considering Asymmetric Tire Stiffness, Speed, and Surface Roughness". In: *Machines* 14.1, p. 120. DOI: 10.3390/machines14010120.

Ventura, R., A. Ghirardi, D. Vetturi, G. Maternini and B. Barabino (2024). "Comparing the vibrational behavior of e-kick scooters and e-bikes: Evidence from Italy". In: *International Journal of Transportation Science and Technology* 16, pp. 68–86. DOI: 10.1016/j.ijtst.2023.10.010.

Virtanen, P., R. Gommers, T. E. Oliphant et al. (2020). “SciPy 1.0: fundamental algorithms for scientific computing in Python”. In: *Nature Methods* 17.3, pp. 261–272. DOI: 10.1038/s41592-019-0686-2.

Voi Technology (Oct. 2026a). *Customer support, personal communication to the author*. Voi customer support chat. Confirmed that the scooter ridden for recording 1 (3 October 2026, vehicle ID SZ22) was a Voiager 5. Model only: wheel sizes were not part of the confirmation. Record kept by the author.

Voi Technology (Oct. 2026b). *Customer support, personal communication to the author*. Voi customer support chat. Confirmed that the scooters ridden for recordings 2 and 3 (7 October 2026, vehicle IDs UPAR and BTLZ) were Voiager 5. Model only: wheel sizes were not part of the confirmation. Record kept by the author.

Welch, P. D. (1967). “The use of fast Fourier transform for the estimation of power spectra: A method based on time averaging over short, modified periodograms”. In: *IEEE Transactions on Audio and Electroacoustics* 15.2, pp. 70–73. DOI: 10.1109/TAU.1967.1161901.

Witte, T. H. and A. M. Wilson (2004). “Accuracy of non-differential GPS for the determination of speed over ground”. In: *Journal of Biomechanics* 37.12, pp. 1891–1898. DOI: 10.1016/j.jbiomech.2004.02.031.

Zang, K., J. Shen, H. Huang, M. Wan and J. Shi (2018). “Assessing and Mapping of Road Surface Roughness based on GPS and Accelerometer Sensors on Bicycle-Mounted Smartphones”. In: *Sensors* 18.3, p. 914. DOI: 10.3390/s18030914.

Appendices at a glance

The appendices give the detail behind the main text. Each can be read on its own.

	Appendix	What it contains	Page
A	Individual recordings	Each ride separately: summary tables, acceleration trace, severity, speed, impact and FixMyStreet maps; FixMyStreet comparison with reports of any date; list of all 100 impacts	29
B	Recording quality and walking detection	How reliable the Sensor Logger data are (sampling, noise, GPS), and how walking was told apart from riding, checked against the rider's references	44
C	Impact spots	Every impact spot with links to street-level imagery and QR codes	51
D	Road bumps	Exploratory, unvalidated extension of the model from potholes to raised bumps	58
E	Further model results	Geometry model, full simulation sweeps, front/rear ratio study, soft pothole edges, fall thresholds without suspension and the sensitivity analysis	65

A Individual recordings

The main text reports the three rides together. This appendix gives each ride on its own. Hotspots here are numbered separately within each ride (impacts within 30 m of each other in that ride); the spots P1–P33 of the main text are formed from the pooled impacts and are numbered differently. Rates per km are left blank (–) where less than 0.2 km was ridden on a road.

Table A1: Impacts per ride.

	Ride 1	Ride 2	Ride 3
Date, start (local)	3 Oct, 18:30	7 Oct, 10:37	7 Oct, 13:38
Vehicle (Voiager 5)	SZ22	UPAR	BTLZ
Distance ridden	2.89 km	1.48 km	1.71 km
Riding time; mean riding speed	9.6 min; 18.1 km/h	5.9 min; 15.1 km/h	6.3 min; 16.4 km/h
Impacts (per km; metres between)	40 (13.8; 72 m)	39 (26.4; 38 m)	21 (12.3; 81 m)
Median peak (interquartile range)	1.19 g (1.13–1.35)	1.34 g (1.19–1.77)	1.67 g (1.13–1.85)
Maximum	2.80 g	2.87 g	6.28 g
≥ 1.5 g; ≥ 2.0 g	6; 4	15; 3	11; 5
Marked wobble; also ≥ 1.5 g	13; 3	11; 5	8; 4
Speed vs peak, Spearman ρ (p)	0.03 (0.88)	−0.63 (< 0.001)	−0.68 (0.001)
Hotspots within the ride (2+ impacts)	11	6	3

Table A2: Impacts per km by road and ride (km ridden in brackets).

Road	Ride 1	Ride 2	Ride 3
Marston Road	11.6 (0.95 km)	15.8 (0.82 km)	9.3 (0.75 km)
Morrell Avenue	12.2 (0.98 km)	– (0.04 km, 3 impacts)	– (0.03 km, 0 impacts)
Old Road	8.2 (0.24 km)	not ridden	not ridden
Headington Road	not ridden	– (0.02 km, 1 impact)	not ridden
Other roads	21.0 (0.72 km)	36.7 (0.60 km)	15.1 (0.93 km)

Table A3: Impacts with a road-surface FixMyStreet report made in the 365 days before the ride within 15, 25 and 50 m, against the same share for all points on the ridden route (baseline).

Radius	Ride 1: impacts / baseline	Ride 2: impacts / baseline	Ride 3: impacts / baseline
15 m	5.0% / 7.7%	15.4% / 20.7%	28.6% / 19.4%
25 m	5.0% / 14.3%	30.8% / 32.7%	42.9% / 34.3%
50 m	17.5% / 28.2%	69.2% / 53.1%	66.7% / 54.0%

Table A4: As Table A3, but with FixMyStreet road-surface reports of any date since 2009. Most route points lie near some report, so these shares measure report density rather than reporting of the impacts; this is why the main comparison uses only the 365 days before each ride.

Radius	Ride 1: impacts / baseline	Ride 2: impacts / baseline	Ride 3: impacts / baseline
15 m	42.5% / 52.3%	84.6% / 75.0%	81.0% / 74.2%
25 m	62.5% / 70.3%	87.2% / 84.9%	100.0% / 87.0%
50 m	72.5% / 85.9%	89.7% / 93.8%	100.0% / 94.1%

Each ride was compared with reports downloaded for its own route's area (ride 1 on 3 October 2026, rides 2 and 3 on 8 October 2026). In ride 1, none of the six impacts of 1.5 g or more and one of its eleven hotspots had a recent report within 25 m.

Notes per ride.

- **Ride 1.** One hotspot (near 51.7522, −1.2377; spot P12 in the main text) was hit on three separate passes, up to 2.68 g. It lies 6 m from a pothole reported in January 2026 and marked "fixed". The two strongest impacts (2.80 g and 2.68 g) came at low speed just after a stop.
- **Ride 2.** Thirteen impacts followed each other every 0.6–1.2 s at about 8 km/h over roughly 20 m (spot P3). The ride ended at 10:46 (Voi rental end); the walk that followed is excluded by the speed rule (Appendix B).
- **Ride 3.** The strongest impact of all rides, 6.28 g at 8.1 km/h, was followed by four more impacts of 1.8–3.0 g within about 4 s (spot P4).

A.1 Ride 1 figures

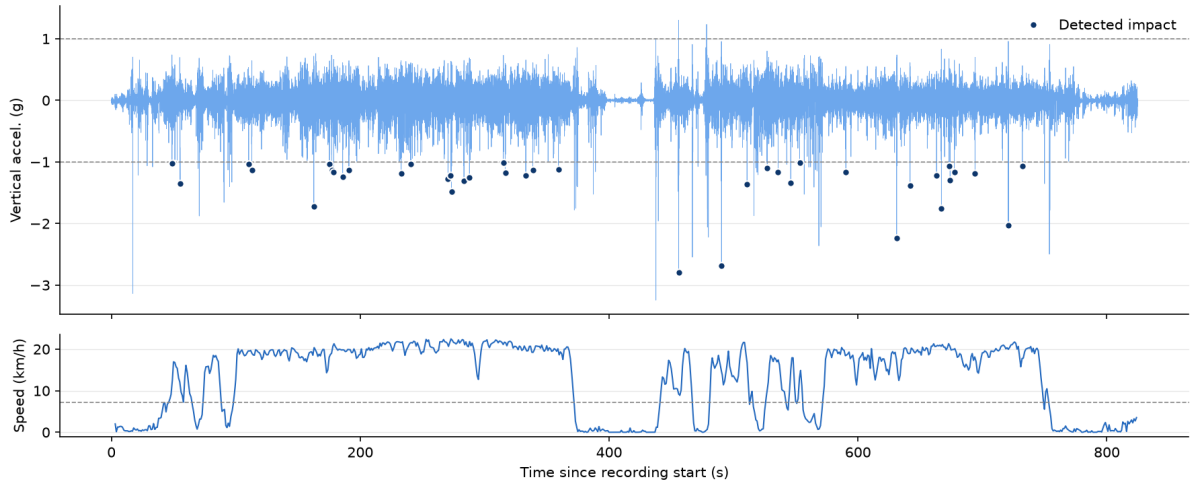


Figure A1: Ride 1: vertical acceleration (top) and GPS speed (bottom) over the recording, with detected impacts. Dashed lines: ±1 g threshold and 7 km/h speed rule.

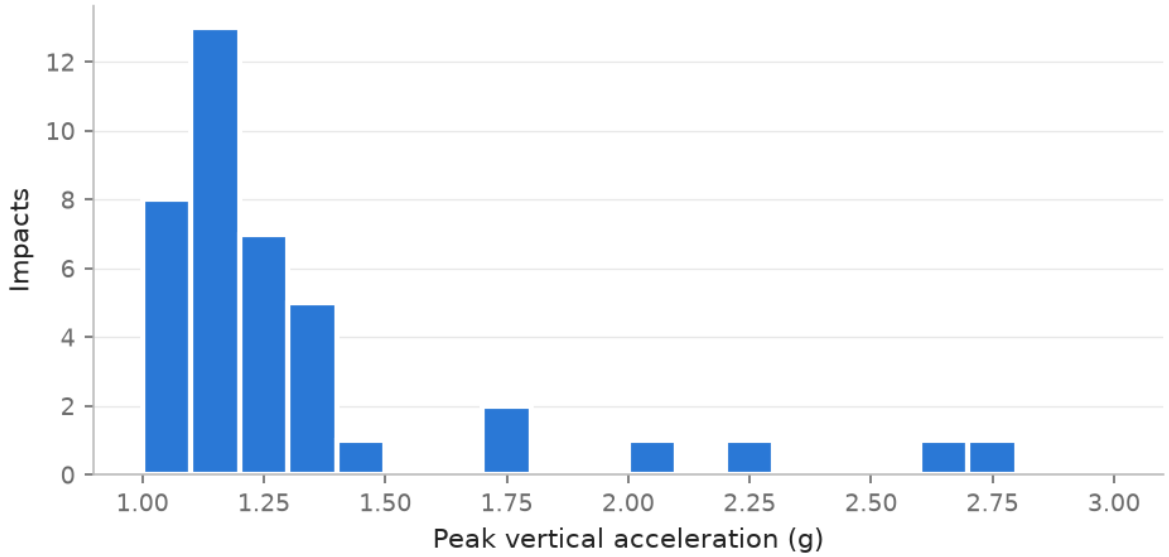


Figure A2: Ride 1: peak vertical acceleration of the 40 impacts.

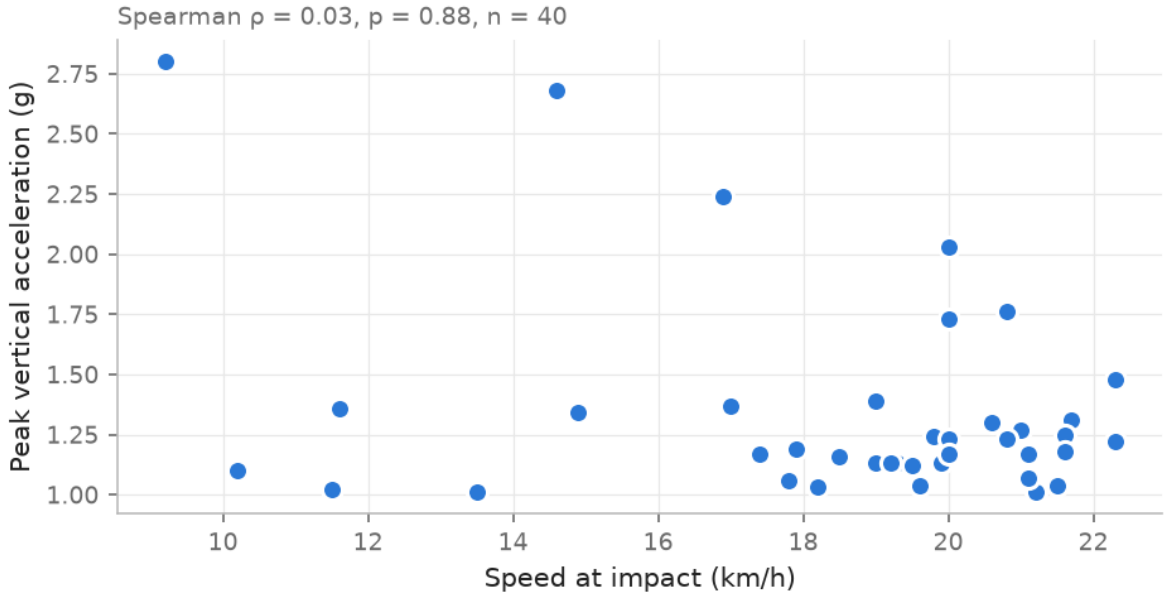


Figure A3: Ride 1: GPS speed against peak vertical acceleration for each impact. One ride; no trend is claimed.

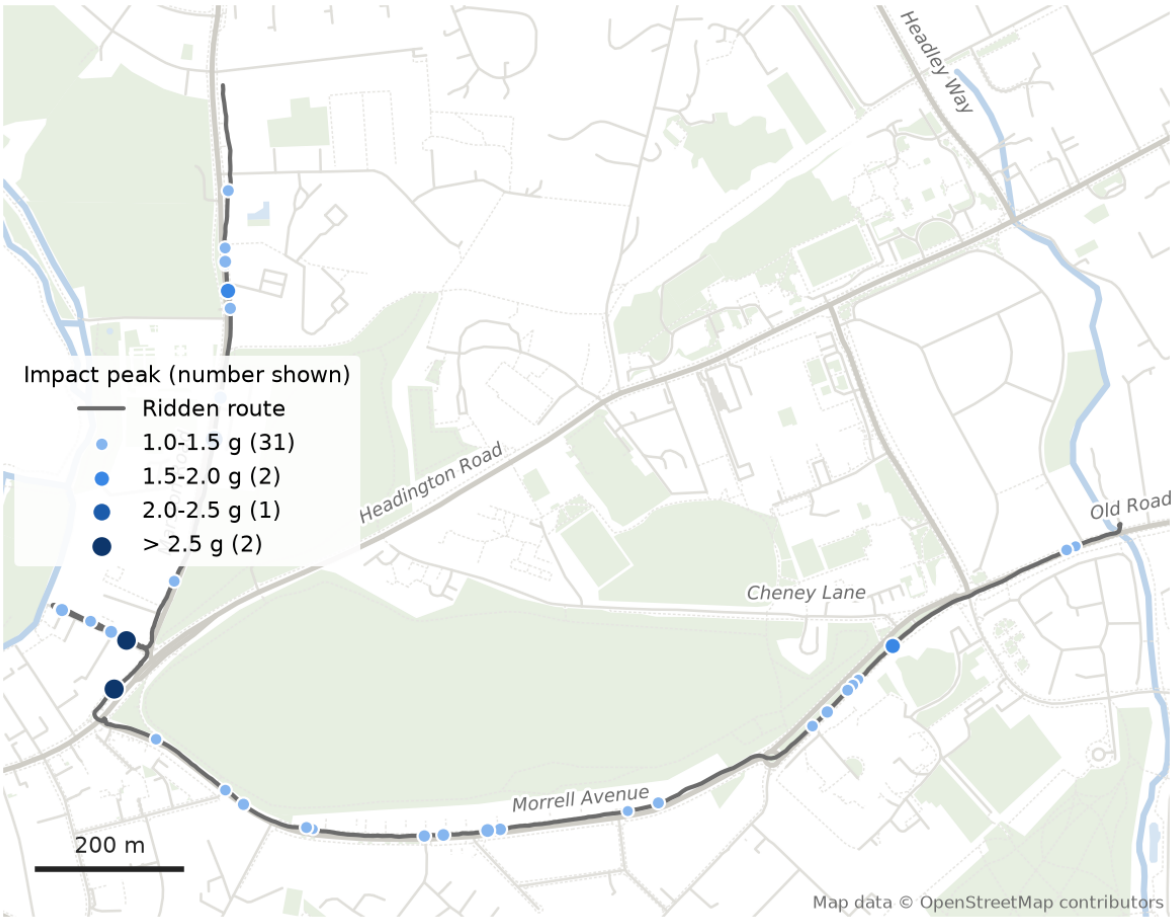


Figure A4: Ride 1: impacts on the ridden route, coloured by severity band (bands are analyst choices). Route ends (200 m) and impacts or reports near them are not drawn, for privacy; such impacts are still counted in Table A1. Map data © OpenStreetMap contributors.

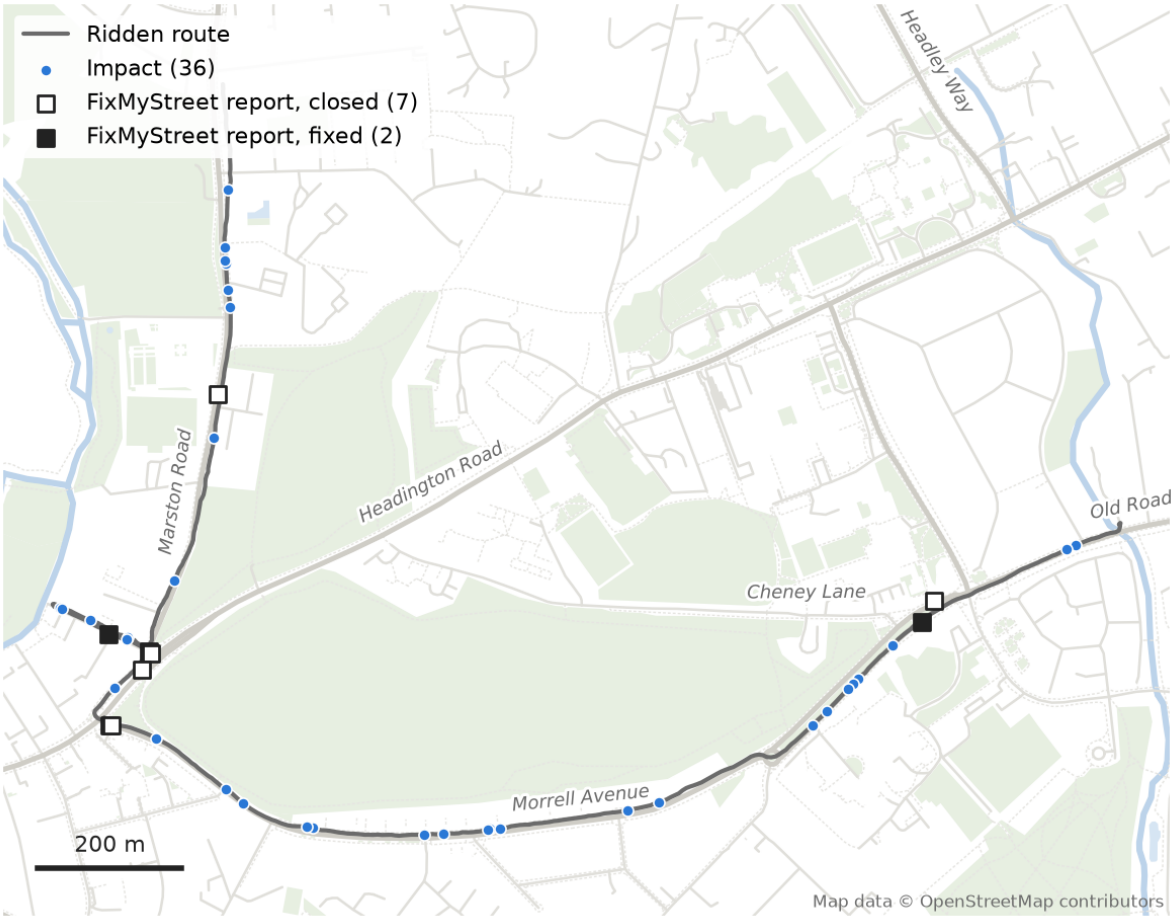


Figure A5: Ride 1: impacts and FixMyStreet road-surface reports made in the 365 days before the ride (hollow: closed; filled: marked fixed). A report near an impact does not show that it refers to the same defect. Route ends (200 m) and impacts or reports near them are not drawn, for privacy; such impacts are still counted in Table A1. Map data © OpenStreetMap contributors.

A.2 Ride 2 figures

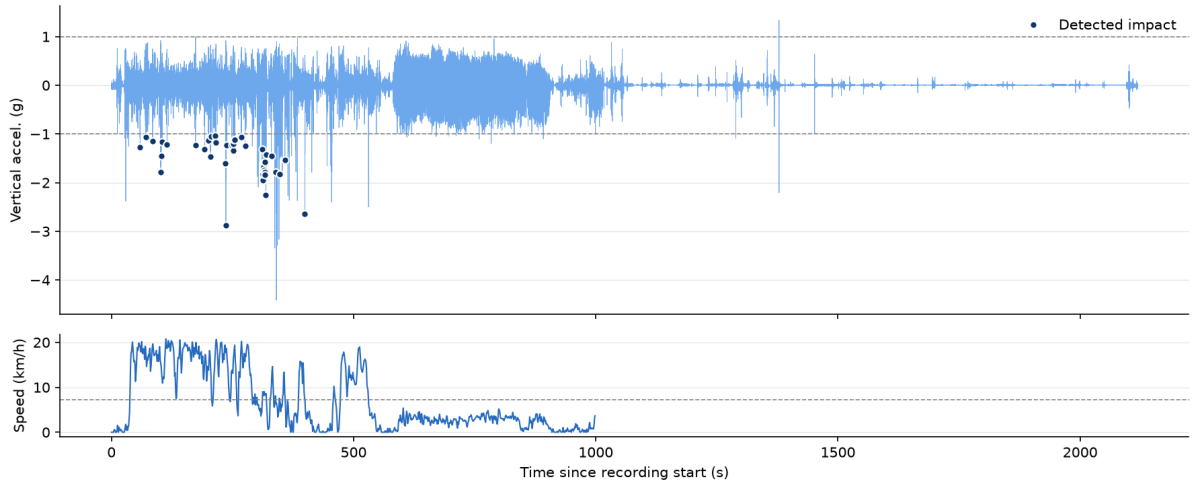


Figure A6: Ride 2: vertical acceleration (top) and GPS speed (bottom) over the recording, with detected impacts. The walk (about 580–900 s) and the phone at rest afterwards produce no impacts because of the speed rule.

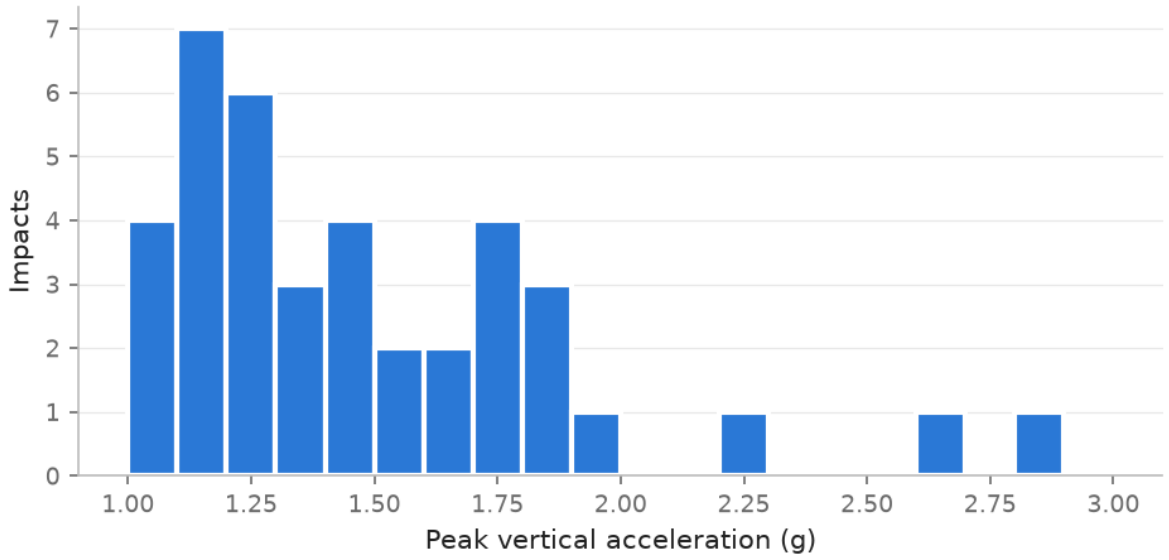


Figure A7: Ride 2: peak vertical acceleration of the 39 impacts.

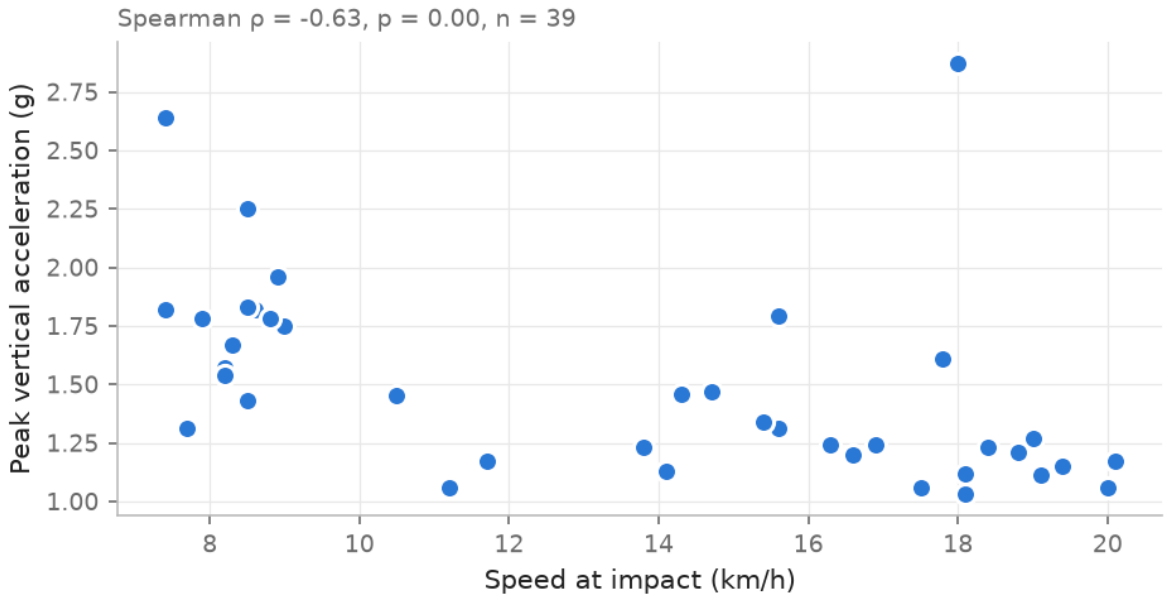


Figure A8: Ride 2: GPS speed against peak vertical acceleration for each impact. One ride; no trend is claimed.

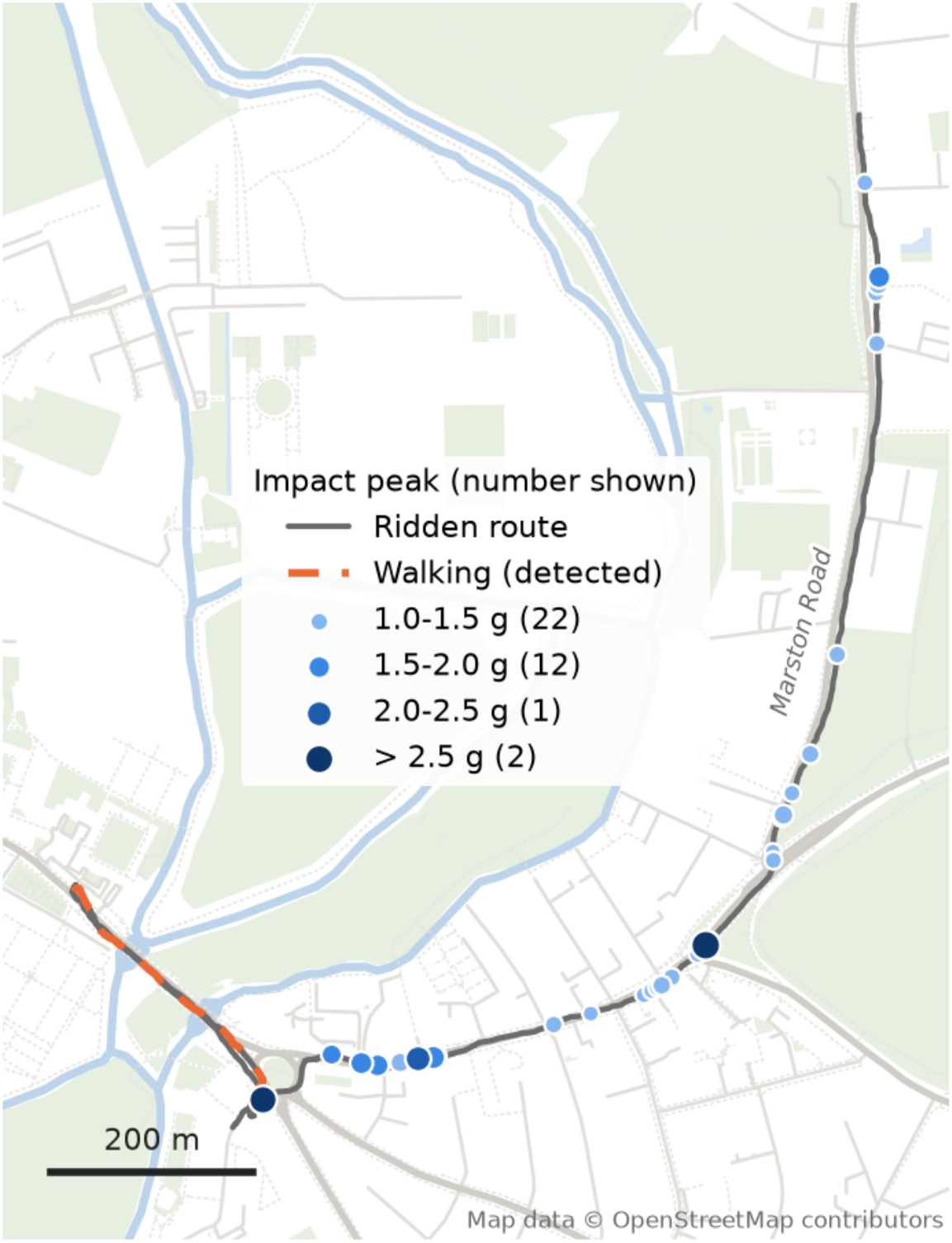


Figure A9: Ride 2: impacts on the ridden route, coloured by severity band (bands are analyst choices). The dashed orange line is the walk detected in Appendix B (not ridden; no impacts are counted there). Route ends (200 m) and impacts or reports near them are not drawn, for privacy; such impacts are still counted in Table A1. Map data © OpenStreetMap contributors.

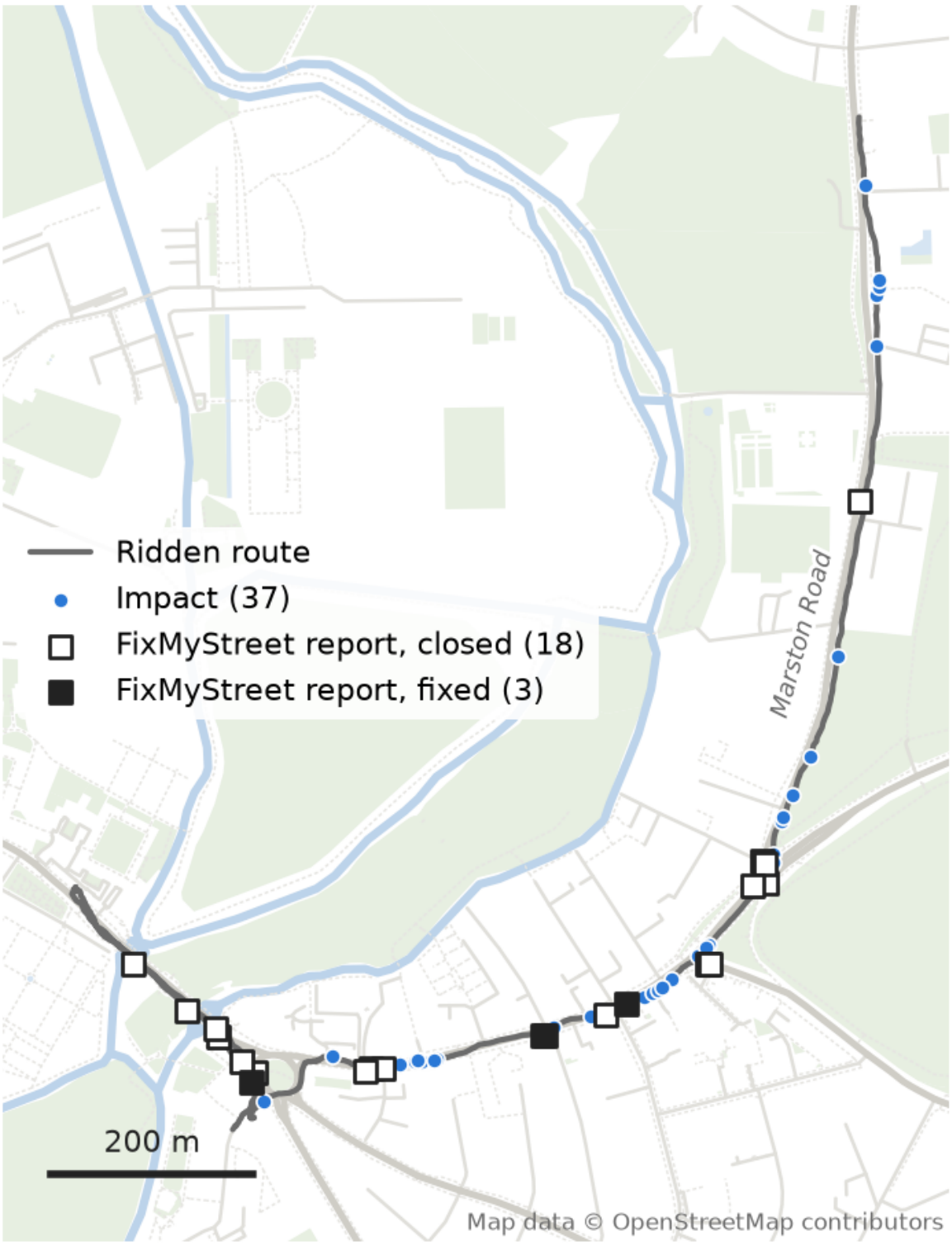


Figure A10: Ride 2: impacts and FixMyStreet road-surface reports made in the 365 days before the ride (hollow: closed; filled: marked fixed). A report near an impact does not show that it refers to the same defect. Route ends (200 m) and impacts or reports near them are not drawn, for privacy; such impacts are still counted in Table A1. Map data © OpenStreetMap contributors.

A.3 Ride 3 figures

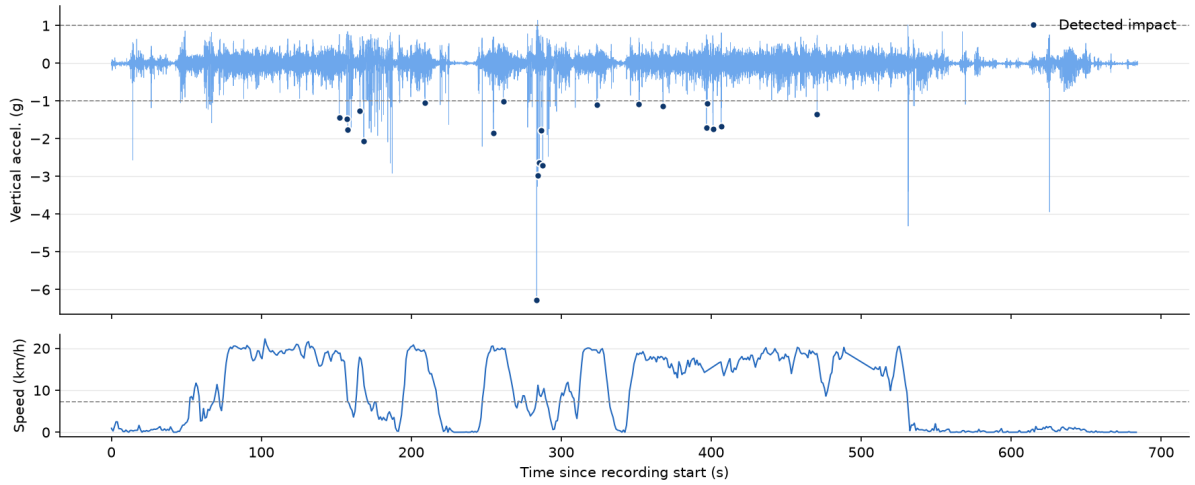


Figure A11: Ride 3: vertical acceleration (top) and GPS speed (bottom) over the recording, with detected impacts.

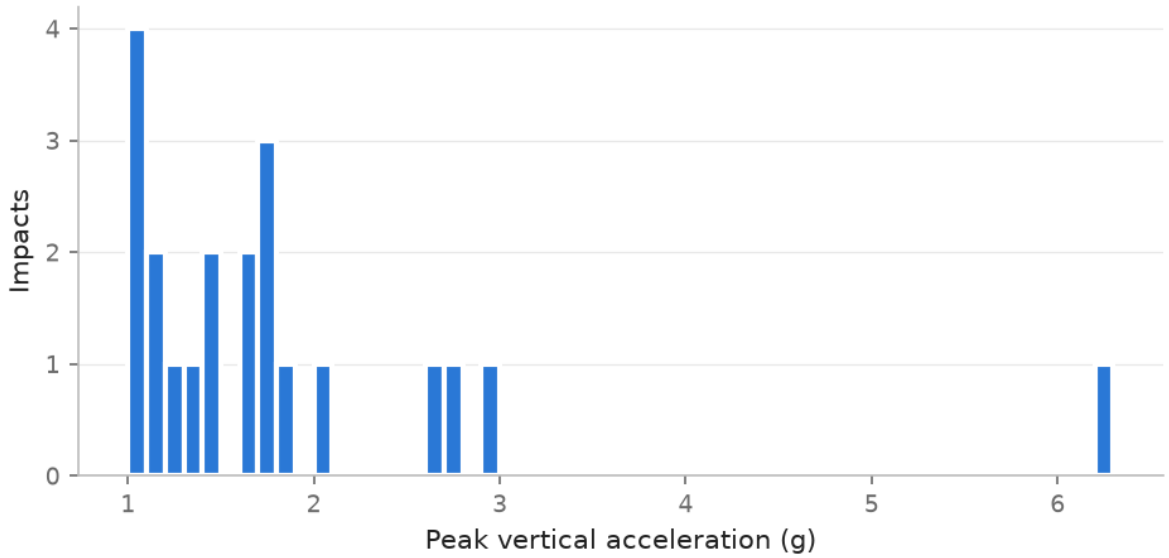


Figure A12: Ride 3: peak vertical acceleration of the 21 impacts.

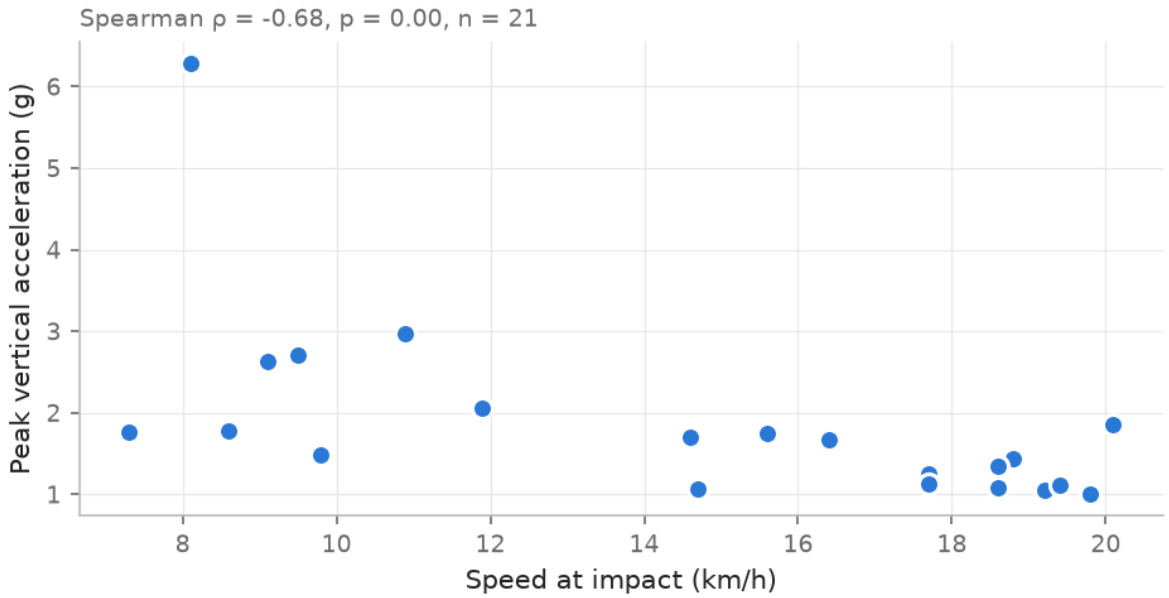


Figure A13: Ride 3: GPS speed against peak vertical acceleration for each impact. One ride; no trend is claimed.

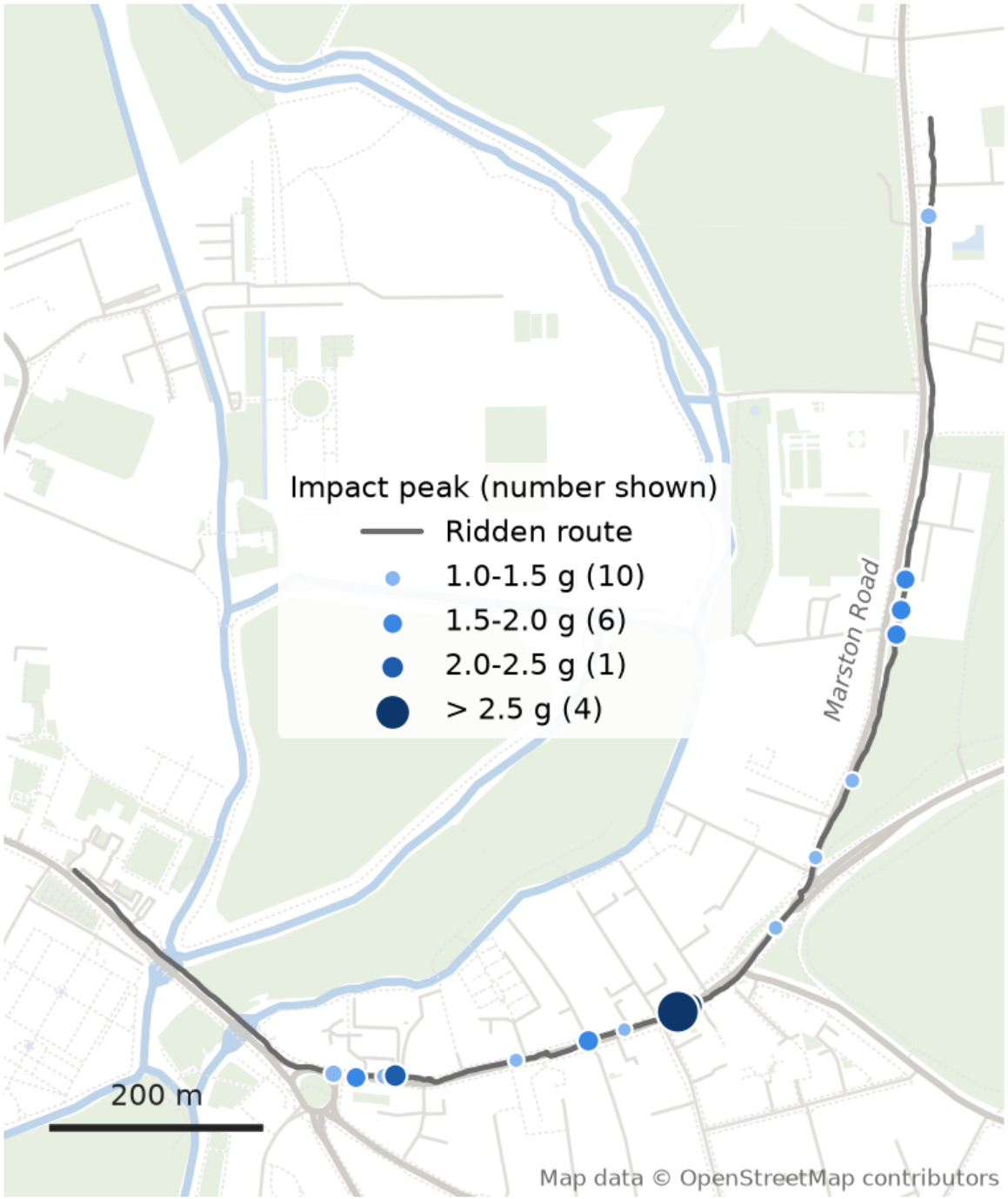


Figure A14: Ride 3: impacts on the ridden route, coloured by severity band (bands are analyst choices). Route ends (200 m) and impacts or reports near them are not drawn, for privacy; such impacts are still counted in Table A1. Map data © OpenStreetMap contributors.

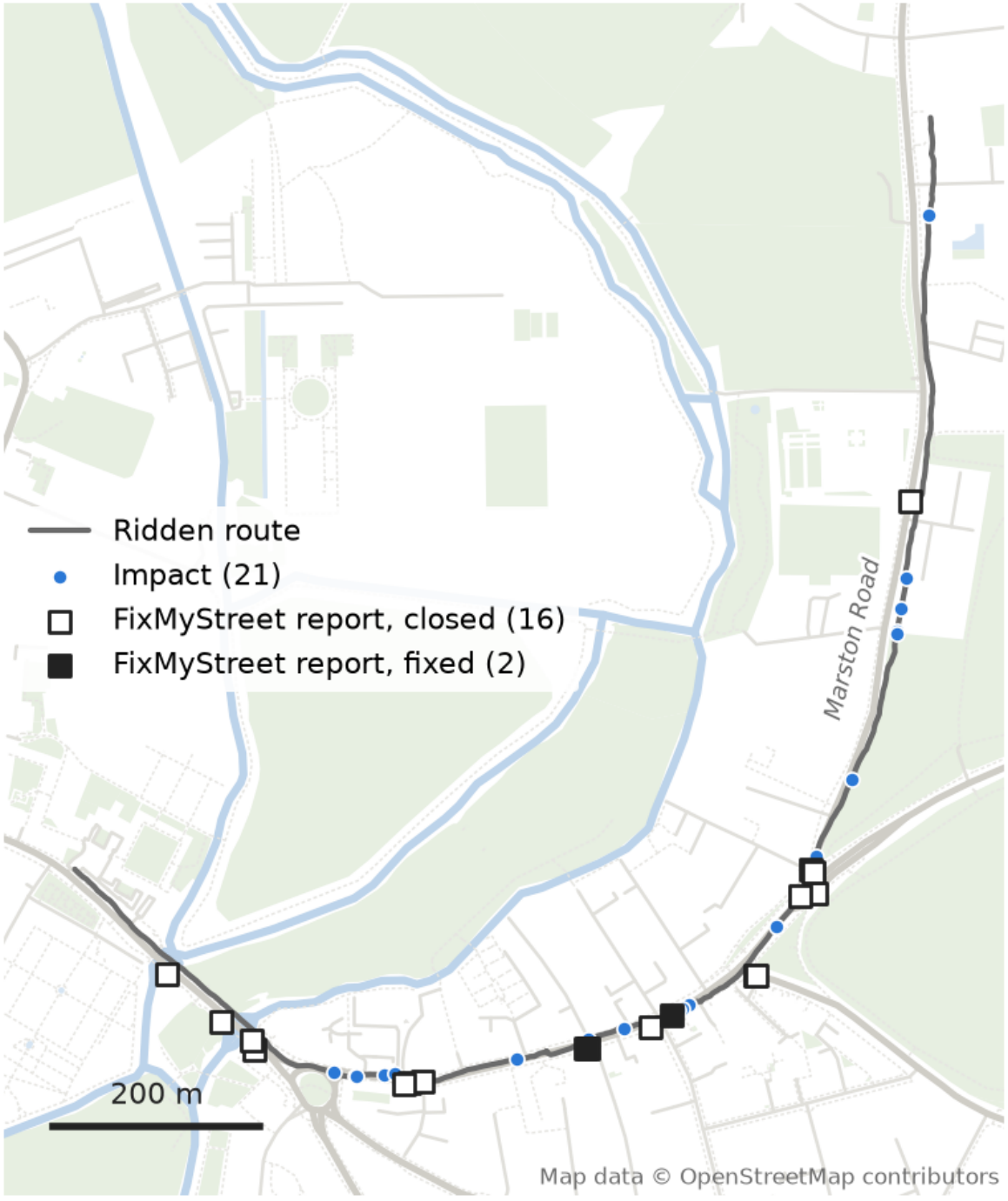


Figure A15: Ride 3: impacts and FixMyStreet road-surface reports made in the 365 days before the ride (hollow: closed; filled: marked fixed). A report near an impact does not show that it refers to the same defect. Route ends (200 m) and impacts or reports near them are not drawn, for privacy; such impacts are still counted in Table A1. Map data © OpenStreetMap contributors.

A.4 All impacts

Table A5: All 100 detected impacts. Time is local (Europe/London); speed is GPS speed at the impact; wobble = gyroscope ratio ≥ 1.5 after vs before (an analyst choice); report = a FixMyStreet road-surface report made in the 365 days before the ride within 25 m; spot = location in Table C1 (spots P9, P16, P30 and P31 lie in the trimmed route ends and have no location there). Coordinates are not listed, for privacy; the anonymised dataset is described under Data availability.

Ride	#	Time	Peak (g)	Speed (km/h)	Wobble	Road	Report ≤25 m	Spot
1	1	18:31:17	1.02	11.5		Other roads		P16
1	2	18:31:24	1.36	11.6	yes	Other roads		P16
1	3	18:32:19	1.04	19.6	yes	Old Road		P20
1	4	18:32:21	1.13	19.0		Old Road		P20
1	5	18:33:11	1.73	20.0		Other roads		P22
1	6	18:33:24	1.03	18.2		Other roads		P14
1	7	18:33:26	1.13	19.3		Other roads		P14
1	8	18:33:27	1.17	21.1		Other roads		P14
1	9	18:33:34	1.24	19.8		Other roads		P18
1	10	18:33:39	1.13	19.9		Other roads		P18
1	11	18:34:21	1.19	21.6		Morrell Avenue		P25
1	12	18:34:29	1.04	21.5		Morrell Avenue		P33
1	13	18:34:59	1.27	21.0	yes	Morrell Avenue		P13
1	14	18:35:01	1.22	22.3	yes	Morrell Avenue		P13
1	15	18:35:02	1.48	22.3		Morrell Avenue		P13
1	16	18:35:12	1.31	21.7	yes	Morrell Avenue		P17
1	17	18:35:16	1.25	21.6		Morrell Avenue		P17
1	18	18:35:44	1.01	21.2	yes	Morrell Avenue		P19
1	19	18:35:45	1.18	21.6		Morrell Avenue		P19
1	20	18:36:02	1.23	20.0		Morrell Avenue		P24
1	21	18:36:07	1.13	19.2		Morrell Avenue		P28
1	22	18:36:28	1.12	19.5		Morrell Avenue		P29
1	23	18:38:05	2.80	9.2	yes	Other roads		P8
1	24	18:38:39	2.68	14.6		Other roads		P12
1	25	18:38:59	1.37	17.0		Other roads		P15
1	26	18:39:16	1.10	10.2		Other roads		P15
1	27	18:39:24	1.17	17.4	yes	Other roads		P26
1	28	18:39:35	1.34	14.9		Other roads	yes	P12
1	29	18:39:42	1.01	13.5		Other roads		P12
1	30	18:40:19	1.16	18.5	yes	Marston Road		P2
1	31	18:41:00	2.24	16.9		Marston Road		P1
1	32	18:41:10	1.39	19.0		Marston Road	yes	P23
1	33	18:41:32	1.23	20.8		Marston Road		P5
1	34	18:41:36	1.76	20.8	yes	Marston Road		P5
1	35	18:41:42	1.07	21.1	yes	Marston Road		P5
1	36	18:41:43	1.30	20.6		Marston Road		P5
1	37	18:41:46	1.17	20.0		Marston Road		P5
1	38	18:42:03	1.19	17.9	yes	Marston Road		P10
1	39	18:42:29	2.03	20.0	yes	Marston Road		P9
1	40	18:42:41	1.06	17.8		Marston Road		P30
2	1	10:38:05	1.27	19.0		Marston Road		P9
2	2	10:38:17	1.06	17.5		Marston Road		P31
2	3	10:38:31	1.15	19.4	yes	Marston Road		P27

Ride	#	Time	Peak (g)	Speed (km/h)	Wobble	Road	Report ≤25 m	Spot
2	4	10:38:47	1.79	15.6	yes	Marston Road		P5
2	5	10:38:49	1.46	14.3		Marston Road		P5
2	6	10:38:51	1.17	11.7		Marston Road		P5
2	7	10:39:01	1.21	18.8		Marston Road		P5
2	8	10:40:00	1.23	18.4		Marston Road		P1
2	9	10:40:18	1.31	15.6	yes	Marston Road		P2
2	10	10:40:27	1.13	14.1		Marston Road		P2
2	11	10:40:31	1.47	14.7	yes	Marston Road		P2
2	12	10:40:32	1.06	11.2		Marston Road		P2
2	13	10:40:40	1.03	18.1	yes	Marston Road	yes	P7
2	14	10:40:42	1.17	20.1		Headington Road	yes	P7
2	15	10:41:02	1.61	17.8	yes	Morrell Avenue	yes	P11
2	16	10:41:02	2.87	18.0	yes	Morrell Avenue	yes	P11
2	17	10:41:04	1.24	16.9		Morrell Avenue	yes	P11
2	18	10:41:15	1.23	13.8		Other roads		P4
2	19	10:41:17	1.34	15.4	yes	Other roads		P4
2	20	10:41:18	1.20	16.6		Other roads		P4
2	21	10:41:19	1.12	18.1		Other roads		P4
2	22	10:41:21	1.11	19.1		Other roads	yes	P4
2	23	10:41:34	1.06	20.0		Other roads	yes	P6
2	24	10:41:42	1.24	16.3	yes	Other roads	yes	P6
2	25	10:42:18	1.31	7.7		Other roads		P3
2	26	10:42:19	1.96	8.9		Other roads		P3
2	27	10:42:19	1.82	8.6		Other roads		P3
2	28	10:42:21	1.67	8.3		Other roads		P3
2	29	10:42:21	1.75	9.0		Other roads		P3
2	30	10:42:22	1.78	8.8		Other roads		P3
2	31	10:42:23	1.83	8.5		Other roads		P3
2	32	10:42:24	1.57	8.2		Other roads		P3
2	33	10:42:25	2.25	8.5		Other roads		P3
2	34	10:42:26	1.43	8.5		Other roads		P3
2	35	10:42:36	1.45	10.5		Other roads	yes	P3
2	36	10:42:45	1.78	7.9		Other roads	yes	P3
2	37	10:42:53	1.82	7.4		Other roads	yes	P3
2	38	10:43:04	1.54	8.2	yes	Other roads		P3
2	39	10:43:45	2.64	7.4	yes	Other roads	yes	P21
3	1	13:41:06	1.44	18.8		Other roads		P3
3	2	13:41:11	1.48	9.8		Other roads		P3
3	3	13:41:12	1.76	7.3		Other roads		P3
3	4	13:41:20	1.25	17.7	yes	Other roads	yes	P3
3	5	13:41:22	2.06	11.9		Other roads	yes	P3
3	6	13:42:03	1.05	19.2		Other roads		P32
3	7	13:42:49	1.85	20.1	yes	Other roads	yes	P6
3	8	13:42:55	1.00	19.8		Other roads		P6
3	9	13:43:18	6.28	8.1	yes	Other roads	yes	P4
3	10	13:43:19	2.97	10.9		Other roads	yes	P4
3	11	13:43:19	2.63	9.1		Other roads	yes	P4
3	12	13:43:21	1.78	8.6		Other roads	yes	P4
3	13	13:43:22	2.71	9.5		Other roads	yes	P4
3	14	13:43:58	1.11	19.4	yes	Other roads		P8

Ride	#	Time	Peak (g)	Speed (km/h)	Wobble	Road	Report ≤25 m	Spot
3	15	13:44:26	1.08	18.6		Marston Road	yes	P7
3	16	13:44:42	1.13	17.7	yes	Marston Road		P2
3	17	13:45:11	1.70	14.6	yes	Marston Road		P1
3	18	13:45:11	1.06	14.7	yes	Marston Road		P1
3	19	13:45:15	1.75	15.6	yes	Marston Road		P1
3	20	13:45:21	1.67	16.4		Marston Road		P1
3	21	13:46:25	1.34	18.6		Marston Road		P10

B Recording quality and walking detection

This appendix gives the full check of the recording method summarised in section 2.2. It tests whether the Sensor Logger data are internally sound and whether riding can be told apart from walking. Neither part needs a reference instrument, and neither proves absolute accuracy, which would need a calibrated accelerometer, survey-grade GNSS or exactly timed events.

B.1 Methods

Data quality. Table B1 lists the checks, applied to all three recordings. Units, as checked: user acceleration and gravity are in m/s² in the device frame (Apple Inc. n.d.[b]); the raw accelerometer is in g (Apple Inc. n.d.[a]) (raw ÷ (user acceleration + gravity) = 1/9.80665 on every axis); the metadata recording time is UTC, so local times were computed from the nanosecond timestamps.

Table B1: Data-quality checks.

Check	How	What it can show
Sampling	Intervals between timestamps: median, 1st and 99th percentile; gaps (more than 1.5 times the nominal interval); out-of-order and duplicate timestamps	Dropped samples, irregular sampling
Clock	Absolute timestamp against elapsed time	Disagreeing time bases
Raw vs fused acceleration	User acceleration plus gravity, in g and interpolated to the raw sample times, compared with the raw accelerometer	Consistency of the iOS fusion output with the raw sensor (not accuracy)
Noise and bias	Quietest 5% of non-overlapping 4 s windows (by user-acceleration SD): acceleration and gyroscope SD, raw magnitude (1 g at rest), calibrated minus uncalibrated gyroscope mean	Noise and static error if the phone was truly still; otherwise upper bounds
Gravity magnitude	Length of the gravity vector	Reported only: fixed at 9.80665 m/s² by the fusion, so not an accuracy test
GPS	Fix interval and gaps; reported horizontal accuracy; reported vs position-derived speed; position scatter while standing	Availability, stated precision, internal consistency

Walking versus riding. The vertical user acceleration (the projection onto the gravity direction, up positive (Mizell 2003)) was analysed in 4 s windows starting every 1 s, with the features in Table B2. The rules, applied in order, were: *stationary* if the vertical SD was below 0.5 m/s² and GPS speed below 2 km/h (or missing); *walking* if step regularity and step-band share were both at least 0.5 and GPS speed below 8 km/h (or missing); *riding* if GPS speed was at least 8 km/h and the step-band share below 0.3; otherwise *unclear* (slow riding, handling the phone, getting on or off). Labels were smoothed with a

centred 5-window majority filter, and a *walking bout* is at least 10 s of walking windows. All thresholds are analyst choices, set from physical reasoning and from inspecting ride 2 before the rider's references were known, not fitted to labels, and not changed afterwards.

Table B2: Window features for walking detection.

Feature	Definition	Rationale
Vertical SD	SD of vertical acceleration (m/s²)	Overall motion; near 0 when the phone is still
Step-band share	Share of vertical-acceleration power (Welch spectrum (Welch 1967), 256-sample segments) in 1.2–2.8 Hz, out of 0.5–20 Hz	Walking concentrates power at the step frequency (measured means 1.8–2.0 Hz (Pachi and Ji 2005); the band is set wider to include slow and brisk walking); scooter vibration is broadband
Step regularity	Highest normalised autocorrelation at lags of one step period (0.36–0.83 s)	Measures step regularity and cadence (Moe-Nilssen and Helbostad 2004); steps repeat, vibration does not
Cadence	1 / that lag (steps/s)	Plausibility check
GPS speed	Median reported speed of the fixes in the window	Separates riding from walking speed

Why two acceleration features. After the walk in ride 2 there are repeated periods with high step regularity (about 0.85) while the phone is almost still. They have essentially no power in the step band (share about 0) and tiny amplitude (SD about 0.1 m/s²): a small periodic vibration, not steps (Figure B3). Step regularity alone would have labelled them walking.

First step of a bout. Vertical acceleration was band-pass filtered (0.8–4 Hz, second-order Butterworth (Butterworth 1930), applied forwards and backwards for zero phase shift; SciPy (Virtanen et al. 2020)). Steps are peaks with a prominence of at least 1 m/s², at least 0.35 s apart. From the first step inside the bout, the method moves back over earlier peaks while successive intervals stay within one step period (0.35–0.85 s); the earliest peak reached is the first step. The *GPS lag* is the time from the first step until GPS speed exceeds 1.5 km/h on three consecutive fixes.

GPS speed against positions. Each riding or walking segment of at least 30 s, trimmed by 15 s at each end, was compared: median reported speed against speed from the summed fix-to-fix distance (inflated by GPS jitter) and from the straight-line distance between the first and last fix (a lower bound on the distance travelled). A reported speed below the straight-line speed under-reads the true speed.

B.2 Data quality

- *Sampling and timing are clean*: no dropped samples, gaps or timestamp faults in any motion stream of any recording.
- *At rest* (ride 2 only), noise is about 0.001 g and the static acceleration magnitude is 0.6% below 1 g, both small against the 1.0 g impact threshold.
- *Raw and fused acceleration* agree most closely in ride 2 (mostly still) and least in ride 1 (riding throughout); likely reasons are fast vibration between samples taken at different instants and filtering in the iOS fusion (not tested further).
- *The gyroscope bias correction* applied by iOS is small and identical across recordings (same phone).
- *GPS horizontal accuracy* (median 11–14 m) is the phone's own estimate, not a measured error; for

Table B3: Data quality per recording. Bold: the only recording in which the phone was truly still (after the walk); in rides 1 and 3 the phone stayed on the handlebar, so the quietest-window values include vibration and are upper bounds.

Measure	Ride 1	Ride 2	Ride 3
Motion sensors: rate	99.72 Hz	99.72 Hz	99.72 Hz
Sampling interval, median (99th percentile)	10.03 (10.03) ms	10.03 (10.03) ms	10.03 (10.03) ms
Gaps, out-of-order, duplicate timestamps	0, 0, 0	0, 0, 0	0, 0, 0
Clock: timestamp vs elapsed time, max difference	0.0 ms	0.0 ms	0.0 ms
Raw vs fused acceleration, median (95th pct.)	0.049 (0.160) g	0.007 (0.092) g	0.013 (0.053) g
Quietest windows: acceleration SD	0.31 m/s²	**0.012 m/s²**	0.19 m/s²
Quietest windows: gyroscope SD	0.16 rad/s	**0.0035 rad/s**	0.095 rad/s
Quietest windows: raw magnitude	1.002 g	**0.994 g** (SD 0.0009)	1.003 g
Gyroscope calibrated – uncalibrated (x, y, z)	0.002, 0.005, −0.002 rad/s	0.002, 0.005, −0.002 rad/s	0.002, 0.005, −0.002 rad/s
GPS fixes; gaps over 1.5 s; longest gap	828; 2; 17.1 s	1160; 144; 15.0 s	686; 0; 1.0 s
GPS reported horizontal accuracy, median (90th pct.)	11.5 (15.1) m	11.3 (15.5) m	13.9 (17.6) m
GPS position scatter while standing, median (90th pct.)	0.7 (2.1) m	0.1 (1.7) m	0.5 (1.6) m
GPS speed: |reported – position-derived|, median	0.38 m/s	0.64 m/s	0.52 m/s

context only, the measured error of an iPhone 6 in an urban area averaged 7–13 m (Merry and Bettinger 2019).

- *GPS gaps*: in ride 2 they start at 10:53:51, after the walk (phone indoors or covered); both gaps in ride 1 are at its very start (the first fix is a cached position from 17 s before recording began).

B.3 Walking detection

Table B4: Time per label (after smoothing).

Recording	Riding	Walking	Stationary	Unclear
Ride 1 (ride only)	564 s	6 s	64 s	187 s
Ride 2 (ride, then walk)	321 s	**330 s**	1103 s	361 s
Ride 3 (ride only)	347 s	8 s	100 s	226 s

Table B5: Walking bouts detected in ride 2. The two bouts are probably one walk with a pause of a few seconds around 10:51:21–25 (not confirmed).

Bout	First step (local time)	Length	Steps	Median cadence	GPS lag at onset
1	**10:46:47.04** (580.6 s into the recording)	275 s	562	2.04 steps/s	12.0 s
2	10:51:25.21	37 s	69	1.89 steps/s	3.8 s

The Voi rental ended at 10:46 (app ride history, 10:46:00–10:46:59), which overlaps the phone-handling period; this is consistent with the rental being ended on the phone then, but the data cannot show it. The 8 s burst at 10:45:58 resembles the confirmed false positives below (short, after riding, step regularity below that of the real walk), so it is not treated as the start of walking.

Table B6: The transition from riding to walking in ride 2 (Figure 2).

Time (approx.)	What the data show
10:45:52–10:45:57	GPS speed falls from riding speed, ending in a sharp acceleration spike
10:45:58–10:46:06	A short walking-like burst (8 s, decaying amplitude)
10:46:06–10:46:46	About 40 s of standing or handling the phone (from about 10:46:16 the gravity vector shows the phone lying or held flat)
from 10:46:47	Steady walking

Feature separation (an independent check). In the 1,126 windows across all three recordings where GPS speed alone was 12 km/h or more (riding, defined without the acceleration features), none met the walking criteria: step regularity had a median of 0.18 (maximum 0.45) and step-band share a median of 0.010 (maximum 0.44). In the main walking bout, step regularity had a median of 0.78 (interquartile range 0.76–0.80) and step-band share 0.83; these walking values come from the detected bout itself, so they are not independent of the method.

Table B7: Walking labels in the two recordings without walking (rider's confirmation).

Recording	Walking labels after smoothing	Before smoothing	Bouts of 10 s+	At GPS 8 km/h+
Ride 1	6 s (0.7% of 821 windows): 18:43:53–55 and 18:44:06–10	9 windows	0	0
Ride 3	8 s (1.2% of 681 windows): 13:49:08–16	9 windows	0	0

All of these came after the ride had ended, at GPS speeds below 1 km/h, with step regularity (0.51–0.71) below the real walk's median of 0.78. In ride 1 they fall in the last 20 s before the recording stopped, with the phone held flat (probably handled); in ride 3 the phone was roughly in its mounted orientation, about 1.5 min after riding ended. Their cause (handling the phone, moving or parking the scooter) is not known. The 10 s minimum bout length was set before the script was first run on rides 1 and 3 and excludes all of them.

Table B8: Detections against the rider's references.

Recording	Reference (source)	Precision	Detected
Ride 1	No walking (rider)	whole recording	0 walking bouts; 6 s of isolated walking labels after the ride ended
Ride 2	Walking "after 10 minutes" = 10:47:06 (rider's recollection)	about a minute	First step of sustained walking 10:46:47.04 (−19 s)
Ride 2	Rental ended 10:46 (Voi app ride history)	one minute	Last riding segment ends 10:45:54; walking starts 10:46:47.04: order consistent
Ride 3	No walking (rider)	whole recording	0 walking bouts; 8 s of isolated walking labels after the ride ended

What the references show: *timing*: two independent references, both with one-minute precision, place the change from riding to walking in the same minute as the detection, which is agreement at minute level, not a timing error in seconds; *specificity*: across about 25 minutes of recordings without walking, no walking bout was detected and no window was labelled walking while riding; *sensitivity*: there is only one walking episode, and it was detected, so no detection rate can be given. The rental end marks the end of the hire, not of riding: the data place riding ending about 6–65 s before it.

B.4 GPS speed

Table B9: GPS-reported speed against speed from GPS positions (km/h; segments of at least 30 s, 15 s trimmed at each end; the second ride-2 riding segment is 26 s after trimming). The straight-line speed is a lower bound on the true speed.

Ride	Activity	Reported median	Straight-line	Summed path
1	riding	20.1	19.0	20.8
1	riding	19.1	19.3	19.7
2	riding	17.9	19.2	19.5
2	riding	12.1	15.3	15.4
2	**walking**	**3.0**	**4.6**	5.5
3	riding	19.5	19.5	19.6
3	riding	17.5	19.1	19.5

- *Walking*: the reported speed is at least about 1.5 km/h (34%) below the position-derived lower bound. With the straight-line speed the implied step length is at least 4.6 km/h ÷ 2.04 steps/s ≈ 0.62 m, close to measured means of 0.67 m (women) and 0.75 m (men) (Pachi and Ji 2005); with the reported speed it would be 0.41 m.
- *Riding*: reported and position-derived speeds are closer, but in 4 of 6 segments the reported speed is below the straight-line lower bound, by 0.2–3.2 km/h.
- *Lag*: GPS speed rose about 12 s after walking began (bout 1).
- *Context*: a dedicated (non-phone) GPS receiver on a bicycle gave speeds within 0.2 m/s of a calibrated speedometer for 45% of values, with larger errors on curved paths through a tendency to underestimate speed (Witte and Wilson 2004); phones and walking speeds were not tested there.
- *Consequences for impact detection*: the speed rule is needed, because walking alone produced 11 peaks of 1.0 g or more (maximum 1.19 g) and handling the phone after the walk 3 more (maximum 2.21 g), none of which were counted. But the rule uses reported speed, so some slow-riding impacts may have been excluded (not quantified).

B.5 Limitations of the check

- One rider, one phone model, one app version, three recordings, one walking episode (about 5 minutes) and about 25 minutes of recordings without walking; other people, gaits or phone positions (pocket, bag, walking while pushing the scooter) may behave differently.
- Thresholds are analyst choices set after looking at ride 2; not fitted to labels, but not blind either. Rides 1 and 3 were not used to set them.
- The references have one-minute precision, and the rider's account that rides 1 and 3 contain no walking is not independently checked.
- No external reference for acceleration, rotation or position: the data-quality results show internal consistency and plausibility, not absolute accuracy.
- Where the phone was during the walk is not known; the gravity direction was steady and similar to the mounted position. Walking with the phone loosely held or in a bag may give weaker step

signals, which could fall under the stationary rule.

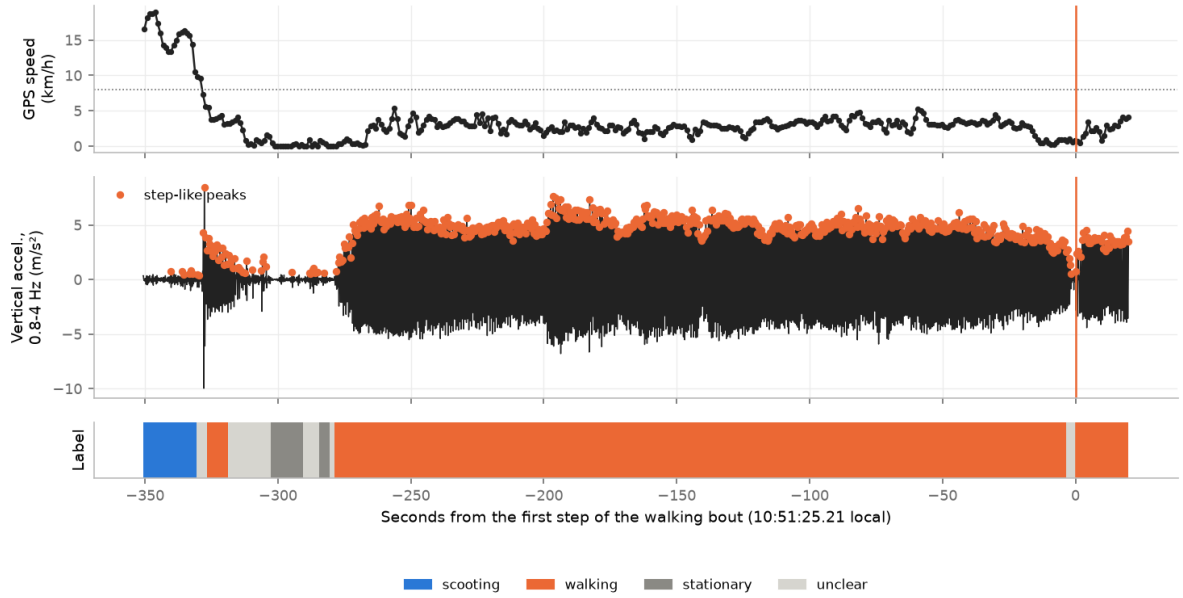


Figure B1: Start of the second walking bout in ride 2 (first step 10:51:25.21), drawn as Figure 2: GPS speed, band-passed vertical acceleration with step-like peaks, and window labels.

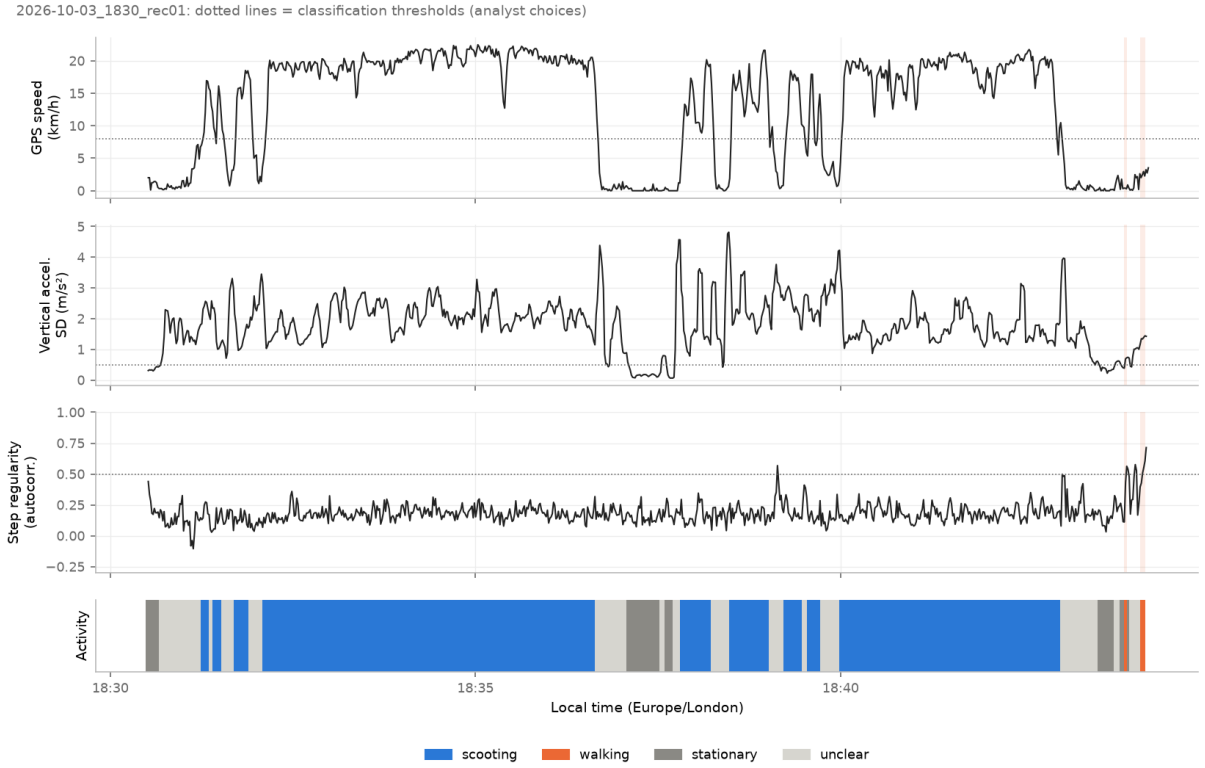


Figure B2: Ride 1: GPS speed, vertical acceleration SD, step regularity and activity labels. Dotted lines: thresholds.

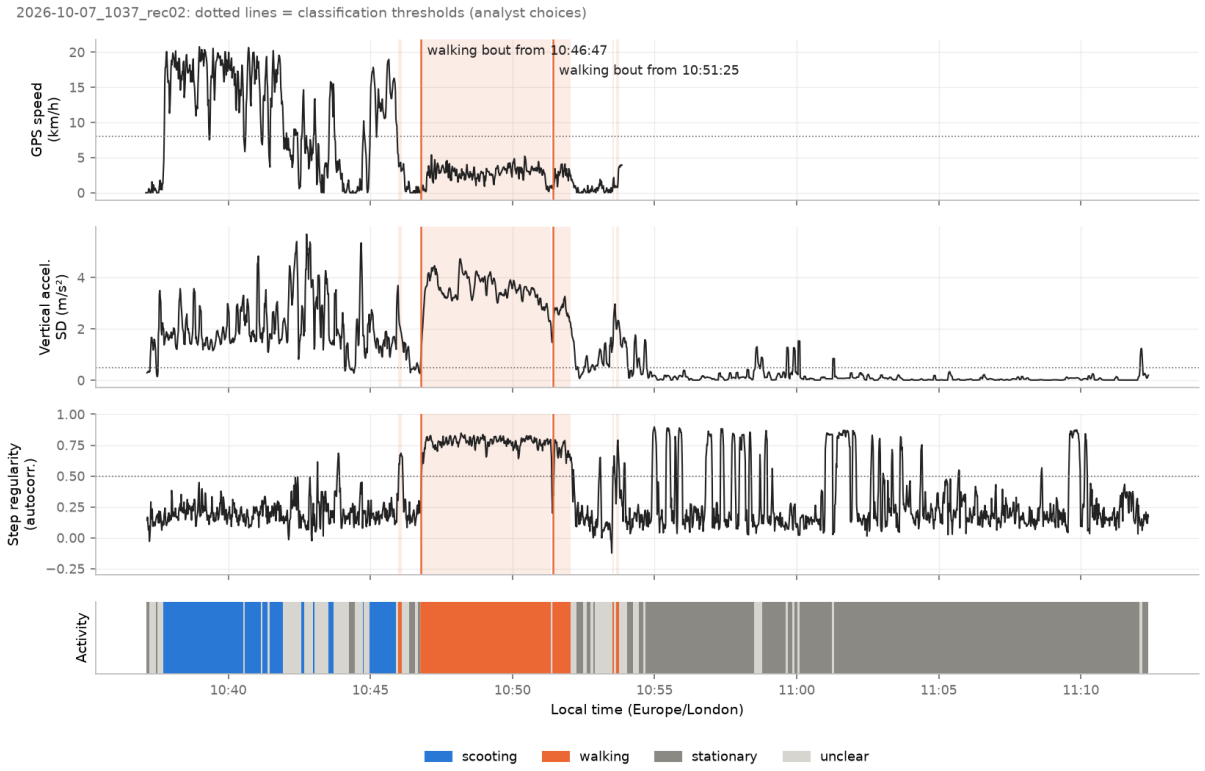


Figure B3: Ride 2: the ride, the walk from 10:46:47 and the phone at rest afterwards. The repeated high step-regularity values after the walk come with almost no power in the step band: a small periodic vibration, which the step-band rule correctly rejects.

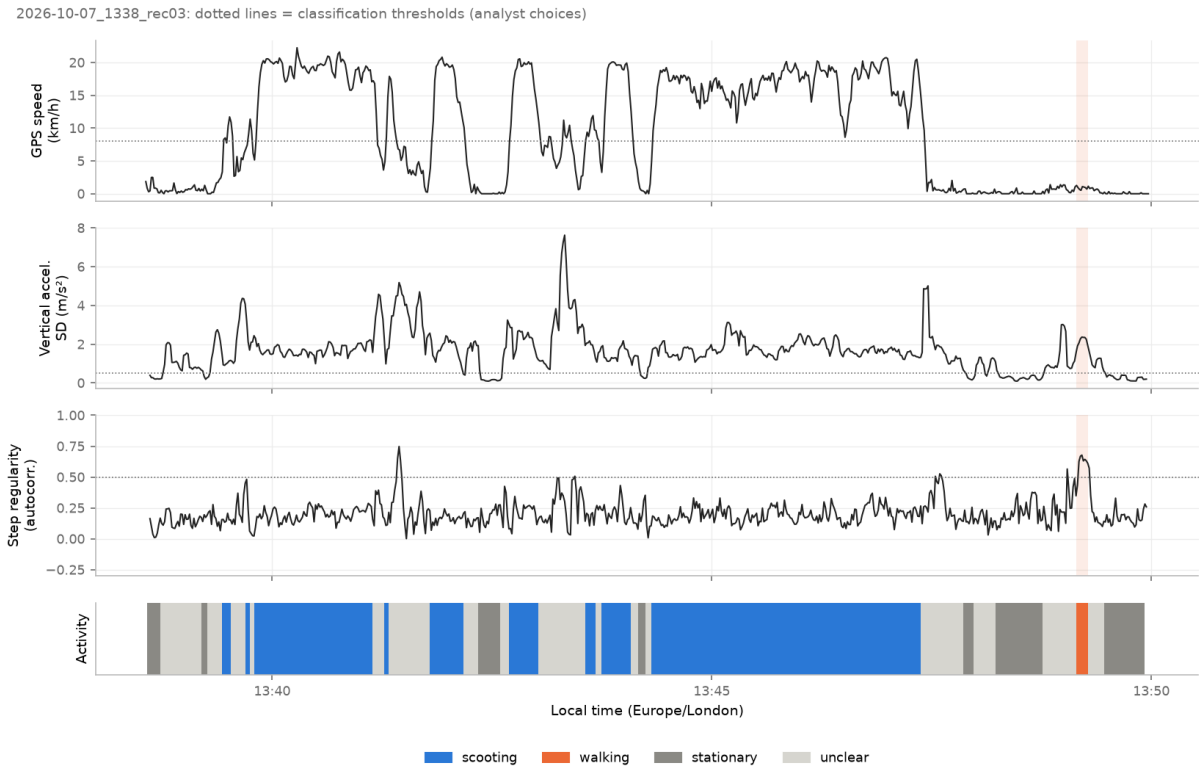


Figure B4: Ride 3.

C Impact spots with links to street-level imagery

Table C1 lists the spots with links that open Google Street View at the spot, facing the direction of travel and tilted 25° towards the road, a Google Maps pin, and Mapillary imagery (the full list of all 33 spots, with the same links, is in the supplementary file `spot_links.csv`). The links use Google's documented Maps URLs, which need no API key (Google n.d.); no imagery is downloaded, and no Street View images are reproduced here, as Google does not permit their use in print. Open imagery was checked first: Panoramax (Panoramax n.d.) had no photos in the study area, and KartaView (KartaView n.d.) had five photos near spot P1, from 2017 and 2021, too old to show the current surface.

Table C1: Impact spots, most important first, as in the project's spot list. Links open Google Street View at the spot facing the direction of travel (tilted towards the road), a Google Maps pin, and Mapillary imagery. On paper, scan the QR code to open the same Street View, or enter the coordinates in any map. FixMyStreet: nearest road-surface report made in the 365 days before the ride (within 25 m, or the nearest). Spots P9, P16, P30, P31 lie in the hidden first or last 200 m of a route and are not listed, for privacy; they are included in all counts and in the tier totals.

Tier 1: Hit on 2+ separate rides (10)

Spot	Impacts	Rides	Max g	Road	Open	FixMyStreet (recent)	Location	Events (recording#impact)	Scan
P1	6	3	2.24	Marston Road	Street View · Map · Mapillary	none ≤ 25 m (nearest 60 m)	51.75427, −1.23588 · facing 10°	rec01#31; rec02#8; rec03#17; rec03#18; rec03#19; rec03#20	
P2	6	3	1.47	Marston Road	Street View · Map · Mapillary	none ≤ 25 m (nearest 44 m)	51.75268, −1.23665 · facing 27°	rec01#30; rec02#9; rec02#10; rec02#11; rec02#12; rec03#16	
P3	19	2	2.25	Other roads	Street View · Map · Mapillary	report 5 m (closed)	51.75024, −1.24253 · facing 261°	rec02#25; rec02#26; rec02#27; rec02#28; rec02#29; rec02#30; rec02#31; rec02#32; rec02#33; rec02#34; rec02#35; rec02#36; rec02#37; rec02#38; rec03#1; rec03#2; rec03#3; rec03#4; rec03#5	
P4	10	2	6.28	Other roads	Street View · Map · Mapillary	report 8 m (fixed)	51.75085, −1.23877 · facing 218°	rec02#18; rec02#19; rec02#20; rec02#21; rec02#22; rec03#9; rec03#10; rec03#11; rec03#12; rec03#13	
P5	9	2	1.79	Marston Road	Street View · Map · Mapillary	none ≤ 25 m (nearest 121 m)	51.75687, −1.23551 · facing 355°	rec01#33; rec01#34; rec01#35; rec01#36; rec01#37; rec02#4; rec02#5; rec02#6; rec02#7	

Spot	Impacts	Rides	Max g	Road	Open	FixMyStreet (recent)	Location	Events (recording#impact)	Scan
P6	4	2	1.85	Other roads	Street View · Map · Mapillary	report 8 m (closed)	51.75059, −1.23993 · facing 249°	rec02#23; rec02#24; rec03#7; rec03#8	
P7	3	2	1.17	Marston Road	Street View · Map · Mapillary	report 8 m (closed)	51.75208, −1.23700 · facing 180°	rec02#13; rec02#14; rec03#15	
P8	2	2	2.80	Other roads	Street View · Map · Mapillary	none ≤ 25 m (nearest 38 m)	51.75155, −1.23772 · facing 52°	rec01#23; rec03#14	
P10	2	2	1.34	Marston Road	Street View · Map · Mapillary	none ≤ 25 m (nearest 87 m)	51.75777, −1.23549 · facing 13°	rec01#38; rec03#21	

Spot	Impacts	Rides	Max g	Road	Open	FixMyStreet (recent)	Location	Events (recording#impact)	Scan

Tier 2: 2+ impacts on one ride (10)

Spot	Impacts	Rides	Max g	Road	Open	FixMyStreet (recent)	Location	Events (recording#impact)	Scan
P11	3	1	2.87	Morrell Avenue	Street View · Map · Mapillary	report 12 m (closed)	51.75124, −1.23796 · facing 224°	rec02#15; rec02#16; rec02#17	
P12	3	1	2.68	Other roads	Street View · Map · Mapillary	report 6 m (fixed)	51.75222, −1.23766 · facing 286°	rec01#24; rec01#28; rec01#29	
P13	3	1	1.48	Morrell Avenue	Street View · Map · Mapillary	none ≤ 25 m (nearest 113 m)	51.74981, −1.23017 · facing 267°	rec01#13; rec01#14; rec01#15	
P14	3	1	1.17	Other roads	Street View · Map · Mapillary	none ≤ 25 m (nearest 118 m)	51.75162, −1.22289 · facing 226°	rec01#6; rec01#7; rec01#8	
P15	2	1	1.37	Other roads	Street View · Map · Mapillary	none ≤ 25 m (nearest 73 m)	51.75256, −1.23889 · facing 293°	rec01#25; rec01#26	
P17	2	1	1.31	Morrell Avenue	Street View · Map · Mapillary	none ≤ 25 m (nearest 178 m)	51.74974, −1.23136 · facing 265°	rec01#16; rec01#17	
P18	2	1	1.24	Other roads	Street View · Map · Mapillary	none ≤ 25 m (nearest 115 m)	51.75120, −1.22356 · facing 224°	rec01#9; rec01#10	

Spot	Impacts	Rides	Max g	Road	Open	FixMyStreet (recent)	Location	Events (recording#impact)	Scan
P19	2	1	1.18	Morrell Avenue	Street View · Map · Mapillary	none ≤ 25 m (nearest 75 m)	51.74983, −1.23387 · facing 285°	rec01#18; rec01#19	
P20	2	1	1.13	Old Road	Street View · Map · Mapillary	none ≤ 25 m (nearest 58 m)	51.75333, −1.21848 · facing 248°	rec01#3; rec01#4	

Spot	Impacts	Rides	Max g	Road	Open	FixMyStreet (recent)	Location	Events (recording#impact)	Scan

Tier 3: Single impact (13)

Spot	Impacts	Rides	Max g	Road	Open	FixMyStreet (recent)	Location	Events (recording#impact)	Scan
P21	1	1	2.64	Other roads	Street View · Map · Mapillary	report 23 m (fixed)	51.74987, −1.24433 · facing 219°	rec02#39	
P22	1	1	1.73	Other roads	Street View · Map · Mapillary	none ≤ 25 m (nearest 52 m)	51.75211, −1.22209 · facing 225°	rec01#5	
P23	1	1	1.39	Marston Road	Street View · Map · Mapillary	report 5 m (closed)	51.75521, −1.23567 · facing 8°	rec01#32	
P24	1	1	1.23	Morrell Avenue	Street View · Map · Mapillary	none ≤ 25 m (nearest 73 m)	51.75013, −1.23521 · facing 308°	rec01#20	
P25	1	1	1.19	Morrell Avenue	Street View · Map · Mapillary	none ≤ 25 m (nearest 115 m)	51.75015, −1.22682 · facing 251°	rec01#11	
P26	1	1	1.17	Other roads	Street View · Map · Mapillary	none ≤ 25 m (nearest 31 m)	51.75242, −1.23829 · facing 111°	rec01#27	
P27	1	1	1.15	Marston Road	Street View · Map · Mapillary	none ≤ 25 m (nearest 249 m)	51.75806, −1.23564 · facing 176°	rec02#3	

Spot	Impacts	Rides	Max g	Road	Open	FixMyStreet (recent)	Location	Events (recording#impact)	Scan
P28	1	1	1.13	Morrell Avenue	Street View · Map · Mapillary	none ≤ 25 m (nearest 68 m)	51.75031, −1.23557 · facing 304°	rec01#21	
P29	1	1	1.12	Morrell Avenue	Street View · Map · Mapillary	none ≤ 25 m (nearest 65 m)	51.75094, −1.23697 · facing 297°	rec01#22	
P32	1	1	1.05	Other roads	Street View · Map · Mapillary	none ≤ 25 m (nearest 68 m)	51.75037, −1.24128 · facing 80°	rec03#6	
P33	1	1	1.04	Morrell Avenue	Street View · Map · Mapillary	none ≤ 25 m (nearest 131 m)	51.75005, −1.22744 · facing 260°	rec01#12	

D Road bumps: an exploratory extension of the model

Status. These are exploratory model results, not findings about real scooters. No bump was measured or identified on the rides, and the model's behaviour on bumps has not been validated. Everything below describes what the simulation produced under its assumed parameters (Table 4) and the analyst-chosen fall criteria (section 3.4).

Why bumps. Raised obstacles, such as a utility cover or patch standing proud of the surface, a lifted slab, a tree-root ridge or a traffic-calming hump, are a second kind of road irregularity a rider may meet. Whether they matter for e-scooter riders in practice is not tested here.

What was simulated. The dynamics model of section 3.4, unchanged except for the road:

- *Sharp-edged rectangular bumps*, 20–100 mm high, 300, 600 and 1000 mm long (the pothole lengths), with MuJoCo's contact. This idealised shape stands for defects: legal UK road humps may not have a vertical face above 6 mm (UK Government 1999).
- *Round-top (circular-arc) humps*, 20–100 mm high, 3.7 m long, the round-top length reported as standard (Department of Transport 1996a), and 600 mm long (shorter than the 900 mm legal minimum, as might be found off the highway), with the profile contact.
- The same sharp bumps with the profile contact, as a cross-check of the contact model.
- Speeds of 15, 20 and 25 km/h, configurations 10/10 (reference), 11/10 (the Voiager 5 that was ridden), 12/10, 12/12 and 14/10, without and with the modelled front spring; and a rear-size check on the 300 mm sharp bump (heights 50–100 mm in 5 mm steps, eight configurations including larger rear wheels).

In the fall attribution, a fall is credited to the rear wheel if the rear wheel had reached the bump's leading face before the fall criterion was met.

Table D1: Number of runs meeting the fall criteria, per configuration (sharp bumps and potholes: of 81 runs; round-top humps: of 54). 11/10 (V5) = the Voiager 5 that was ridden; 10/10 (ref.) = smaller reference wheels.

Obstacle	Fork	10/10 (ref.)	11/10 (V5)	12/10	12/12	14/10
Pothole (sharp)	none	15	15	14	14	11
Sharp bump	none	22	18	13	14	4
Round-top hump	none	0	0	0	0	0
Pothole (sharp)	front spring	9	9	6	6	1
Sharp bump	front spring	1	2	2	1	2
Round-top hump	front spring	0	0	0	0	0

Table D2: Rear-size check, 300 mm sharp bump: lowest bump height (mm, 5 mm steps) at which the model met the fall criteria; – = not met up to 100 mm.

Fork	Speed	10/10	11/10 (V5)	12/10	14/10	10/12	12/12	10/14	14/14
none	15 km/h	85	90	95	–	85	95	85	–
none	20 km/h	80	85	90	100	80	90	80	100
none	25 km/h	80	85	85	100	80	85	80	95
front spring	15 km/h	85	85	85	85	90	95	–	100
front spring	20, 25 km/h	–	–	–	–	–	–	–	–

Observations (model only).

1. *Without the front spring*, the current 11/10 configuration met the fall criteria in more sharp-bump runs than pothole runs of the same dimensions (18 against 15 of 81; 10/10 reference: 22 against 15; Table D1). A plausible geometric explanation, not tested separately, is that a wheel meets the full height of a step but drops only part way into a short pothole. Fewer falls occurred with larger front wheels (22, 18, 13 and 4 for 10, 11, 12 and 14 in fronts); changing the rear size alone did not change the falling heights (Table D2). Without the spring, part of the bump falls in every configuration (6 of the 18 for 11/10) were credited to the rear wheel by the rule above. Figure D1 shows one case.

2. *With the front spring*, bump falls were rare: one or two runs per configuration, all on the 300 mm bump at 15 km/h. In each, the fall criteria were met after the rear wheel struck the bump's leading edge (Figure D2). In the rear-size check, a larger rear wheel raised the falling height or prevented the fall (10/12: 90 mm; 12/12: 95 mm; 14/14: 100 mm; 10/14: none up to 100 mm), while a larger front wheel alone did not (85 mm for 10/10, 12/10 and 14/10). This rests on very few runs (one bump length, one speed, differences of 5–15 mm, close to the 5 mm step) and on a model not validated for bumps. It is reported as an open question, whether rear-wheel size or rear suspension matters on scooters with a sprung front, not as evidence that larger rear wheels are more stable. The Voiager 5 (11/10) behaved like the 10/10 reference here (85 mm, 2 falls in the main bump sweep, both after the rear-wheel strike). It does indicate that the working hypothesis of this paper (the front wheel alone) may not hold for every type of obstacle.

3. No round-top hump run met the fall criteria in any configuration, consistent with, but not evidence for, guidance that humps can be ridden safely by cyclists and motorcyclists (Department of Transport 1996b); that guidance was not written for e-scooters. The short 600 mm hump gave large modelled speed losses at 15 km/h (up to 47% with the fork).

4. The front spring changed the bump results much more than the pothole results (11/10: 18 to 2 falls on bumps, 15 to 9 on potholes; 10/10: 22 to 1 and 15 to 9), so the bump results depend strongly on the assumed fork stiffness, damping and travel, which were not varied for bumps.

Figures D3–D10 give the full bump results behind Table D1.

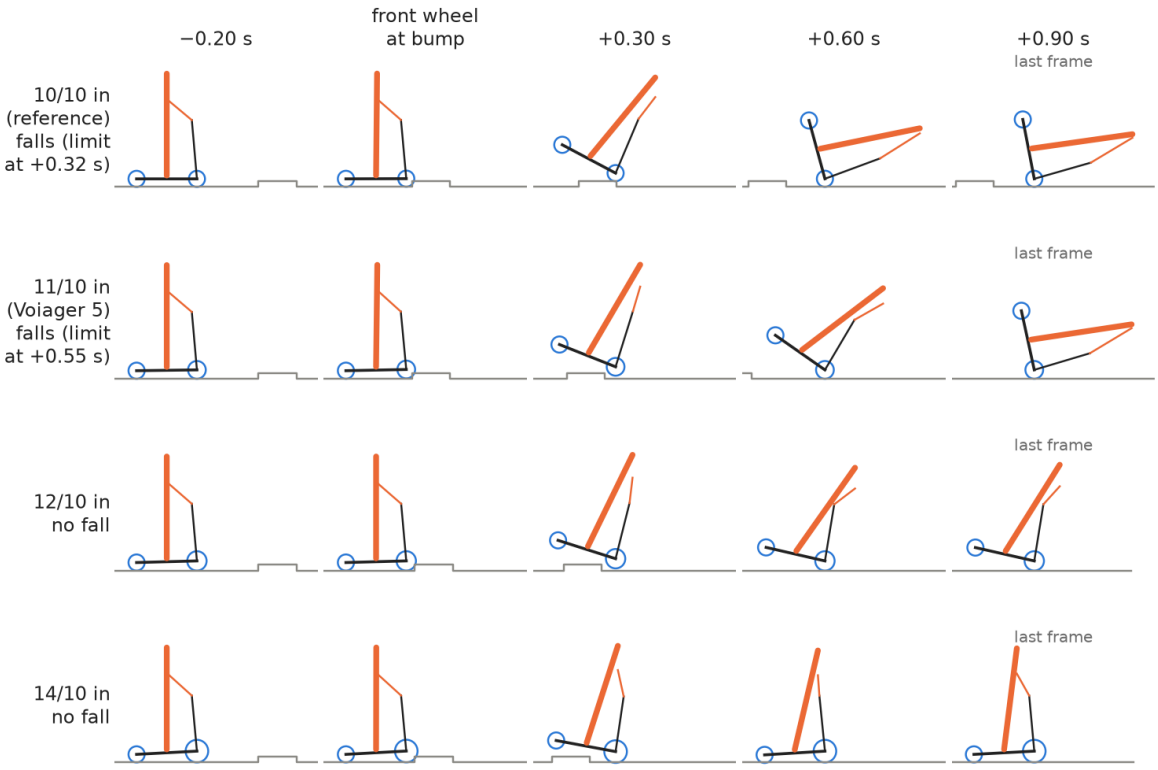


Figure D1: Key frames, sharp-edged bump 90 mm high and 600 mm long at 20 km/h, model without front suspension. Columns: the same moments in each row, relative to the front wheel reaching the bump. The 10/10 reference and the Voiager 5 (11/10) pitch forward after the front-wheel strike (the Voiager 5 more slowly); with 12 in and 14 in front wheels the model rolls over the bump. Drawn as in Figure 8.

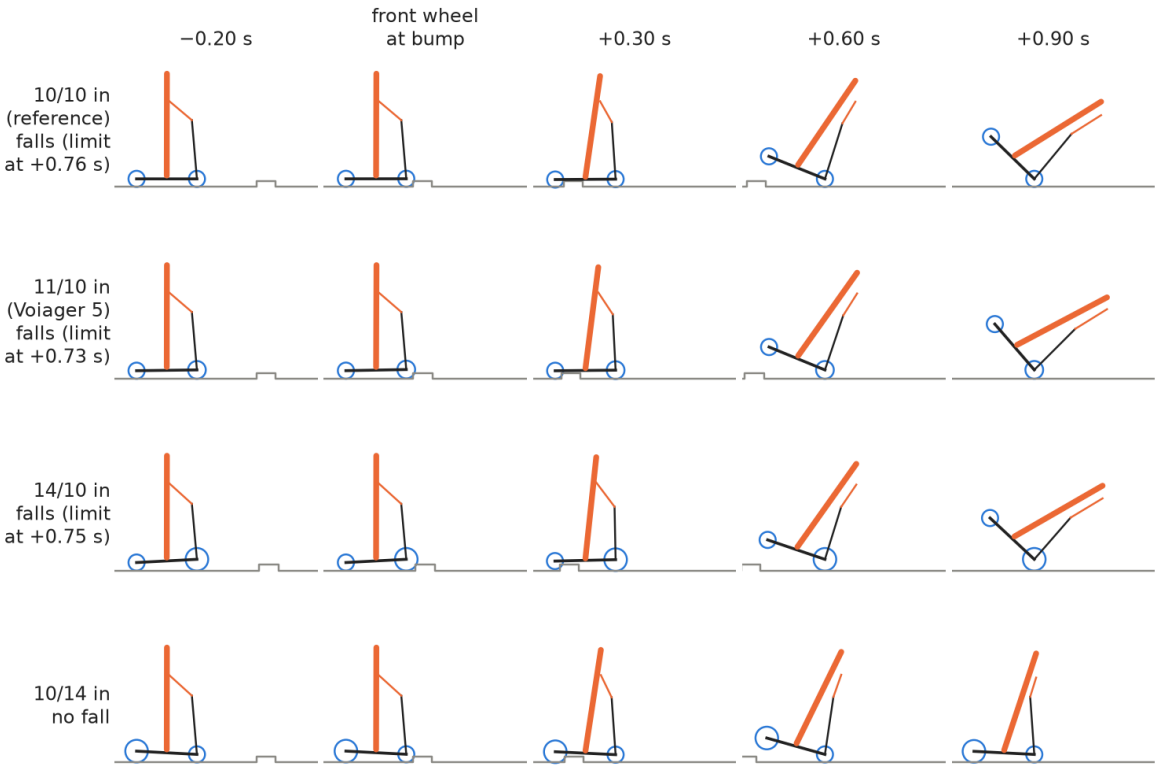


Figure D2: Key frames, short sharp-edged bump 90 mm high and 300 mm long at 15 km/h, model with the front spring. The front wheels cross; about 0.3 s later the rigid rear wheel strikes the edge and, with 10 in rear wheels (10/10, the Voiager 5's 11/10 and 14/10), the scooter pitches forward. With a 14 in rear wheel (10/14) the model does not fall. One run per configuration; an open question, not a result about rear-wheel size (text).

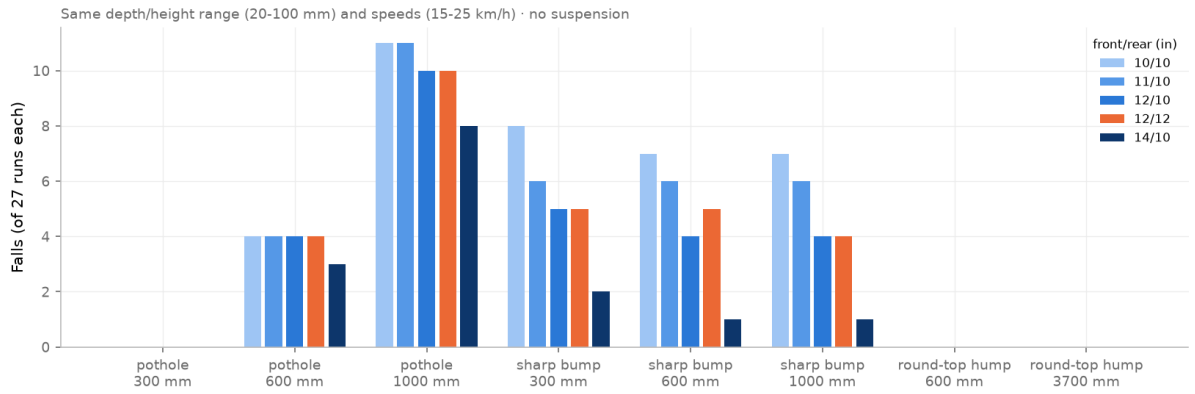


Figure D3: Falls by obstacle type and length, model without front suspension, per wheel configuration (same depth or height range, 20–100 mm, and speeds, 15–25 km/h). Exploratory; bump results are not validated against measurements.

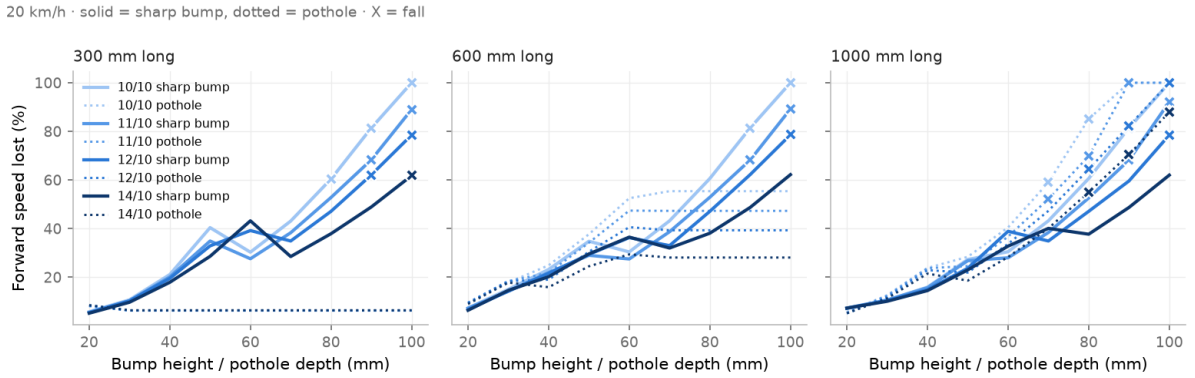


Figure D4: Forward speed lost on sharp-edged bumps (solid) and potholes of the same size (dotted), 20 km/h, model without front suspension; X marks a fall. Exploratory; bump results are not validated against measurements.

Uncertainties and what these results cannot show.

- *Contact model.* On sharp bumps the two contact models agreed on the fall outcome in 370 of 405 runs without the fork and 384 of 405 with it (potholes: 30 of 30), with disagreements in both directions; the cause is not established.
- *No validation.* Modelled jolts on sharp bumps (for example 3.3–8.5 g at 20–40 mm, 600 mm long, 20 km/h, with the fork, across the five configurations) exceed the 1.0–2.97 g measured for 99 of the 100 impacts in Part A, and no bumps were identified on the rides to compare with.
- *No sensitivity analysis* for bumps (tyre, rider bracing, fork parameters), and rear sizes were varied only on the 300 mm bump.
- *Simplifications* as for potholes: planar model, no steering, coasting, rigid rider pinned at the feet, idealised shapes, nominal wheel sizes from the Voiager 5's tyre markings.
- *Not shown:* that real bumps cause falls, that bumps are more or less dangerous than potholes on the road, any injury risk, or that any wheel size is safe.

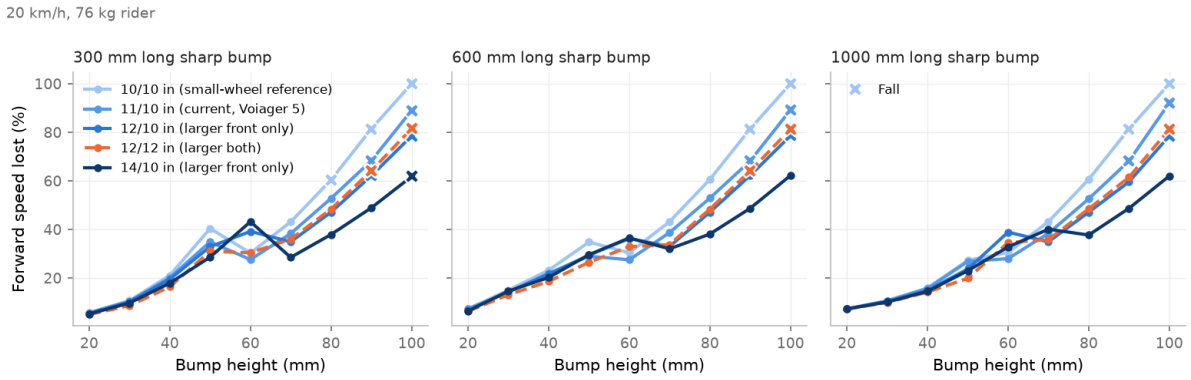


Figure D5: Forward speed lost on bumps by height, length and wheel configuration, model without front suspension. Exploratory; bump results are not validated against measurements.

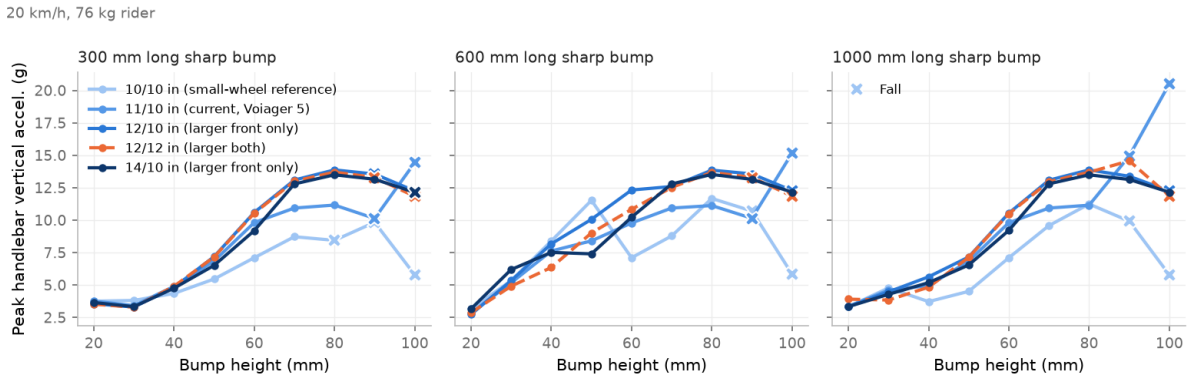


Figure D6: Simulated handlebar peak on bumps, model without front suspension. Values above the measured 1.0–2.97 g range are not supported by any measurement. Exploratory; bump results are not validated against measurements.

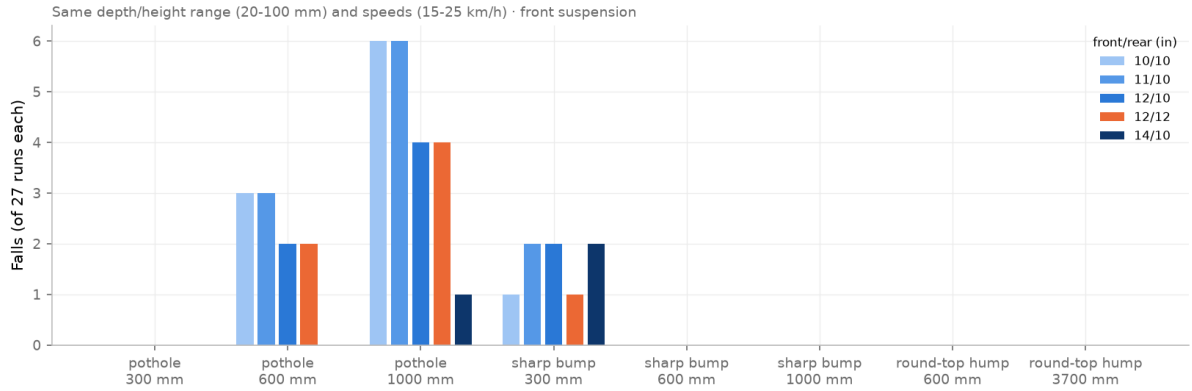


Figure D7: Falls by obstacle type and length, model with the front spring, per wheel configuration (same depth or height range, 20–100 mm, and speeds, 15–25 km/h). Exploratory; bump results are not validated against measurements.

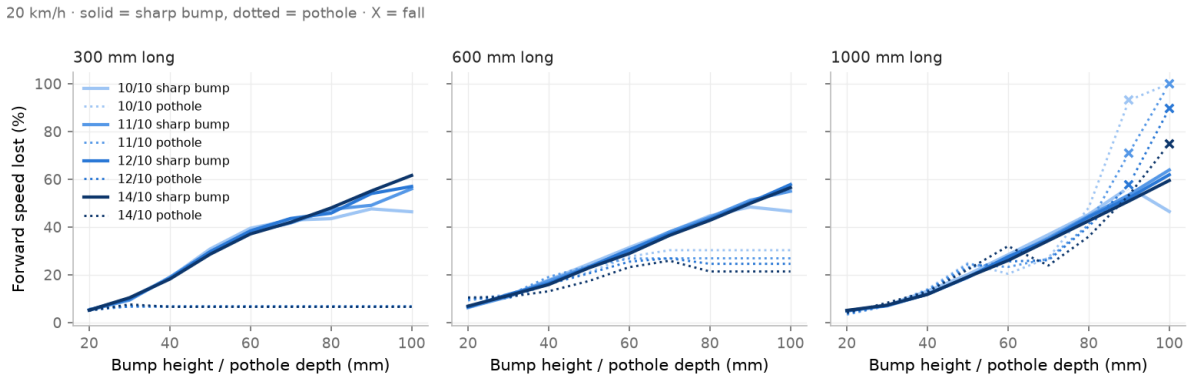


Figure D8: Forward speed lost on sharp-edged bumps (solid) and potholes of the same size (dotted), 20 km/h, model with the front spring; X marks a fall. Exploratory; bump results are not validated against measurements.

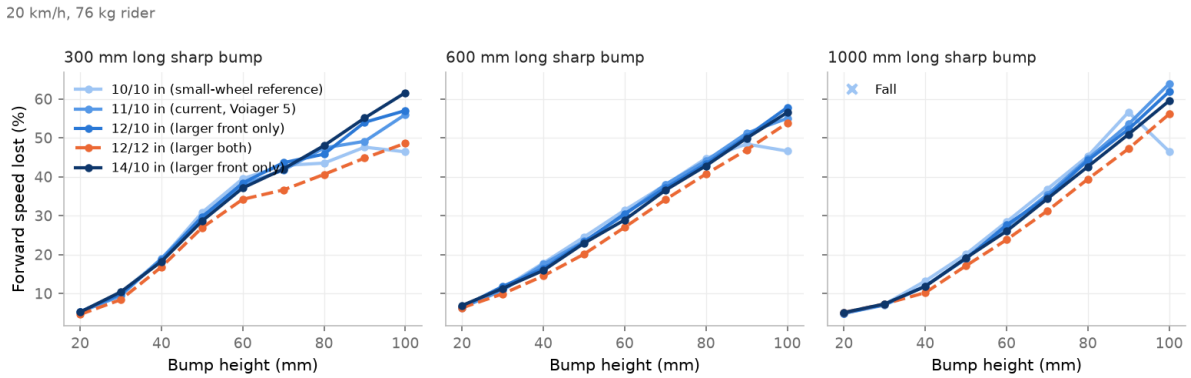


Figure D9: Forward speed lost on bumps by height, length and wheel configuration, model with the front spring. Exploratory; bump results are not validated against measurements.

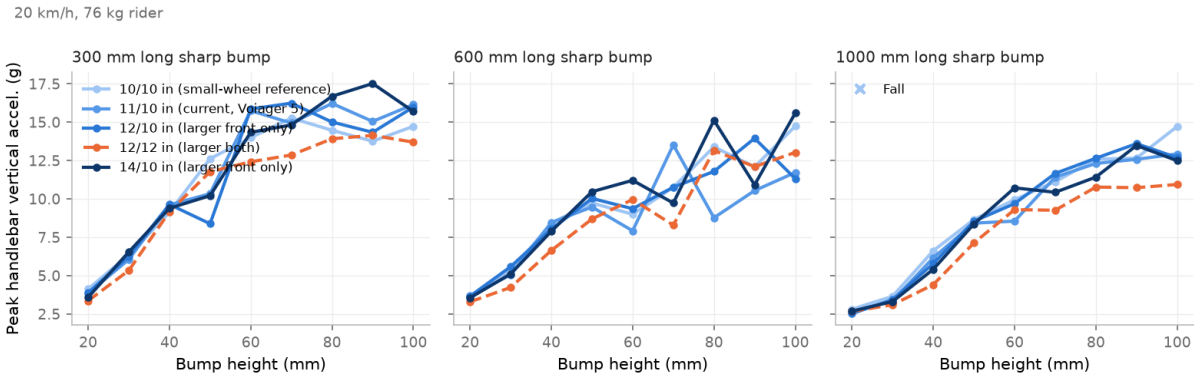


Figure D10: Simulated handlebar peak on bumps, model with the front spring. Values above the measured 1.0–2.97 g range are not supported by any measurement. Exploratory; bump results are not validated against measurements.

E Further model results

These figures give the full results behind the tables and text of the model sections. All come from the same planar model or geometry model with assumed parameters (Table 4); they compare wheel configurations and are not real-world thresholds. “With fork” is the main model of the Voiager 5; “no fork” is the rigid version kept for comparison. In every figure, 11/10 is the Voiager 5 that was ridden and 10/10 a smaller reference.

E.1 Geometry model

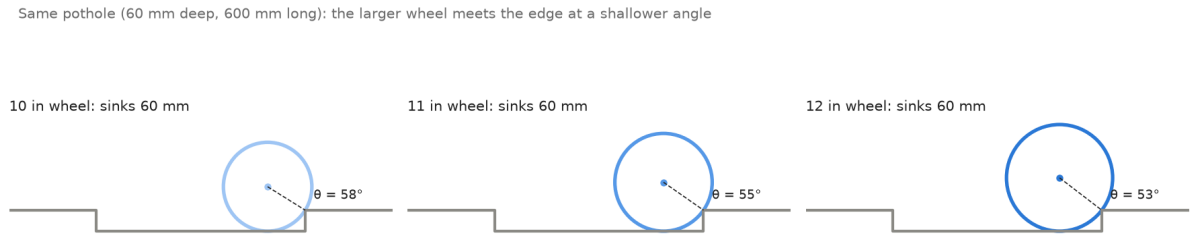


Figure E1: Geometry model schematic: 10 in (reference), 11 in (Voiager 5 front) and 12 in rigid wheels at the far edge of the same 60 mm deep, 600 mm long pothole, drawn sunk to full depth for illustration. Larger wheels meet the edge at a shallower angle θ.

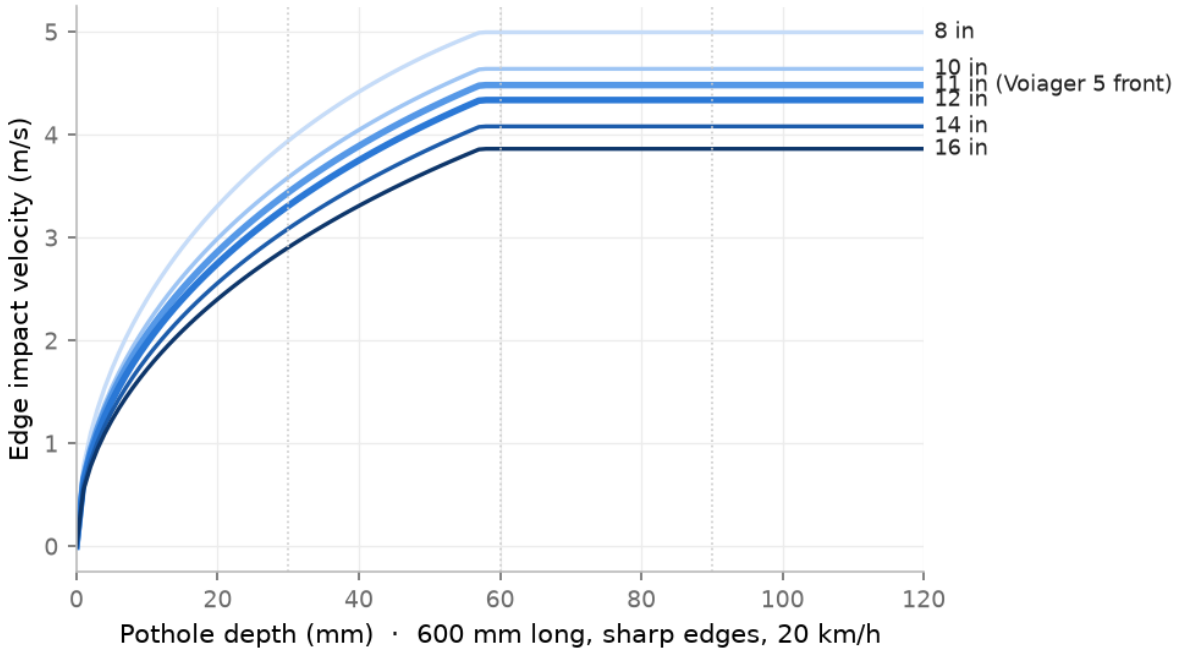


Figure E2: Geometry model: front-wheel edge impact velocity against pothole depth by wheel diameter (600 mm long, 20 km/h).

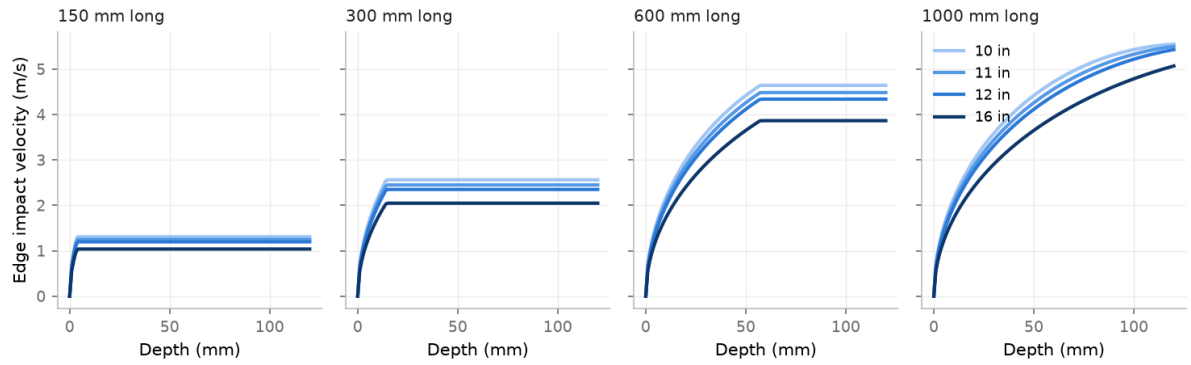


Figure E3: Geometry model: edge impact velocity by pothole depth and length, 20 km/h. Short potholes limit how far the wheel can drop at speed.

E.2 Main sweep

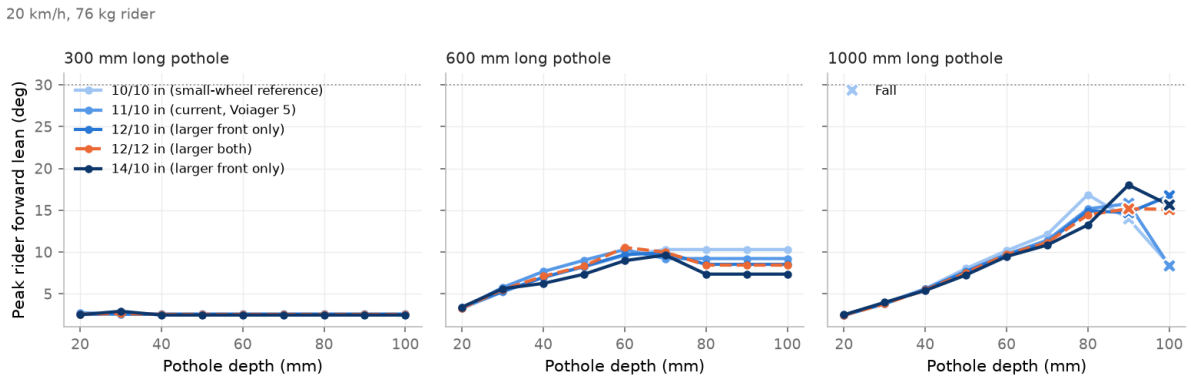


Figure E4: Main sweep, with front suspension (main model): peak forward lean of the rider by pothole depth and length at 20 km/h; X marks a fall. Planar model with assumed parameters (Table 4); relative comparison of wheel configurations only, not real-world thresholds.

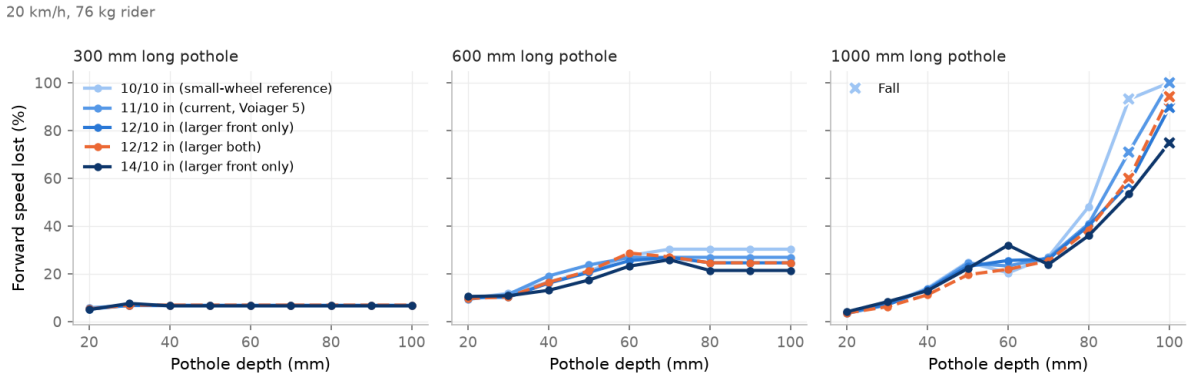


Figure E5: Main sweep, with front suspension (main model): forward speed lost crossing the pothole. Planar model with assumed parameters (Table 4); relative comparison of wheel configurations only, not real-world thresholds.

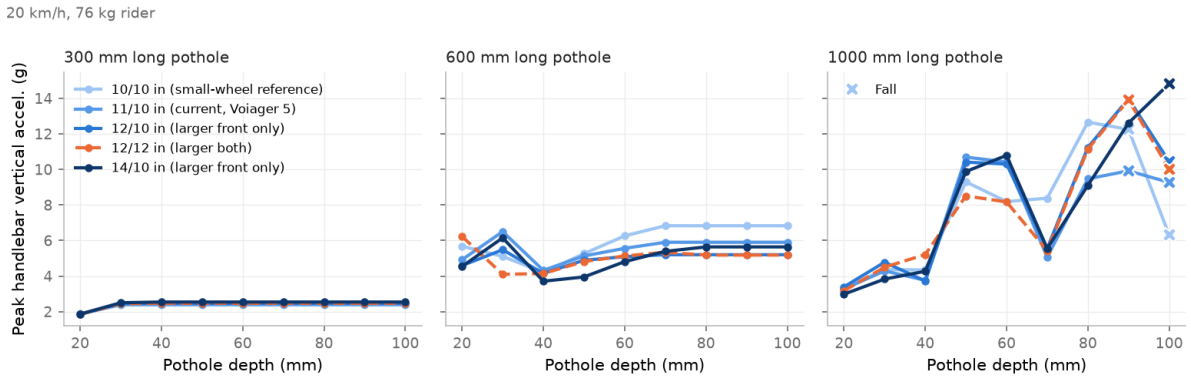


Figure E6: Main sweep, with front suspension (main model): simulated handlebar peak acceleration. Absolute values are consistent with the rides only for short potholes with the fork (Figure 10). Planar model with assumed parameters (Table 4); relative comparison of wheel configurations only, not real-world thresholds.

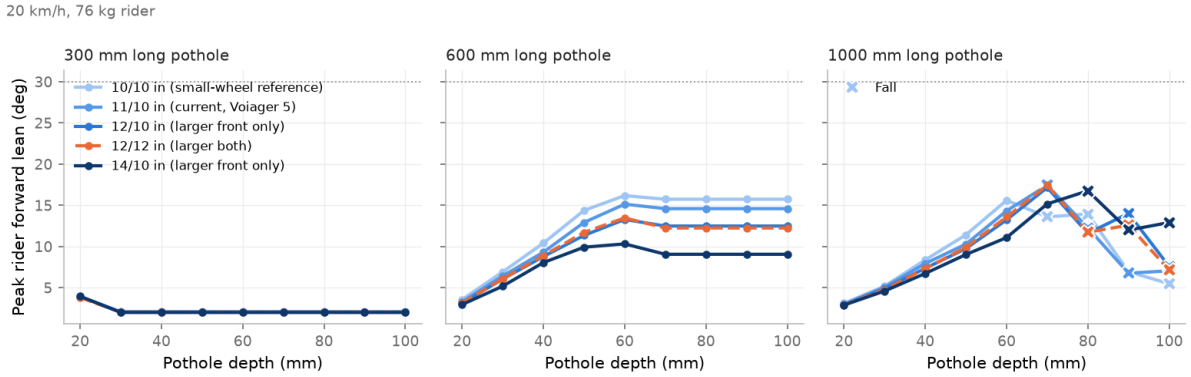


Figure E7: Main sweep, without suspension: peak forward lean of the rider by pothole depth and length at 20 km/h; X marks a fall. Planar model with assumed parameters (Table 4); relative comparison of wheel configurations only, not real-world thresholds.

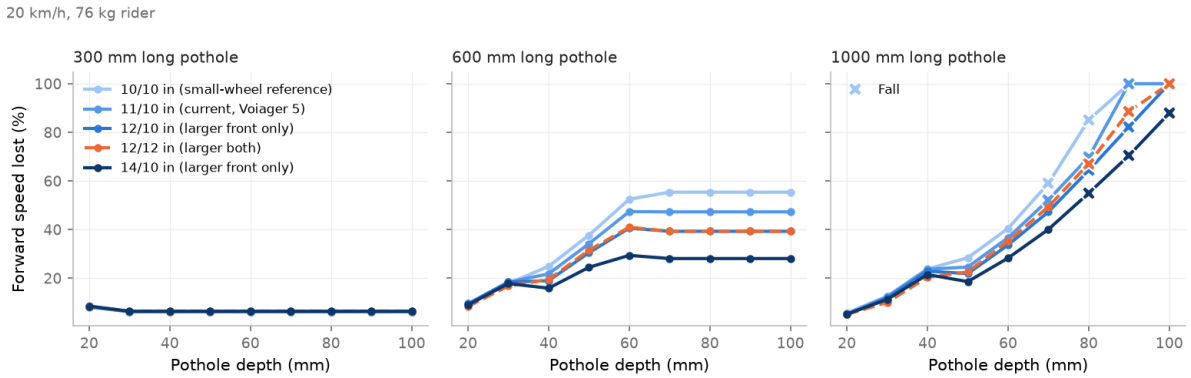


Figure E8: Main sweep, without suspension: forward speed lost crossing the pothole. Planar model with assumed parameters (Table 4); relative comparison of wheel configurations only, not real-world thresholds.

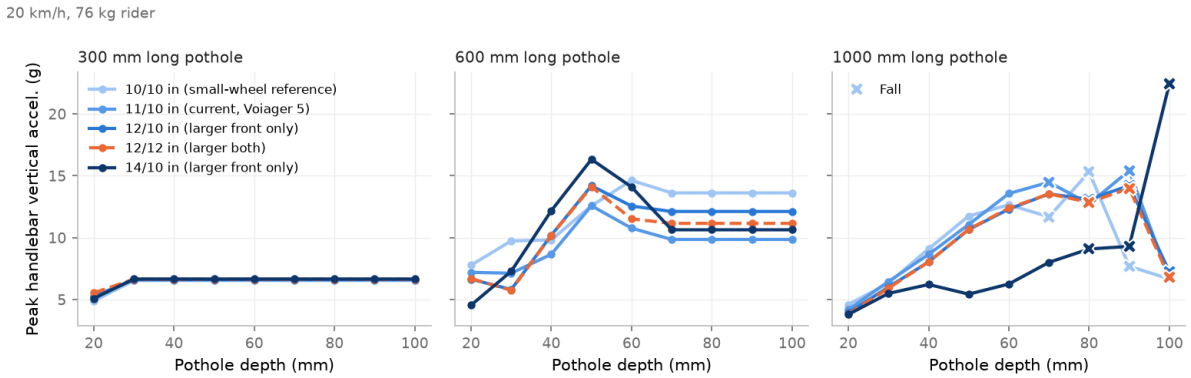


Figure E9: Main sweep, without suspension: simulated handlebar peak acceleration. Absolute values are consistent with the rides only for short potholes with the fork (Figure 10). Planar model with assumed parameters (Table 4); relative comparison of wheel configurations only, not real-world thresholds.

E.3 Front/rear ratio study

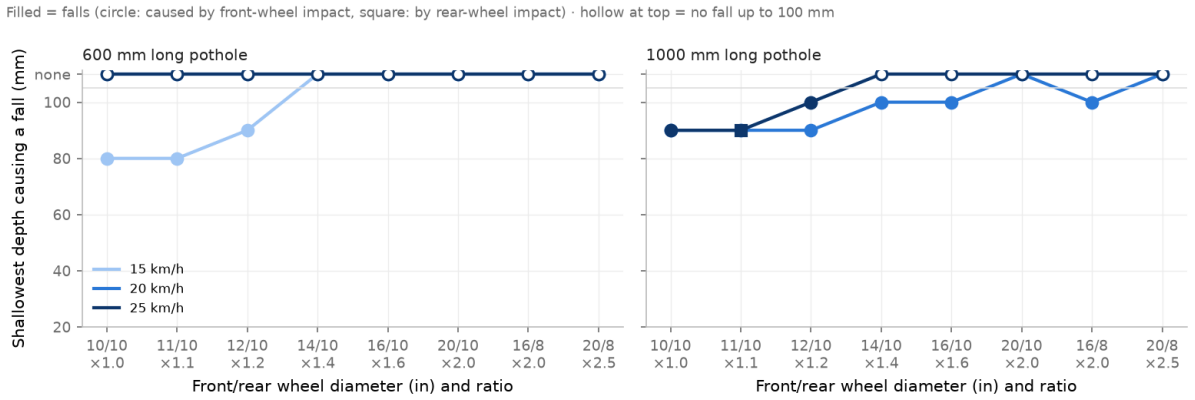


Figure E10: Ratio study, with front suspension: shallowest pothole depth (10 mm steps) at which the model fell, by front/rear diameter and speed; "none" = no fall up to 100 mm. No "too large" front wheel appeared in this planar model, which omits steering, frame geometry and cost. Planar model with assumed parameters (Table 4); relative comparison of wheel configurations only, not real-world thresholds.

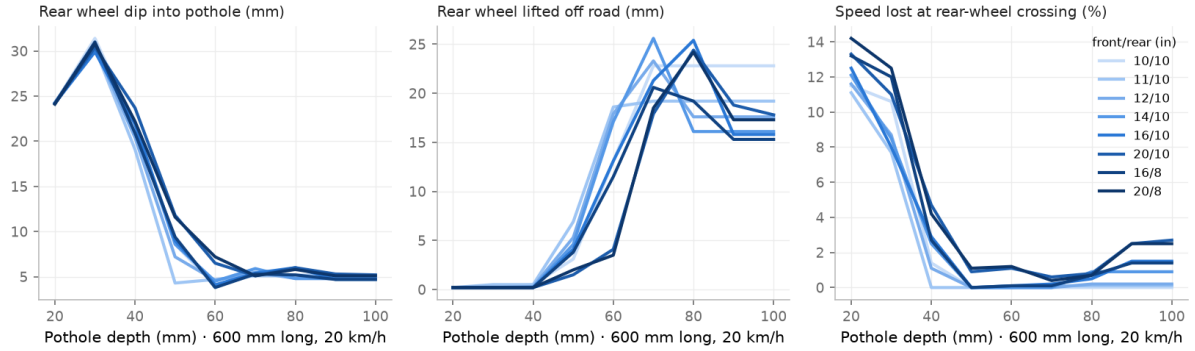


Figure E11: Ratio study, with front suspension: rear-wheel behaviour on a 600 mm pothole at 20 km/h (dip into the pothole, lift off the road, speed lost at the rear crossing). With the fork the configurations differ little, and the rear wheel is lifted by at most about 25 mm. Planar model with assumed parameters (Table 4); relative comparison of wheel configurations only, not real-world thresholds.

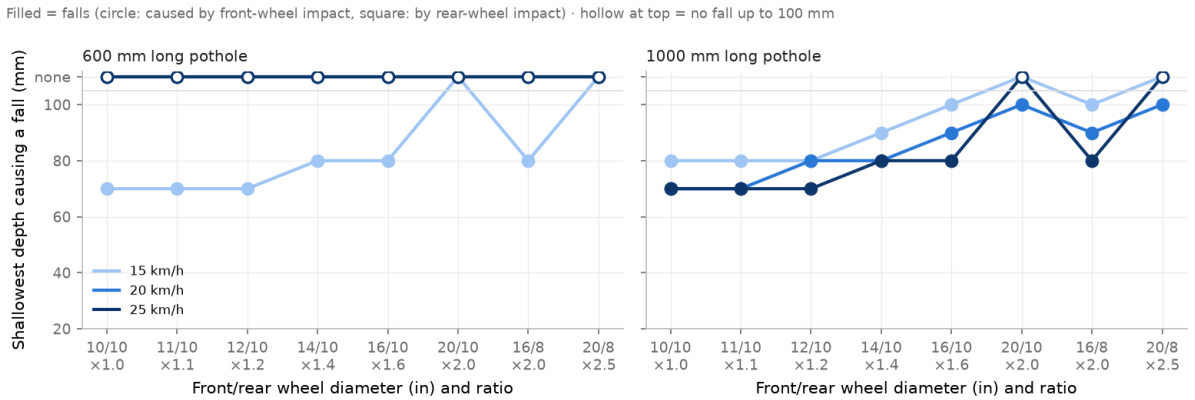


Figure E12: Ratio study, without suspension: shallowest pothole depth (10 mm steps) at which the model fell, by front/rear diameter and speed; "none" = no fall up to 100 mm. No "too large" front wheel appeared in this planar model, which omits steering, frame geometry and cost. Planar model with assumed parameters (Table 4); relative comparison of wheel configurations only, not real-world thresholds.

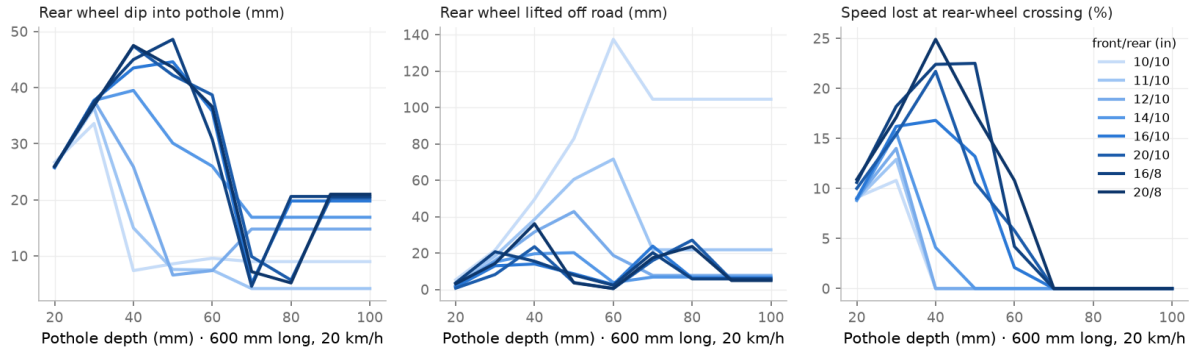


Figure E13: Ratio study, without suspension: rear-wheel behaviour on a 600 mm pothole at 20 km/h (dip into the pothole, lift off the road, speed lost at the rear crossing). With small front wheels the rear is lifted clear; with large ones it drops in and gives a second jolt, which caused no falls here. Planar model with assumed parameters (Table 4); relative comparison of wheel configurations only, not real-world thresholds.

E.4 Soft pothole edges

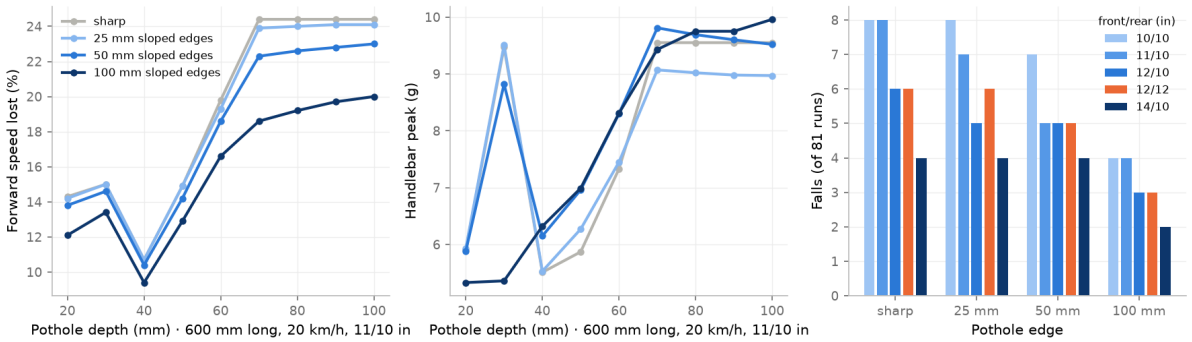


Figure E14: Sloped (raised-cosine) pothole edges, with front suspension, profile contact model: speed lost and handlebar peak for the Voiager 5 (11/10; 600 mm, 20 km/h), and falls of 81 runs per configuration by edge width. The profile contact's absolute accelerations are too high (no tyre enveloping) and are used only for comparison between edge widths. Planar model with assumed parameters (Table 4); relative comparison of wheel configurations only, not real-world thresholds.

E.5 Summary tables

The tables below give the fall thresholds without suspension and the sensitivity analysis behind the statements in the model sections. Depths are on a 10 mm grid: "80" means a first fall between 70 and 80 mm. – = no fall up to the deepest case simulated (100 mm). Rows in which no configuration fell are left out. All values come from the planar model with assumed parameters (Table 4) and compare configurations; they are not real-world thresholds.

Table E1: Shallowest pothole depth (mm) causing a fall, model without suspension (counterpart of Table 7). 11/10 (V5) = the Voi Voiager 5 that was ridden; 10/10 (ref.) = smaller reference wheels.

Length, speed	10/10 (ref.)	11/10 (V5)	12/10	12/12	14/10
600 mm, 15 km/h	70	70	70	70	80
1000 mm, 15 km/h	80	80	80	80	90
1000 mm, 20 km/h	70	70	80	80	80
1000 mm, 25 km/h	70	70	70	70	80
Falls (of 81 runs)	**15**	**15**	**14**	**14**	**11**

Table E2: Sensitivity analysis, with front suspension (main model): first fall depth (mm) on a 1000 mm pothole at 20 km/h / forward speed lost on a 60 mm deep, 600 mm pothole at 20 km/h. One parameter changed at a time from the baseline. 11/10 (V5) = the Voi Voiager 5 that was ridden; 10/10 (ref.) = smaller reference wheels.

Setting	10/10 (ref.)	11/10 (V5)	12/10	12/12	14/10
baseline	90 / 27.6%	90 / 26.8%	90 / 25.5%	90 / 28.7%	100 / 23.2%
stiffer tyre (τ 0.015 s)	90 / 30.2%	90 / 26.8%	90 / 25.1%	90 / 25.7%	90 / 26.3%
softer tyre (τ 0.035 s)	80 / 27.4%	90 / 26.7%	90 / 23.5%	90 / 23.7%	90 / 20.6%
weaker bracing (1000 N m/rad)	90 / 27.8%	90 / 29.9%	90 / 26.4%	90 / 24.7%	90 / 21.3%
stronger bracing (3000 N m/rad)	90 / 28.1%	90 / 28.3%	90 / 25.4%	90 / 25.5%	100 / 22.2%
softer fork (15 kN/m)	80 / 26.5%	80 / 24.1%	90 / 25.8%	90 / 24.6%	90 / 22.1%
stiffer fork (40 kN/m)	90 / 32.7%	90 / 25.8%	90 / 26.1%	90 / 26.9%	100 / 21.3%
less fork damping (ratio 0.15)	80 / 24.8%	80 / 24.9%	80 / 24.0%	80 / 25.4%	80 / 22.4%
more fork damping (ratio 0.5)	90 / 31.1%	90 / 29.0%	90 / 25.5%	100 / 25.4%	100 / 22.4%

Table E3: Sensitivity analysis, without suspension: first fall depth (mm) on a 1000 mm pothole at 20 km/h / forward speed lost on a 60 mm deep, 600 mm pothole at 20 km/h. One parameter changed at a time from the baseline. 11/10 (V5) = the Voi Voiager 5 that was ridden; 10/10 (ref.) = smaller reference wheels.

Setting	10/10 (ref.)	11/10 (V5)	12/10	12/12	14/10
baseline	70 / 52.5%	70 / 47.4%	80 / 40.6%	80 / 41.0%	80 / 29.4%
stiffer tyre (τ 0.015 s)	60 / 69.9%	70 / 60.4%	70 / 52.4%	70 / 52.9%	80 / 36.2%
softer tyre (τ 0.035 s)	70 / 42.2%	80 / 38.1%	80 / 33.7%	80 / 33.4%	90 / 26.0%
weaker bracing (1000 N m/rad)	70 / 52.7%	80 / 47.9%	80 / 41.0%	80 / 42.9%	90 / 29.9%
stronger bracing (3000 N m/rad)	70 / 51.5%	70 / 45.7%	80 / 38.7%	80 / 40.0%	80 / 26.2%